

30-DAY COACHING ACADEMY EDITION

Computer Science XI

The Bilingual Teacher's Edition rebuilt as a 30-day academy batch — 90-minute classes, board work, timed drills, homework stamps, weekly chapter exams, mock paper and repair clinic.

Bilingual support: English + اردو

تیس دن کی کوچنگ اکیڈمی — نوے منٹ کلاس، بورڈ، ڈرل، ہوم ورک

01 Computer Systems

02 Computational Thinking & Algorithm

03 Programming Fundamentals

04 Data and Analysis

05 Application and Impacts of Computing

06 Digital Literacy

XI

30-day academy calendar

One 90-minute class a day. Tick the day after homework is stamped. Do not skip chapter-exam, mock, or Golden-blitz days.

روزانہ ایک نوے منٹ کی کلاس۔ باب کے امتحان، ماک اور گولڈن بلٹز نہ چھوڑیں۔

Day	Focus	Kind	★ Golden today
Week 1 · academy batch			
Day 01	Computer Systems (1/5)	Teach	Analog and Digital Signals
Day 02	Computer Systems (2/5)	Teach	Logic Gates (Basic Gates: AND, OR, NOT)
Day 03	Computer Systems (3/5)	Teach	Logic Diagrams, Karnaugh Maps (K-Maps)
Day 04	Computer Systems (4/5)	Teach	Software Development Life Cycle (SDLC), The Waterfall Model
Day 05	Computer Systems (5/5)	Chapter exam	The Agile Model, Communication Models and the OSI 7-Layer Model
Week 2 · academy batch			
Day 06	Computational Thinking & Algorithm (1/3)	Teach	Algorithm, Algorithm vs Pseudocode, Decomposition
Day 07	Computational Thinking & Algorithm (2/3)	Teach	Sorting Algorithms: Bubble Sort, Sorting Algorithms: Selection Sort
Day 08	Computational Thinking & Algorithm (3/3)	Chapter exam	Searching Algorithms: Binary Search
Day 09	Programming Fundamentals (1/5)	Teach	—
Day 10	Programming Fundamentals (2/5)	Teach	Fundamentals of Python Programming, Variables, Data Types in Python, Operators and Operands: Arithmetic and Assignment
Week 3 · academy batch			
Day 11	Programming Fundamentals (3/5)	Teach	Relational, Logical and Membership Operators, Bitwise Operators, Control Structures, Selection Statements: if and if-else

Day	Focus	Kind	★ Golden today
Day 12	Programming Fundamentals (4/5)	Teach	Selection Statements: if-elif-else and Nested if, Repetition Statements: the for Loop and range(), The while Loop, break and continue, Nested Loops and for vs while
Day 13	Programming Fundamentals (5/5)	Chapter exam	—
Day 14	Data and Analysis (1/3)	Teach	Keys and Integrity Constraints
Day 15	Data and Analysis (2/3)	Teach	Entity Relationship Model (ER-Model), Referential Integrity
Week 4 · academy batch			
Day 16	Data and Analysis (3/3)	Chapter exam	Creating Queries in MS Access
Day 17	Application and Impacts of Computing (1/3)	Teach	Artificial Intelligence (AI), Internet of Things (IoT) & Components
Day 18	Application and Impacts of Computing (2/3)	Teach	Information Sources (Primary, Secondary, Tertiary)
Day 19	Application and Impacts of Computing (3/3)	Chapter exam	Assistive Technologies
Day 20	Digital Literacy (1/4)	Teach	Types of Data, Data-Collection Strategies, Interviews and Surveys
Week 5 · academy batch			
Day 21	Digital Literacy (2/4)	Teach	Prototypes, Observation and Simulations, Primary and Secondary Data
Day 22	Digital Literacy (3/4)	Teach	Case Study: Digital Inquiry Project, Steps 1 – 3: Advanced Search and Methodology, Steps 4 – 6: Survey Preparation and Primary Data Collection
Day 23	Digital Literacy (4/4)	Chapter exam	Steps 7 – 9: Secondary Data and Spreadsheet Organisation, Steps 10 – 12: Data Analysis, Conclusion and Recommendations, The Final Digital Artefact
Day 24	★ only · Chapters 1–2	Golden blitz	14 Golden cards
Week 6 · academy batch			
Day	Focus	Kind	★ Golden today
Day 25	★ only · Chapters 3–4	Golden blitz	16 Golden cards
Day 26	★ only · Chapters 5–6	Golden blitz	15 Golden cards
Day 27	Say every chapter recap aloud, books closed	Recap	All chapters
Day 28	2 hours 30 minutes · 75 marks · no notes	Mock	Whole syllabus
Day 29	Mark to the key. List every ★ Golden the batch missed.	Mark	Missed Goldens
Day 30	Redo the Day 29 list only. Dismiss by 6 pm.	Repair	Missed Goldens only

Batch register — starter marks

Three 1-mark oral or written questions from yesterday. Stamp homework full / half / zero in the last column. Re-teach any Golden that more than a third of the batch missed.

Day	Present	Q1	Q2	Q3	Mean /10	HW stamp
Day 01						
Day 02						
Day 03						
Day 04						
Day 05						
Day 06						
Day 07						
Day 08						
Day 09						
Day 10						
Day 11						
Day 12						
Day 13						
Day 14						
Day 15						
Day 16						
Day 17						
Day 18						
Day 19						
Day 20						
Day 21						
Day 22						
Day 23						
Day 24						
Day 25						
Day 26						
Day 27						
Day 28						
Day 29						
Day 30						

حاضری، تین اسٹارٹر نمبر، اور ہوم ورک کی مہر اس جدول میں لگائیں۔

How to run this 30-Day Coaching Academy Edition

This book is the **Bilingual Teacher's Edition** rebuilt as a **30-day coaching academy batch**. It is not a 40-minute school period and not a 3-hour self-study crash. A standard academy class is **90 minutes**. The coach stands at the board. Students copy, drill, and take homework home. ★ Golden topics are never skipped.

یہ کتاب تیس دن کی کوچنگ اکیڈمی بیچ کے لیے ہے۔ ہر کلاس نوے منٹ کی ہے۔ استاد بورڈ پر پڑھاتا ہے، طلبہ کاپی کرتے ہیں، ڈرل حل کرتے ہیں، ہوم ورک گھر لے جاتے ہیں۔ گولڈن موضوع نہ چھوڑیں۔

The 30 days

- **Days 1–23 — Teach the whole grade.** Every topic from the teacher's notes appears as a board card. The last day of each chapter is a 75-mark closed-book exam in the academy hall.
- **Days 24–26 — ★ Golden blitz.** No new non-Golden topic. Error clinic on high-yield board wording.
- **Day 27 — Oral recap.** Students say each chapter recap at the board, books closed.
- **Day 28 — Mock morning.** 2 hours 30 minutes. Hall conditions. No notes, no phones.
- **Day 29 — Mark and repair list.** Mark to the sealed key. Every missed Golden goes on the list.
- **Day 30 — Repair clinic, then stop.** Redo only the list. Dismiss by 6 pm.

The 90-minute academy class (Days 1–23 teach days)

1. **0–10 min — Starter test.** Three oral or written 1-mark questions from yesterday. Marks go in the batch register in this book.
2. **10–35 min — Concept lock.** Write the definition, table and diagram on the whiteboard. Students copy. Phones in the box at the door.
3. **35–55 min — Board working.** Do the worked example. Hide it. Two students redo it on the board.
4. **55–75 min — Academy drill.** Timed. 1-mark, then 3-mark, then 5-mark. Golden days also take an 8-mark. Mark at once with the recipe.
5. **75–85 min — Error clinic.** Collect the three most common wrong answers. Correct them in one box on the board.
6. **85–90 min — Homework out.** One 5-mark (or 8-mark if Golden) + redraw the board from memory. Collect next class. Stamp full / half / zero.

Chapter-exam days

On the last day of a chapter the class is a hall paper: 75 marks · 1 hour 15 minutes · 15 MCQ + 10 short + any 3 long. Use the leftover minutes to collect scripts. Do not mark in front of the batch. Re-teach any Golden below 60% in the next starter.

باب کے آخری دن پچھتر نمبر کا پیپر ہال میں کرائیں۔ اگلی کلاس کے اسٹارٹر میں کمزور گولڈن دوبارہ پڑھائیں۔

How academy answers score marks

- **1 mark:** one exact definition or one named fact.
- **2 marks:** definition + one local example (JazzCash, school portal, NADRA, load-shedding).
- **3 marks:** definition + two clear points + example.
- **5 marks:** definition, a four-point table or explanation, example, and a labelled diagram if the topic has one.
- **8–10 marks:** 5-mark structure + full working (trace, K-map, code, ER diagram) + a one-line conclusion.

Train this wording every class: *Define. Explain. Example. Diagram. Working.* An answer with no working is a half-answer.

Batch rules

- One 90-minute class a day, six days a week, for thirty days. No extra unpaid “make-up” class unless a Golden exam mark is below 60%.
- If the batch falls behind, drop non-Golden cards that day. Never skip a chapter exam, the mock, or a Golden blitz day.
- Homework is collected. Uncollected homework is a zero in the register, not a warning.

What the extra academy boxes mean

COACH RUN-SHEET

How to run this day: starter, board, two-student redo, drill, error clinic.

FULL-MARK ANSWER RECIPE

The sentence pattern that scores 1, 3, 5 and 8 marks today.

ACADEMY DRILL

Timed class questions with answers — not for homework peeking.

BATCH HOMEWORK

What the batch takes home, and how you stamp it tomorrow.

ACADEMY CALENDAR

The 30-day calendar and the batch register.

GOLDEN TOPIC

★ **Golden** — never skip, never rush, test it every week until the board.

Chapter 1: Computer Systems

Computer Systems (1/5) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Discrete and Continuous Quantities · Introduction to Digital Systems · Analog and Digital Signals · Boolean Algebra

★ **Must lock:** Analog and Digital Signals

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Define the last Golden topic in one exact sentence.
2. Redraw yesterday's table or diagram with labels.
3. Give one local example (JazzCash, NADRA, school portal, load-shedding).

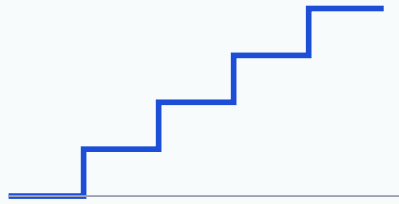
کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

1.1.1 Discrete and Continuous Quantities

Everything around you that can be measured or counted is a quantity.

- Quantities come in two kinds.
- Discrete quantities are separate, countable items with specific values.
- You can count them one by one: 1, 2, 3 ... There is nothing "in between".

Discrete (counted)	Continuous (measured)
Number of students in a classroom	Temperature of water
Number of cars in a parking lot	Speed of a moving car
Has fixed, separate values	Has any value in a range



Discrete (Stairs)



Continuous (Ramp)

Stairs have fixed steps (discrete). A ramp lets you stand at any point (continuous).

Remember: Stairs vs Ramp. On stairs you can only stand on step 1, 2 or 3 — that is discrete data. On a ramp you can stand at any height — that is continuous data. Ask...

منفصل (Discrete) مقداریں وہ ہیں جنہیں ایک ایک کر کے گنا جا سکے، جیسے کلاس میں طلبہ کی تعداد۔

1.1.2 Introduction to Digital Systems

A digital system is an electronic system that uses only two values: 0 and 1.

- 0 represents OFF, LOW, or FALSE.
- 1 represents ON, HIGH, or TRUE.

Advantage	Detail
Reliable	Less affected by noise and interference.
Accurate	No loss of quality when copying data.
Easy to design	Only two states (0 and 1) to handle.
Programmable	Can be easily controlled by software.

Remember: Think of a bit as a light switch: it is either ON (1) or OFF (0). A digital system is a huge panel of such switches.

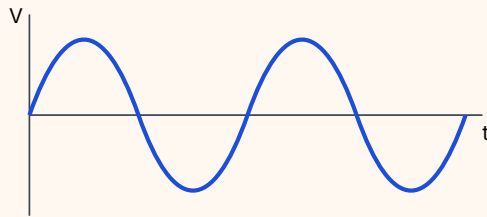
ڈیجیٹل نظام صرف دو قیمتیں استعمال کرتا ہے: 0 اور 1، جنہیں بٹ (Bit) کہتے ہیں۔

1.1.3 Analog and Digital Signals ★ Golden

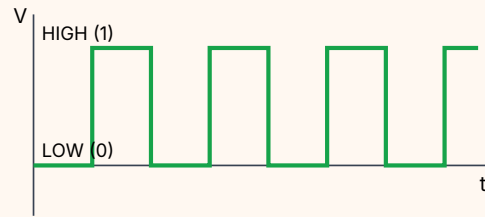
A signal is a changing voltage (or other quantity) that carries information.

- Analog signals are continuous signals that change smoothly over time, like a sine wave.
- Examples: the human voice, a traditional mercury thermometer.
- Digital signals have only two levels: HIGH (1) and LOW (0).

Feature	Analog data	Digital data
Nature	Continuous flow.	Discrete, separate values (bits).
Wave shape	Sine wave (smooth curve).	Square wave (sudden jumps).
Reliability	Low (affected by noise).	High (resistant to noise).
Examples	Human voice, traditional thermometer.	Computer data.



Analog Signal (Sine Wave)



Digital Signal (Square Wave)

An analog signal changes smoothly; a digital signal jumps between only two levels.

Remember: Analog = A smooth curve. Digital = Doorsteps. Analog looks like ocean waves; digital looks like a castle wall (square teeth).

Exam: Common questions: "Differentiate between analog and digital signals" (3–4 marks) and "Draw an analog and a digital signal." Write the comparison as a table with at least...

اینا لگ سگنل وقت کے ساتھ ہموار انداز میں بدلتا ہے، جیسے سائن ویو (Sine Wave)؛ انسانی آواز اس کی مثال ہے۔

1.1.4 Boolean Algebra

Boolean algebra was developed by the English mathematician George Boole.

- It is a kind of algebra that deals with only two values: TRUE (1) and FALSE (0).
- It is the foundation of digital electronics and computer programming — every if condition in a program is a small piece of Boolean algebra.

Concept	Description	Example
Variables	Letters that hold the value 0 or 1.	A, B, C, X, Y
Operations	Actions performed on variables.	AND, OR, NOT
Expressions	Combinations of variables and operations.	A AND B, $Y = A + B$

Trap: In Boolean algebra "+" means OR, not addition.

بولین الجبرا جارج بول (George Boole) نے بنایا۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Discrete and Continuous Quantities:** Students sometimes say "money is continuous because amounts are large".
- **Analog and Digital Signals:** Do not draw a digital signal with sloping sides.
- **Boolean Algebra:** In Boolean algebra "+" means OR, not addition.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Analog and Digital Signals**. Write: A signal is a changing voltage (or other quantity) that carries information.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Analog signals are continuous signals that change smoothly over time, like a sine wave. (2) Examples: the human voice, a traditional mercury thermometer. — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + a fully labelled diagram copied from the board + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. What shape of wave does a digital signal make?

3 marks. Explain Analog and Digital Signals with two clear points and one example.

5 marks. Common questions: "Differentiate between analog and digital signals" (3–4 marks) and "Draw an analog and a digital signal." Write the comparison as a table with at least three features (nature, wave...

8 marks. Analog and Digital Signals: 5-mark answer plus a fully labelled diagram copied from the board.

Answers: 1. A square wave 3. Analog signals are continuous signals that change smoothly over time, like a sine wave.; Examples: the human voice, a traditional mercury thermometer. 5. Use the recipe in the blue box.

Model opening: A signal is a changing voltage (or other quantity) that carries information. 8. Same 5-mark body + a fully labelled diagram copied from the board + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Analog and Digital Signals** using the recipe in this topic. Include a fully labelled diagram copied from the board. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 1: Computer Systems

Computer Systems (2/5) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Boolean Operations · Truth Tables · Logic Gates (Basic Gates: AND, OR, NOT) · Universal Logic Gates (NAND and NOR)

★ **Must lock:** Logic Gates (Basic Gates: AND, OR, NOT)

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. What shape of wave does a digital signal make?
2. Is the number of pages in a book discrete or continuous?
3. What does a 1 represent in a digital system?

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

1.1.5 Boolean Operations

There are three basic Boolean operations. Every other operation can be made from them.

- Every other operation can be made from them.
- AND operation ($\cdot$) — gives output 1 only when ALL inputs are 1.
- Expression: $Y = A \cdot B$ (often written just AB).

Operation	Real-life analogy
AND	A door with two locks. The door opens only when both locks are unlocked.
OR	A room with two doors. You can enter if either door is open.
NOT	A room light switch. When the switch is ON the room is lit; when it is OFF the room is dark — the state is the opposite of "dark".

Remember: AND is "strict" — everyone must agree (all 1s). OR is "easy-going" — one 1 is enough. NOT is "the rebel" — it always says the opposite.

AND آپریشن تب ہی 1 دیتا ہے جب تمام ان پٹ 1 ہوں (دو تالوں والا دروازہ).

1.1.6 Truth Tables

A truth table shows all possible input combinations and the output for each one, for a logic gate or a circuit.

- Count the number of inputs (n).
- Calculate total rows = 2^n .
- List all input combinations in binary counting order (0, 1, 2, 3 ... in binary).
- Calculate the output for each combination.

A	B	Combination
0	0	Row 1
0	1	Row 2
1	0	Row 3
1	1	Row 4

Remember: Quick fill trick for 3 inputs: In column C write 0,1,0,1... (change every row). In column B write 0,0,1,1... (change every 2 rows). In column A write...

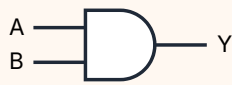
ٹروٹھ ٹیبل (Truth Table) کسی گیٹ یا سرکٹ کے تمام ممکنہ ان پٹ اور ان کے آؤٹ پٹ دکھاتا ہے۔

1.1.7 Logic Gates (Basic Gates: AND, OR, NOT) ★ Golden

Logic gates are electronic circuits that perform Boolean operations.

- They take one or more input bits and produce one output bit.
- They are the building blocks of all digital systems — a processor contains billions of them.
- The three basic gates match the three basic operations: the AND gate, the OR gate and the NOT gate (also called an inverter).

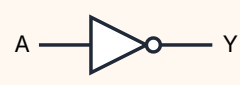
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



$$\text{AND: } Y = A \cdot B$$



$$\text{OR: } Y = A + B$$



$$\text{NOT: } Y = A'$$

Symbols of the basic gates. AND has a flat back and round front ("D" shape); OR has a curved back and pointed front; NOT is a triangle with a small circle (bubble).

Remember: Shortcut: AND output is 0 if any input is 0. OR output is 1 if any input is 1. Learn just the "special row": AND is 1 only at 11; OR is 0 only at 00.

Exam: You are often asked to "draw the symbol, write the expression and truth table" of a gate (2–3 marks each). Draw neat symbols with labelled inputs A, B and output Y....

لاجک گیٹس الیکٹرانک سرکٹ ہیں جو بولین آپریشن انجام دیتے ہیں اور تمام ڈیجیٹل نظاموں کی بنیادی اینٹیں ہیں۔

1.1.7 Universal Logic Gates (NAND and NOR)

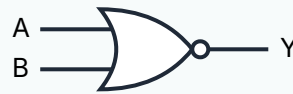
NAND means "NOT AND": it is an AND gate followed by a NOT.

- NOR means "NOT OR": an OR gate followed by a NOT.
- On the symbol, the NOT is shown as a small bubble at the output.
- NAND and NOR are called universal gates because any logic circuit in the world can be built using only NAND gates (or only NOR gates).

A	B	A·B	Y
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0



$$\text{NAND: } Y = (A \cdot B)'$$



$$\text{NOR: } Y = (A + B)'$$

NAND = AND symbol + bubble. NOR = OR symbol + bubble.

Remember: NAND is the exact opposite of AND (0 only at 11). NOR is the exact opposite of OR (1 only at 00). Just flip the AND/OR output column.

NAND کا مطلب ہے NOT AND اور NOR کا مطلب ہے NOT OR.

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Truth Tables:** Do not list rows in a random order — you may miss or repeat a combination.
- **Logic Gates (Basic Gates: AND, OR, NOT):** Do not mix up the AND and OR symbols.
- **Universal Logic Gates (NAND and NOR):** Do not forget the bubble.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Logic Gates (Basic Gates: AND, OR, NOT)**. Write: Logic gates are electronic circuits that perform Boolean operations.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) They take one or more input bits and produce one output bit. (2) They are the building blocks of all digital systems — a processor contains billions of them. — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + a complete truth table and a labelled gate / logic diagram + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. What is the output of an OR gate when $A = 0$ and $B = 0$?

3 marks. Explain Logic Gates (Basic Gates: AND, OR, NOT) with two clear points and one example.

5 marks. You are often asked to "draw the symbol, write the expression and truth table" of a gate (2–3 marks each). Draw neat symbols with labelled inputs A, B and output Y. Write the full truth table in...

8 marks. Logic Gates (Basic Gates: AND, OR, NOT): 5-mark answer plus a complete truth table and a labelled gate / logic diagram.

Answers: 1. 0 3. They take one or more input bits and produce one output bit.; They are the building blocks of all digital systems — a processor contains billions of them. 5. Use the recipe in the blue box. Model opening: Logic gates are electronic circuits that perform Boolean operations. 8. Same 5-mark body + a complete truth table and a labelled gate / logic diagram + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Logic Gates (Basic Gates: AND, OR, NOT)** using the recipe in this topic. Include a complete truth table and a labelled gate / logic diagram. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 1: Computer Systems

Computer Systems (3/5) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Advanced Logic Gates (XOR and XNOR) · Logic Expressions, Minterms and Maxterms · Logic Diagrams · Karnaugh Maps (K-Maps)

★ **Must lock:** Logic Diagrams, Karnaugh Maps (K-Maps)

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. What is the output of an OR gate when $A = 0$ and $B = 0$?
2. What symbol is used for the OR operation?
3. How many rows does a truth table with 4 inputs have?

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

1.1.7 Advanced Logic Gates (XOR and XNOR)

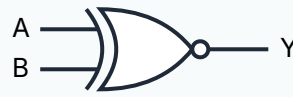
XOR (Exclusive OR) gives output 1 when the inputs are different.

- In terms of basic operations, $A \oplus B = A'B + AB'$.
- XNOR (Exclusive NOR) is XOR followed by NOT.
- It gives output 1 when the inputs are the same.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0



$$\text{XOR: } Y = A \oplus B$$



$$\text{XNOR: } Y = (A \oplus B)'$$

XOR = OR symbol with an extra curve at the back. XNOR = XOR symbol + bubble.

Remember: Memory trick: XOR outputs 1 when inputs are "eXclusively" different. XNOR outputs 1 when inputs are the same. XNOR is sometimes called the "equality gate".

XOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ مختلف ہوں، اور XNOR گیٹ تب 1 دیتا ہے جب دونوں ان پٹ ایک جیسے ہوں۔

1.1.8 – 1.1.10 Logic Expressions, Minterms and Maxterms

A logic expression (Boolean expression) combines variables with operations, for example $Y = A + B' \cdot C$.

1. Parentheses () → 2. NOT (') → 3. AND (·) → 4. OR (+)

- To evaluate it correctly you must follow the order of precedence, just like BODMAS in maths: 1.

A	B	C	B'	B'·C	Y = A + B'·C
0	0	0	1	0	0
0	0	1	1	1	1
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	1	1
1	1	0	0	0	1
1	1	1	0	0	1

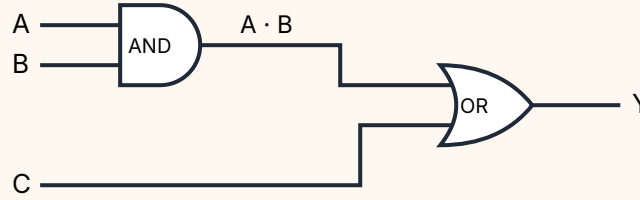
Remember: SOP = 1s, POS = 0s. Minterm: 0 → bar, 1 → plain. Maxterm: the reverse (1 → bar, 0 → plain). Think "Min likes 1s, Max likes 0s".

ایکسپریشن حل کرتے وقت ترتیب یہ ہے: پہلے قوسین، پھر NOT، پھر AND اور آخر میں OR.

1.1.11 Logic Diagrams ★ Golden

A logic diagram shows how logic gates are connected to create a circuit.

- It is a "picture" of a Boolean expression.
- To draw one, use the order of precedence: the operation with the highest priority is drawn first (nearest the inputs), and the last...



$Y = A \cdot B + C$: AND first, then OR.

Remember: Read the expression and ask "What is done last?" That gate goes on the far right, next to Y. Work backwards to the inputs. Count gates: $Y = A \cdot B + C$ needs 2...

Exam: Typical question: "Draw the logic diagram of $Y = \dots$ " and sometimes "also make its truth table" (4–5 marks). Write the precedence steps in one line, draw neat gate symbols...

لاچک ڈایاگرام دکھاتا ہے کہ گیٹس آپس میں کیسے جڑ کر سرکٹ بناتے ہیں۔

1.1.13 Karnaugh Maps (K-Maps) ★ Golden

A Karnaugh map (K-map) is a graphical tool used to simplify Boolean expressions.

- It is a grid of cells — one cell for each row of the truth table.
- The cells are arranged so that neighbouring (adjacent) cells differ by only ONE bit.
- This ordering is called Gray code.

	B = 0	B = 1
A = 0	Cell 0 ($A'B'$)	Cell 1 ($A'B$)
A = 1	Cell 2 (AB')	Cell 3 (AB)

Remember: Gray code song: "zero-zero, zero-one, one-one, one-zero" — only one bit flips at each step.

Imagine the K-map is drawn on a tube, so the left and right edges...

Exam: K-maps are a favourite long question (5–6 marks): "Simplify $F(A,B,C) = \sum m(\dots)$ using a K-map." Draw the map with labelled Gray-code headings, fill every cell (write the 0s...

کارنو میپ (K-Map) بولین ایکسپریشن کو آسان بنانے کا تصویری طریقہ ہے۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Advanced Logic Gates (XOR and XNOR):** XOR is not the same as OR.
- **Logic Expressions, Minterms and Maxterms:** Two classic errors: (1) writing minterms for the 0-rows — minterms come only from rows where $Y = 1$; (2) forgetting precedence — in $A + B \cdot C$...
- **Logic Diagrams:** Do not draw a separate wire for a variable each time it appears and then leave one unconnected.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Logic Diagrams**. Write: A logic diagram shows how logic gates are connected to create a circuit.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) It is a "picture" of a Boolean expression. (2) To draw one, use the order of precedence: the operation with the highest priority is drawn first (nearest the inputs), and the last... — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + a complete truth table and a labelled gate / logic diagram + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Karnaugh Maps (K-Maps)**. Write: A Karnaugh map (K-map) is a graphical tool used to simplify Boolean expressions.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) It is a grid of cells — one cell for each row of the truth table. (2) The cells are arranged so that neighbouring (adjacent) cells differ by only ONE bit. — + example.

5 marks: 3-mark structure + table: $B = 0 / B = 1 / A = 0 / A = 1$ + the worked example from this topic.

8–10 marks: 5-mark structure + a completed K-map with legal groups (1, 2, 4 or 8) and the simplified Boolean expression + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. In $Y = A + B \cdot C$, which gate is drawn first?

3 marks. Explain Logic Diagrams with two clear points and one example.

5 marks. Typical question: "Draw the logic diagram of $Y = \dots$ " and sometimes "also make its truth table" (4–5 marks). Write the precedence steps in one line, draw neat gate symbols with labelled intermediate...

8 marks. Logic Diagrams: 5-mark answer plus a complete truth table and a labelled gate / logic diagram.

Answers: 1. The AND gate (for $B \cdot C$) 3. It is a "picture" of a Boolean expression.; To draw one, use the order of precedence: the operation with the highest priority is drawn first (nearest the inputs), and the last... 5. Use the recipe in the blue box. Model opening: A logic diagram shows how logic gates are connected to create a circuit. 8. Same 5-mark body + a complete truth table and a labelled gate / logic diagram + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Logic Diagrams** using the recipe in this topic. Include a complete truth table and a labelled gate / logic diagram. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 1: Computer Systems

Computer Systems (4/5) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: LogiSim Evolution: Simulation Tool · Building a Circuit in LogiSim · Software Development Life Cycle (SDLC) · The Waterfall Model

★ **Must lock:** Software Development Life Cycle (SDLC), The Waterfall Model

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. In $Y = A + B \cdot C$, which gate is drawn first?
2. In a 3-variable K-map, which column is next to $BC = 01$ on the right?
3. What is $1 \oplus 1$?

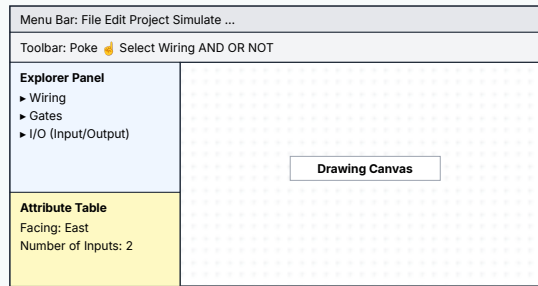
کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

LogiSim Evolution: Simulation Tool

LogiSim Evolution is a free graphical tool for designing and simulating logic circuits.

- "Simulating" means the computer pretends to be the real circuit: you can switch inputs on and off and watch the output change.
- It lets you see the truth tables and Boolean expressions you learn in theory.

Area	What it does
Menu Bar (top)	File, Edit, Project, Simulate menus — open/save files and control the simulation.
Toolbar	Quick access to standard tools: Wiring, Poke, and AND, OR, NOT gates.
Explorer Panel (left)	Contains the component libraries: Gates, I/O, Wiring.
Attribute Table (bottom left)	Adjust the properties of the selected gate, e.g. Number of Inputs or Facing direction.
Drawing Canvas (right)	The dotted grid where you place and wire the gates.



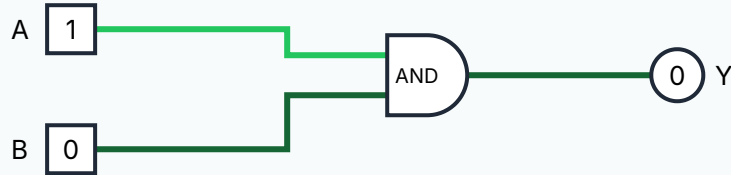
The main areas of the LogiSim Evolution window.

Remember: Colours like a traffic light: Green means "working" (bright = 1, dark = 0). Blue means "the wire is lonely — not connected". Red means "stop, something is..."

لاجی سیم ایوولوشن (LogiSim Evolution) لاجک سرکٹ بنانے اور چلا کر دیکھنے کا سافٹ ویئر ہے۔

Building a Circuit in LogiSim

Poke A	Poke B	Y shows	Output wire colour
0	0	0	Dark green
0	1	0	Dark green
1	0	0	Dark green
1	1	1	Bright green



Bright green = 1, dark green = 0

Testing with the Poke Tool: A = 1 (bright green wire), B = 0 (dark green), so $Y = 1 \cdot 0 = 0$.

Remember: Input pin = square (you push it). Output pin = circle (it shows a light). Poke = finger — you "poke" inputs to change them.

لاجی سیم میں $Y = A \cdot B$ بنانے کے لیے AND گیٹ رکھیں، بائیں جانب دو ان پٹ پن (چوکور) اور دائیں جانب ایک آؤٹ پٹ پن (گول) رکھیں، پھر وائر...

1.2.1 – 1.2.3 Software Development Life Cycle (SDLC) ★ Golden

The Software Development Life Cycle (SDLC) is a structured process that guides software development from beginning to end.

1. Requirement Analysis → 2. System Design → 3. Implementation → 4. Testing → 5. Deployment → 6. Maintenance

- It ensures clear goals, better planning, risk reduction and high quality.
- Without it, teams forget features, go over budget and deliver buggy software.
- SDLC has 6 phases: 1.

Phase	Activities and details	Deliverable (what it produces)
1. Requirement Analysis	Meet clients, understand their needs, list the features. This is the most important phase — mistakes here spread to every later phase.	SRS Document (Software Requirements Specification)
2. System Design	Create the High-Level Design (overall architecture), the Low-Level Design (database, UI screens) and flowcharts.	Design Document
3. Implementation	Write the actual code in programming languages (Python, Java, etc.) based on the design documents.	Working Source Code

Remember: Remember the order with "Real Developers Implement, Test, Deploy, Maintain" (R-D-I-T-D-M). Black box = a closed box (you can't see inside). White box = a glass...

Exam: "Explain the phases of SDLC" is a very common long question (5–8 marks). Name all 6 phases in order, write 1–2 lines of activities for each, and state the deliverable of...

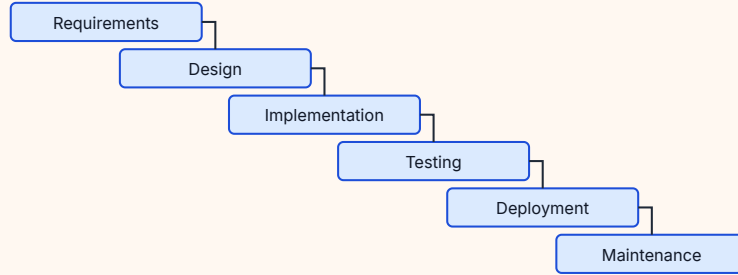
ایس ڈی ایل سی (SDLC) سافٹ ویئر بنانے کا منظم طریقہ ہے جس کے چھ مراحل ہیں: ضروریات کا تجزیہ، سسٹم ڈیزائن، کوڈنگ (Implementation)، ...

1.2.4 The Waterfall Model ★ Golden

The Waterfall model is a linear, sequential approach to the SDLC.

- Each phase must be completed before the next one begins, like water falling down steps.
- There is no easy way to go back to a previous phase.

Advantages (Pros)	Disadvantages (Cons)
Simple to understand and manage.	Not flexible to changes.
Well documented at every step.	Testing happens very late in the process.



The Waterfall model: each phase "falls" into the next, in one direction only.

Remember: Waterfall = water only flows downwards. Once water has fallen it cannot climb back up — once a phase is finished you do not go back.

Exam: Expect "Explain the Waterfall model with its advantages and disadvantages" or "When should the Waterfall model be used?" Draw the step diagram, list the three "when to..."

واثر فال (آبشار) ماڈل ایک سیدھا اور ترتیب وار طریقہ ہے؛ ہر مرحلہ مکمل ہونے کے بعد ہی اگلا شروع ہوتا ہے اور پیچھے جانا مشکل ہوتا ہے۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **LogiSim Evolution: Simulation Tool:** If your output wire is blue, the circuit is not wrong in logic — a wire is simply not touching a pin.
- **Building a Circuit in LogiSim:** Students often try to poke inputs while the Wiring or Select tool is active — then clicking moves the pin instead of changing it.
- **Software Development Life Cycle (SDLC):** Do not say the SDLC ends at deployment.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Software Development Life Cycle (SDLC)**. Write: The Software Development Life Cycle (SDLC) is a structured process that guides software development from beginning to end.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) It ensures clear goals, better planning, risk reduction and high quality. (2) Without it, teams forget features, go over budget and deliver buggy software. — + example.

5 marks: 3-mark structure + table: Phase / Activities and details / Deliverable (what it produces) + the worked example from this topic.

8–10 marks: 5-mark structure + a comparison table plus a named local case (school portal / NADRA) + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **The Waterfall Model**. Write: The Waterfall model is a linear, sequential approach to the SDLC.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Each phase must be completed before the next one begins, like water falling down steps. (2) There is no easy way to go back to a previous phase. — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + a comparison table plus a named local case (school portal / NADRA) + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. Which phase produces the SRS document?

3 marks. Explain Software Development Life Cycle (SDLC) with two clear points and one example.

5 marks. "Explain the phases of SDLC" is a very common long question (5–8 marks). Name all 6 phases in order, write 1–2 lines of activities for each, and state the deliverable of each (SRS, Design Document,...)

8 marks. Software Development Life Cycle (SDLC): 5-mark answer plus a comparison table plus a named local case (school portal / NADRA).

Answers: 1. Requirement Analysis 3. It ensures clear goals, better planning, risk reduction and high quality.; Without it, teams forget features, go over budget and deliver buggy software. 5. Use the recipe in the blue box. Model opening: The Software Development Life Cycle (SDLC) is a structured process that guides software development from beginning to end. 8. Same 5-mark body + a comparison table plus a named local case (school portal / NADRA) + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Software Development Life Cycle (SDLC)** using the recipe in this topic. Include a comparison table plus a named local case (school portal / NADRA). Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 1: Computer Systems

Computer Systems (5/5) · Chapter exam

TODAY'S 90 MINUTES

90 minutes. 0–10 min recap yesterday ★ on the board · 10–85 min sit the 75-mark chapter exam (1 h 15 min, hall conditions) · 85–90 min collect scripts. Do not mark in front of the batch.

Board today: The Agile Model · Waterfall vs Agile: Comparison and Case Studies · Communication Models and the OSI 7-Layer Model · TCP/IP Model vs OSI Model

★ **Must lock:** The Agile Model, Communication Models and the OSI 7-Layer Model

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Hall paper today. Recap the cards for ten minutes, then scripts out. Re-teach any Golden below 60% in tomorrow's starter.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Which phase produces the SRS document?
2. Why is the Waterfall model called "sequential"?
3. Where do you change the number of inputs of a gate?

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

1.2.4 The Agile Model ★ Golden

The Agile model is an iterative approach: the software is built in small, repeated cycles called sprints (usually 2–4 weeks long).

Plan → Design → Build → Test → Review & feedback → Next sprint ↻

- Each sprint plans, builds and tests a small part of the software, and then shows it to the customer.
- Customer feedback is gathered continuously and used in the next sprint.
- Plan → Design → Build → Test → Review & feedback → Next sprint ↻

Advantages (Pros)	Disadvantages (Cons)
Highly flexible to changes.	Less documentation.
Early delivery of working software.	Unpredictable end-time and cost.

Remember: Agile = agile athlete: quick and able to change direction. Think of making a pizza where the customer tastes each slice and you adjust the next one.

Exam: Questions ask you to define Agile, explain sprints, list pros/cons, or say when to use it. Always mention the words "iterative", "sprints (2–4 weeks)" and "continuous..."

ایجائل (Agile) ماڈل میں سافٹ ویئر چھوٹے چھوٹے ادوار میں بنتا ہے جنہیں اسپرینٹ (Sprint) کہتے ہیں، جو عموماً دو سے چار ہفتے کے ہوتے ہیں۔

1.2.5 – 1.2.7 Waterfall vs Agile: Comparison and Case Studies

Both models follow the same SDLC phases. The difference is how they move through them: Waterfall goes through each phase once, in a straight line; Agile goes through all phases many times, in small loops.

- The difference is how they move through them: Waterfall goes through each phase once, in a straight line; Agile goes through all phases...

Dimension	Waterfall model	Agile model
Flexibility	Rigid. Hard to make changes later.	Very flexible. Adapts to changes easily in the next sprint.
Customer involvement	Only at the start (requirements) and the end (delivery).	Continuous throughout the project (after every sprint).
Delivery of software	Single delivery at the very end.	Frequent, partial deliveries after every sprint.
Testing	Done late, in a dedicated testing phase.	Continuous testing within every sprint.
Documentation	Heavy documentation required.	Light documentation. Focus is on working software.

Remember: One question decides it: "Will the requirements change?" No → Waterfall. Yes / not sure → Agile.

وائرفال ماڈل سخت (Rigid) ہے، گاہک صرف شروع اور آخر میں شامل ہوتا ہے، سافٹ ویئر ایک ہی بار آخر میں ملتا ہے اور دستاویزات زیادہ ہوتی ہیں۔

1.3.1 – 1.3.4 Communication Models and the OSI 7-Layer Model ★ Golden

A communication model explains how data travels from one device to another through a network.

- Because sending data is complicated, the job is split into layers; each layer does one part of the job and passes the data to the next...
- The OSI (Open Systems Interconnection) model was developed by ISO (International Organization for Standardization) in 1984.
- It divides communication into 7 layers.

Layer	Function	Protocols / examples
7. Application	Provides network services directly to user applications (browser, email program).	HTTP, HTTPS, FTP, SMTP, DNS
6. Presentation	Data translation, encryption (security) and compression.	SSL/TLS, JPEG, MP3, ASCII
5. Session	Opens, maintains and closes communication sessions.	NetBIOS, RPC
4. Transport	Reliable end-to-end delivery, error checking, port numbering.	TCP (reliable), UDP (fast)



The 7 layers of the OSI model with the mnemonic "Please Do Not Throw Sausage Pizza Away".

Remember: Mnemonic (bottom to top, 1 → 7): "Please Do Not Throw Sausage Pizza Away" = Physical, Data Link, Network, Transport, Session, Presentation, Application. (Top...

Exam: A guaranteed long question: "Explain the seven layers of the OSI model." Draw the 7-layer stack (numbered, layer 7 on top), then write one function and one or two...

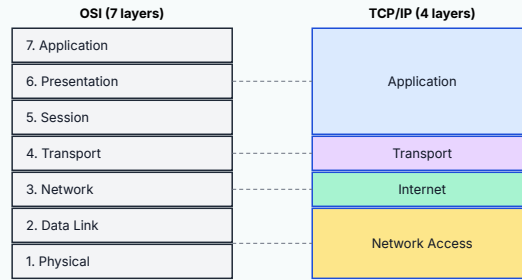
کمیونیکیشن ماڈل بتاتا ہے کہ ڈیٹا نیٹ ورک کے ذریعے ایک ڈیوائس سے دوسری تک کیسے پہنچتا ہے۔

1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

The OSI model is mainly a reference model used for teaching and design.

- The TCP/IP model is the practical model used on the real Internet.
- It condenses the 7 OSI layers into just 4 layers.

TCP/IP model (4 layers)	Corresponds to OSI layers
1. Application	Application, Presentation, Session (7, 6, 5)
2. Transport	Transport (4)
3. Internet	Network (3)
4. Network Access	Data Link, Physical (2, 1)



How the 7 OSI layers map onto the 4 TCP/IP layers.

Remember: Count it as 3-1-1-2: the TCP/IP Application layer takes 3 OSI layers, Transport takes 1, Internet takes 1, Network Access takes 2. Total $3+1+1+2 = 7$.

ٹی سی پی/آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔

Computer Systems

20 single-page lessons · 8 Golden · read, close, say aloud, then sit the exam

Chapter in one look

- Discrete quantities are counted (students, cars); continuous quantities are measured and can take any value (temperature, speed). Stairs vs ramp.
- A digital system uses only two values, 0 (OFF/LOW/FALSE) and 1 (ON/HIGH/TRUE), called bits. It is reliable, accurate, easy to design and programmable.
- Analog signals change smoothly (sine wave) and are affected by noise; digital signals jump between HIGH and LOW (square wave) and resist noise.
- Boolean algebra, by George Boole, works with only TRUE (1) and FALSE (0) using variables, operations and expressions.
- AND (·) gives 1 only when all inputs are 1; OR (+) gives 1 when any input is 1; NOT (!) inverts its single input (unary).
- A truth table lists all 2^n input combinations in binary order with their outputs.
- Logic gates are the building blocks of digital systems: AND, OR, NOT (basic); NAND, NOR (universal); XOR, XNOR (advanced).
- NAND and NOR are universal: any circuit can be built using only one of them.
- XOR gives 1 when inputs are different; XNOR gives 1 when inputs are the same.
- Order of precedence: Parentheses → NOT → AND → OR. A logic diagram is drawn in the same order.
- Minterms (SOP) come from rows with output 1; maxterms (POS) come from rows with output 0.
- K-maps use Gray code (00, 01, 11, 10); group 1s in sizes 1, 2, 4, 8, as large as possible; overlap and wrap-around are allowed; changing variables are eliminated.
- LogiSim Evolution simulates circuits: input pins are squares, output pins circles; the Poke Tool toggles inputs; bright green = 1, dark green = 0, blue = unconnected, red = error.
- SDLC has 6 phases: Requirement Analysis (SRS), Design, Implementation, Testing (black box/white box), Deployment (direct/phased/pilot), Maintenance.
- Waterfall is linear and suits fixed requirements (AI-Noor Library); Agile is iterative with 2–4 week sprints and suits changing requirements (AI-Noor Student App).
- The OSI model (ISO, 1984) has 7 layers — "Please Do Not Throw Sausage Pizza Away"; the practical TCP/IP model has 4 layers: Application, Transport, Internet, Network Access.

Key terms

Term	Meaning	اردو
Discrete quantity	A countable quantity with separate, fixed values.	منفصل مقدار
Continuous quantity	A quantity that can take any value within a range.	مسلسل مقدار
Bit	A binary digit, 0 or 1.	بائنری ہندسہ
Digital system	An electronic system that uses only the values 0 and 1.	ڈیجیٹل نظام
Analog signal	A continuous signal that changes smoothly, like a sine wave.	اینالوگ سگنل (مسلسل اشارہ)
Digital signal	A signal with only two levels, HIGH and LOW (square wave).	ڈیجیٹل سگنل
Noise	Unwanted interference that disturbs a signal.	شور (مداخلت)
Boolean algebra	Algebra of two values, TRUE (1) and FALSE (0), developed by George Boole.	بولین الجبرا
Truth table	A table of all input combinations and their outputs.	تروٹھ ٹیبل (صداقت جدول)
Logic gate	An electronic circuit that performs a Boolean operation.	منطقی گیٹ
Universal gate	A gate (NAND or NOR) from which any circuit can be built.	عالمگیر گیٹ
Unary operation	An operation with only one operand, e.g. NOT.	یک عددی عمل

Computer Systems

75 marks · 1 hour 15 minutes · no notes · no phones

CHAPTER EXAM RULES

Instructions. Sit this exam after Lesson 20 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

Marks. $15 + 30 + 30 = 75$. A 5-mark / 10-mark answer needs Define · Explain · Example · Diagram · Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

Section A — Multiple choice (15 × 1 = 15)

1. Which of the following is a continuous quantity?
 - a. Number of students in a class
 - b. Number of cars in a parking lot
 - c. Temperature of water
 - d. Number of books on a shelf
2. In a digital system, the value 1 represents:
 - a. OFF
 - b. LOW
 - c. FALSE
 - d. ON / HIGH / TRUE
3. A square wave represents a:
 - a. Digital signal
 - b. Analog signal
 - c. Sine wave
 - d. Noise signal
4. Boolean algebra was developed by:
 - a. Charles Babbage
 - b. George Boole
 - c. Alan Turing
 - d. Maurice Karnaugh
5. How many rows does a truth table with 4 inputs have?
 - a. 4
 - b. 8
 - c. 16
 - d. 32
6. Which gate gives output 1 only when all inputs are 1?
 - a. OR
 - b. AND
 - c. NOR
 - d. XOR
7. Which pair of gates are called universal gates?
 - a. AND and OR
 - b. XOR and XNOR
 - c. NOT and AND
 - d. NAND and NOR
8. What is the output of a NOR gate when $A = 0$ and $B = 0$?
 - a. 1
 - b. 0
 - c. Unknown
 - d. Depends on the gate size
9. An XOR gate gives output 1 when:
 - a. Both inputs are 1
 - b. Both inputs are 0
 - c. The inputs are different
 - d. The inputs are the same
10. After parentheses, which operation has the highest precedence?
 - a. OR
 - b. NOT
 - c. AND
 - d. XOR
11. The minterm for the row $A = 1, B = 0$ is:
 - a. AB'
 - b. $A'B$
 - c. $A + B'$
 - d. $A' + B$
12. The columns of a 3-variable K-map (BC) are written in the order:
 - a. 00, 01, 10, 11
 - b. 00, 10, 01, 11
 - c. 11, 10, 01, 00
 - d. 00, 01, 11, 10
13. In black box testing, the tester:
 - a. Reads the source code line by line
 - b. Does not see the code and tests inputs/outputs
 - c. Is always the developer
 - d. Tests only the internal structure
14. Which OSI layer handles logical (IP) addressing and routing?
 - a. Data Link
 - b. Transport
 - c. Network
 - d. Session
15. The TCP/IP Network Access layer corresponds to which OSI layers?
 - a. 7, 6 and 5
 - b. 4
 - c. 3
 - d. 2 and 1

Section B — Short questions (10 × 3 = 30)

1. Define discrete and continuous quantities. Give two examples of each.
2. State four advantages of digital systems.
3. Differentiate between analog and digital signals (any three points).
4. Why is NOT called a unary operation? Write its truth table.
5. How many rows are needed in a truth table for 3 inputs? List all the input combinations.
6. Why are NAND and NOR called universal gates? Show how a NAND gate can be used as a NOT gate.
7. A 2-input truth table has $Y = 1$ only for rows $A=0, B=1$ and $A=1, B=0$. Write the SOP expression and name the gate.
8. Name the deliverable of each SDLC phase.
9. Differentiate between black box testing and white box testing.
10. When should TCP be used and when should UDP be used? Give one example of each.

Section C — Long questions (attempt any 3 × 10 = 30)

1. Draw the symbol, write the Boolean expression and give the truth table for each of these gates: AND, OR, NOT, NAND, NOR, XOR and XNOR.
2. A function is given as $F(A, B, C) = \sum m(1, 3, 5, 6, 7)$. (a) Make its truth table. (b) Simplify it using a 3-variable K-map, explaining each group. (c) Draw the logic diagram of the simplified expression.
3. For $Y = A \cdot B' + C$: (a) state the order in which the operations are done; (b) make the complete truth table; (c) draw the logic diagram.
4. Explain the six phases of the Software Development Life Cycle with the activities and deliverable of each phase. Use the house-building analogy.
5. Compare the Waterfall and Agile models under five headings. Explain why the Al-Noor Library Management System used Waterfall and the Al-Noor Student Mobile Application used Agile.
6. Explain the seven layers of the OSI model with the function and one protocol/device for each layer. Then show how they map onto the four layers of the TCP/IP model.

Answer key. Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

Check yourself — answers

1. **1.1.1 Discrete and Continuous Quantities.** Answers: 1. Discrete (you count pages). 2. Continuous (height is measured and can be any value). 3. The ramp.
2. **1.1.2 Introduction to Digital Systems.** Answers: 1. ON, HIGH or TRUE. 2. Any two of: reliable (less affected by noise), accurate (no loss when copying), easy to design (two states), programmable (controlled by software).
3. **1.1.3 Analog and Digital Signals.** Answers: 1. A square wave. 2. Analog. 3. Human voice (or a traditional thermometer reading).
4. **1.1.4 Boolean Algebra.** Answers: 1. George Boole. 2. 1 (because + means OR).
5. **1.1.5 Boolean Operations.** Answers: 1. "+". 2. Because it works on only one operand (one input). 3. $B' = 0$, so $0 + 0 = 0$.
6. **1.1.6 Truth Tables.** Answers: 1. $2^4 = 16$. 2. 101.
7. **1.1.7 Logic Gates (Basic Gates: AND, OR, NOT).** Answers: 1. 0. 2. Inverter. 3. Only $A = 1$, $B = 1$.
8. **1.1.7 Universal Logic Gates (NAND and NOR).** Answers: 1. Any logic circuit can be built using only NAND gates or only NOR gates. 2. 0. 3. Join both inputs together and feed A into them: $(A \cdot A)' = A'$.
9. **1.1.7 Advanced Logic Gates (XOR and XNOR).** Answers: 1. 0. 2. XNOR. 3. $A \oplus B = A'B + AB'$.
10. **1.1.8 – 1.1.10 Logic Expressions, Minterms and Maxterms.** Answers: 1. AND. 2. $AB'C$. 3. $A + B'$.
11. **1.1.11 Logic Diagrams.** Answers: 1. The AND gate (for $B \cdot C$). 2. Two: one NOT gate (for A') and one AND gate.
12. **1.1.13 Karnaugh Maps (K-Maps).** Answers: 1. $BC = 11$. 2. Cells 2 and 3 are the row $A = 1$; B changes $\rightarrow Y = A$. 3. The whole row $A = 1$ is a group of 4 $\rightarrow Y = A$.
13. **LogiSim Evolution: Simulation Tool.** Answers: 1. In the Attribute Table (bottom left). 2. An error or short circuit.
14. **Building a Circuit in LogiSim.** Answers: 1. Square (output pin is the circle). 2. The Poke Tool (pointing finger).
15. **1.2.1 – 1.2.3 Software Development Life Cycle (SDLC).** Answers: 1. Requirement Analysis. 2. Developers. 3. Direct, Phased and Pilot release.
16. **1.2.4 The Waterfall Model.** Answers: 1. Because each phase must be completed before the next begins, in a fixed order. 2. Because its requirements were 100% fixed before coding began.
17. **1.2.4 The Agile Model.** Answers: 1. A short development cycle that delivers a working part of the software; usually 2–4 weeks. 2. Less documentation, or unpredictable end-time and cost.
18. **1.2.5 – 1.2.7 Waterfall vs Agile: Comparison and Case Studies.** Answers: 1. Waterfall. 2. Agile. 3. Waterfall (fixed, clear requirements).
19. **1.3.1 – 1.3.4 Communication Models and the OSI 7-Layer Model.** Answers: 1. Presentation layer (6). 2. Data Link layer (2). 3. UDP, because speed matters more than perfect accuracy.
20. **1.3.5 – 1.3.6 TCP/IP Model vs OSI Model.** Answers: 1. Four. 2. Network Access. 3. Application.

Answer key

Multiple choice

1. c
2. d
3. a
4. b
5. c
6. b
7. d
8. a
9. c
10. b
11. a
12. d
13. b
14. c
15. d

Short questions — model answers

1. Discrete quantities are separate, countable items with specific values (number of students in a classroom, number of cars in a parking lot). Continuous quantities can take any value within a range and flow without breaks (temperature of water, speed of a moving car).
2. Reliable (less affected by noise and interference); accurate (no loss of quality when copying); easy to design (only two states, 0 and 1); programmable (easily controlled by software).
3. Nature: analog is a continuous flow; digital has discrete values (bits, HIGH/LOW). Shape: analog is a sine wave; digital is a square wave. Reliability: analog is low (affected by noise); digital is high (resistant to noise). Examples: human voice vs computer data.
4. NOT works on only one operand (one input), so it is unary. Truth table: $A = 0 \rightarrow Y = 1$; $A = 1 \rightarrow Y = 0$.
5. $2^3 = 8$ rows: 000, 001, 010, 011, 100, 101, 110, 111.
6. Any logic circuit can be built using only NAND gates, or only NOR gates. NAND as NOT: join both inputs and feed A: $Y = (A \cdot A)' = A'$. Check: $A = 0 \rightarrow 1$, $A = 1 \rightarrow 0$.
7. Minterms: $A'B$ (row 01) and AB' (row 10). $Y = A'B + AB'$. This is the XOR gate, $Y = A \oplus B$.
8. Requirement Analysis $\rightarrow$ SRS document; System Design $\rightarrow$ Design Document; Implementation $\rightarrow$ Working Source Code; Testing $\rightarrow$ Test Reports / bug-free software; Deployment $\rightarrow$ Live Software; Maintenance $\rightarrow$ Patches / Updated Versions.
9. Black box: tester does not see the code, tests functionality (inputs/outputs), done by dedicated testers. White box: tester can see the code, tests internal logic/structure, done by developers.
10. Use TCP when accuracy matters, e.g. web browsing or email. Use UDP when speed matters more than perfection, e.g. video streaming or online gaming.

Long questions — model answers

- Symbols as drawn in 1.1.7 (AND = D shape; OR = curved back, pointed front; NOT = triangle + bubble; NAND = AND + bubble; NOR = OR + bubble; XOR = OR with extra back curve; XNOR = XOR + bubble). Expressions: AND $Y = A \cdot B$; OR $Y = A + B$; NOT $Y = A'$; NAND $Y = (A \cdot B)'$; NOR $Y = (A + B)'$; XOR $Y = A \oplus B$; XNOR $Y = (A \oplus B)'$. Outputs for inputs 00, 01, 10, 11: AND 0,0,0,1; OR 0,1,1,1; NAND 1,1,1,0; NOR 1,0,0,0; XOR 0,1,1,0; XNOR 1,0,0,1. NOT: 0 $\rightarrow$ 1, 1 $\rightarrow$ 0.
- (a) Truth table ($ABC \rightarrow F$): 000 $\rightarrow$ 0, 001 $\rightarrow$ 1, 010 $\rightarrow$ 0, 011 $\rightarrow$ 1, 100 $\rightarrow$ 0, 101 $\rightarrow$ 1, 110 $\rightarrow$ 1, 111 $\rightarrow$ 1.

(b) K-map:

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0	0	1	1	0
A = 1	0	1	1	1

Group 1 (size 4): the two middle columns, m_1, m_3, m_5, m_7 . A changes and B changes; C = 1 stays $\rightarrow$ C. Group 2 (size 2): m_7 and m_6 in row A = 1. C changes; A = 1, B = 1 stay $\rightarrow$ AB (m_7 is shared — overlapping is allowed). $F = C + AB$. Check: C + AB is 1 for rows 1, 3, 5, 7 (C = 1) and 6, 7 (AB = 1) = {1, 3, 5, 6, 7}. ✓

(c) Logic diagram: an AND gate with inputs A and B gives AB; an OR gate takes AB and C and gives F.

- (a) First NOT (B'), then AND ($A \cdot B'$), then OR ($+ C$).

(b) Truth table:

A	B	C	B'	A · B'	Y
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1

(c) Logic diagram: input B goes through a NOT gate to give B' . A and B' go into an AND gate to give $A \cdot B'$. $A \cdot B'$ and C go into an OR gate to give Y. (3 gates.)

- Requirement Analysis: meet clients, understand needs, list features; most important phase; deliverable SRS (house: asking the family what rooms they need). System Design: high-level design (architecture), low-level design (database, UI screens), flowcharts; deliverable Design Document (architect draws blueprints). Implementation: write code in Python, Java etc. from the design; deliverable Working Source Code (builders lay bricks). Testing: find and fix bugs using black box and white box testing; deliverable Test Reports / bug-free software (inspector checks the house). Deployment: release to end-users by direct, phased or pilot release; deliverable Live Software (family moves in). Maintenance: support, updates, fixing post-release issues; deliverable Patches / Updated Versions (repairs and extensions).
- Comparison: Flexibility — Waterfall rigid, Agile very flexible. Customer involvement — Waterfall only at start and end, Agile continuous after every sprint. Delivery — Waterfall single delivery at the end, Agile frequent partial deliveries. Testing — Waterfall late in a dedicated phase, Agile continuous in every sprint. Documentation — Waterfall heavy, Agile light. The Al-Noor Library system used Waterfall because its requirements were 100% fixed before coding began, so a simple, well-documented linear process fitted. The Al-Noor Student App used Agile because teachers, parents and students kept requesting new features, so short sprints with continuous feedback were needed.
- OSI (ISO, 1984), top to bottom: 7 Application — network services to user applications (HTTP, HTTPS, FTP, SMTP, DNS); 6 Presentation — translation, encryption, compression (SSL/TLS, JPEG, MP3, ASCII); 5 Session — opens, maintains, closes sessions (NetBIOS, RPC); 4 Transport — reliable end-to-end delivery, error checking, port numbers (TCP, UDP); 3 Network — logical IP addressing and routing (IP, ICMP, routers); 2 Data Link — physical MAC addressing and error detection in frames (Ethernet, switches); 1 Physical — raw bits over cables/wireless (cables, hubs, Wi-Fi radio). Mnemonic bottom to top: "Please Do Not Throw Sausage Pizza Away". TCP/IP mapping: Application = OSI 7, 6, 5; Transport = 4; Internet = 3; Network Access = 2, 1.

Self-assessment: I can...

- ☐ ...tell discrete from continuous quantities and analog from digital signals, and draw both waves.
- ☐ ...evaluate AND, OR and NOT expressions and follow the order of precedence.
- ☐ ...draw all seven gate symbols and write their truth tables from memory.
- ☐ ...write the SOP (minterms) and POS (maxterms) from a truth table.
- ☐ ...simplify expressions with 2- and 3-variable K-maps, including overlap and wrap-around groups.
- ☐ ...explain the six SDLC phases and choose between Waterfall and Agile for a project.
- ☐ ...list four advantages of digital systems.
- ☐ ...build a complete truth table for 2 or 3 inputs without missing a row.
- ☐ ...explain why NAND and NOR are universal gates.
- ☐ ...draw a logic diagram from an expression.
- ☐ ...build and test a simple circuit in LogiSim and read the wire colours.
- ☐ ...name the 7 OSI layers in order, state their functions, and map them to the 4 TCP/IP layers.

Chapter 2: Computational Thinking & Algorithm

Computational Thinking & Algorithm (1/3) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Computational Thinking · Algorithm · Algorithm vs Pseudocode · Decomposition · Pattern Recognition

★ **Must lock:** Algorithm, Algorithm vs Pseudocode, Decomposition

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. What is a sprint and how long does it usually last?
2. Which OSI layer is responsible for encryption and compression?
3. In which model is the customer involved only at the start and the end?

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

2.1 Computational Thinking

Computational Thinking (CT) is a systematic approach to solving problems.

Problem → Decomposition → Pattern recognition → Abstraction → Algorithm → Program

- Decomposition — breaking a big problem down into smaller parts.
- Pattern recognition — identifying things that repeat or follow a rule.
- Abstraction — hiding (ignoring) details that do not matter.
- Algorithms — developing clear step-by-step instructions to solve the problem.

Remember: Remember the four parts of CT with the phrase "Don't Pick Any Apple": Decomposition, Pattern recognition, Abstraction, Algorithm.

کمپیوٹیشنل تھنکنگ (Computational Thinking) مسائل حل کرنے کا ایک منظم طریقہ ہے۔

2.2 Algorithm ★ Golden

An algorithm is a step-by-step logical procedure used to solve a problem.

- Input — the data it starts with.
- Clear steps — each step has only one meaning.
- A definite end — it must finish after a limited number of steps.
- Output — the result it produces.

Reason	What it means for you
Structured approach	Algorithms provide a structured approach to problem-solving, making complex tasks manageable.
Optimisation	They optimise time and memory usage, which is crucial for large-scale applications (for example, searching millions of records).
Reusability	They can be reused for similar problems, saving time and effort. A sorting algorithm for marks can also sort prices.
Accuracy and consistency	They ensure accuracy and consistency, reducing human error. The same steps always give the same result.
Blueprint	They act as a blueprint (a plan) before actual code implementation, just like an architect's map before building a house.

Remember: Remember the five reasons as S-O-R-A-B: Structured, Optimised, Reusable, Accurate, Blueprint.

Exam: Very common short question: "Define algorithm and write its importance." Write the definition word for word — a step-by-step logical procedure used to solve a problem;...

الگورتھم (Algorithm) کسی مسئلے کو حل کرنے کا مرحلہ وار منطقی طریقہ ہے۔

2.2.2 Algorithm vs Pseudocode ★ Golden

Both algorithms and pseudocode are used in problem-solving, but they serve different purposes in the design phase.

Problem → Algorithm (what) → Pseudocode (how) → Program (code) → Output

- The algorithm says what must be done.
- The pseudocode shows how it could be implemented, using programming-like keywords such as IF, FOR and WHILE, but it still cannot be run on...

Point	Algorithm	Pseudocode
Definition	A step-by-step logical procedure to solve a problem.	A human-readable representation of an algorithm using plain English & keywords.
Nature	Conceptual and language-independent.	Semi-formal and structured, but not executable code.
Purpose	To define what needs to be done logically.	To present how it can be implemented in a readable format.
Form	Usually described in plain steps, bullets or numbered points.	Written using structured constructs (IF, FOR, WHILE) like programming.
Focus	Focuses on the logic and steps of the problem.	Focuses on representing steps clearly before coding.
Audience	For anyone trying to understand the logic (including non-programmers).	Mainly for programmers or students preparing for coding.
Format rules	No fixed format; can be informal.	Follows standard pseudocode conventions for clarity.

Remember: Teacher's analogy: an algorithm is like a recipe ("Bake a cake at 350 degrees"). Pseudocode is like a structured kitchen manual using standard verbs (SET...

Exam: "Differentiate between algorithm and pseudocode" is a favourite 4–5 mark question. Draw a two-column table and give at least five points (definition, nature, purpose,...

الگورتھم بتاتا ہے کہ کیا کرنا ہے، جبکہ سوڈو کوڈ (Pseudocode) بتاتا ہے کہ اسے کیسے کیا جا سکتا ہے۔

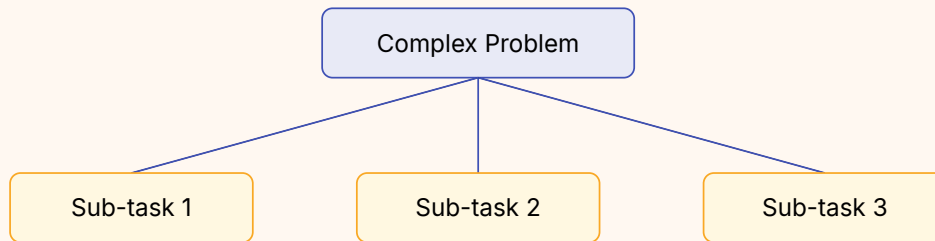
2.3.1 Decomposition ★ Golden

Decomposition is a problem-solving technique in which a complex task or system is broken down into smaller, manageable parts.

Vocabulary → Grammar rules → Tenses → Sentence building

- By dividing a big problem into simpler steps, it becomes easier to understand, develop and solve.
- Each small part is called a sub-task (or sub-problem).
- You solve each sub-task separately, then join the solutions to solve the whole problem.

Sub-task	Description
Learn vocabulary	Learn the words for people, actions and things (e.g., "I", "eat", "apple").
Grammar rules	Learn the sentence structure: Subject → Verb → Object.
Learn tenses	Present, past, future (e.g., "I eat" vs. "I ate").
Sentence building	Combine words using the rules to create valid sentences: "I ate an apple."



Decomposition: one complex problem is split into smaller sub-tasks.

Remember: Think of eating a big roti: you cannot eat it in one bite, so you break it into small pieces.
Decomposition = "break the roti".

Exam: Expect "Define decomposition with a real-life example" or "Explain the benefits of decomposition". Give the definition, draw the small tree (Complex Problem → Sub-task...)

ڈیکمپوزیشن (Decomposition) کا مطلب ہے کسی بڑے اور پیچیدہ مسئلے کو چھوٹے اور آسان حصوں میں تقسیم کرنا۔

2.3.2 Pattern Recognition

Pattern recognition is the process of identifying regularities, repetitions or rules in data.

- What is repeating?
- What changes in a predictable way?
- Can I generalise this (write one rule that works for every case)?

Row	Stars
1	1
2	2
3	3
...	...
N	N

```
print("*")
print("**")
print("***")
print("****")
print("*****")
```

Remember: Whenever you see "the same thing again and again, with a small change each time", think loop. Pattern recognition is the habit of spotting that.

پیٹرن ریکگنیشن (Pattern Recognition) کا مطلب ہے ڈیٹا میں دہرائی جانے والی چیزوں یا اصولوں کو پہچاننا۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Computational Thinking:** Students sometimes think computational thinking means "writing code".
- **Algorithm:** Do not write "an algorithm is a program".
- **Algorithm vs Pseudocode:** Pseudocode is not executable.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Algorithm**. Write: An algorithm is a step-by-step logical procedure used to solve a problem.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Clear steps — each step has only one meaning. (2) A definite end — it must finish after a limited number of steps. — + example.

5 marks: 3-mark structure + table: Reason / What it means for you + the worked example from this topic.

8–10 marks: 5-mark structure + working code with correct indentation and a short trace of one example + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Algorithm vs Pseudocode**. Write: Both algorithms and pseudocode are used in problem-solving, but they serve different purposes in the design phase.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) The algorithm says what must be done. (2) The pseudocode shows how it could be implemented, using programming-like keywords such as IF, FOR and WHILE, but it still cannot be run on... — + example.

5 marks: 3-mark structure + table: Point / Algorithm / Pseudocode + the worked example from this topic.

8–10 marks: 5-mark structure + the full comparison table with every cell filled + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Decomposition**. Write: Decomposition is a problem-solving technique in which a complex task or system is broken down into smaller, manageable parts.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) By dividing a big problem into simpler steps, it becomes easier to understand, develop and solve. (2) Each small part is called a sub-task (or sub-problem). — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + the layer table with one protocol or device on each layer + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. Why is an algorithm called "language-independent"?

3 marks. Explain Algorithm with two clear points and one example.

5 marks. Very common short question: "Define algorithm and write its importance." Write the definition word for word — a step-by-step logical procedure used to solve a problem; conceptual and independent of...

8 marks. Algorithm: 5-mark answer plus working code with correct indentation and a short trace of one example.

Answers: 1. Because it describes only the logic; it is not tied to Python, C++ or any other language 3. Clear steps — each step has only one meaning.; A definite end — it must finish after a limited number of steps. 5. Use the recipe in the blue box. Model opening: An algorithm is a step-by-step logical procedure used to solve a problem. 8. Same 5-mark body + working code with correct indentation and a short trace of one example + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Algorithm** using the recipe in this topic. Include working code with correct indentation and a short trace of one example. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 2: Computational Thinking & Algorithm

Computational Thinking & Algorithm (2/3) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Abstraction · Sorting and Searching Algorithms · Sorting Algorithms: Bubble Sort · Sorting Algorithms: Selection Sort

★ **Must lock:** Sorting Algorithms: Bubble Sort, Sorting Algorithms: Selection Sort

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Why is an algorithm called "language-independent"?
2. Which one — algorithm or pseudocode — uses keywords like FOR and WHILE?
3. What is each smaller part of a decomposed problem called?

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

2.3.3 Abstraction

Abstraction means hiding unnecessary features and focusing only on the essential ones needed to solve a problem.

Boil water → Add tea → Steep → Pour → Serve

- Necessary steps — the core logic that never changes.
- Changes based on the situation — these become variables (inputs) in the algorithm.
- Abstraction items (noise) — details you can ignore completely.

1. Necessary steps (core logic)	2. Changes based on situation (variables)	3. Abstraction items (ignore / noise)
1. Boil water 2. Add tea leaves/bag 3. Steep for a few minutes 4. Pour into cup 5. Serve	Ingredients (black, green, milk) Additions (sugar, lemon) Tools (stove vs kettle)	Brand of kettle Cup colour or design Where tea is made (kitchen, office)

Remember: Abstraction = "Keep the skeleton, drop the decoration." Ask: "If I remove this detail, does the solution still work?" If yes, it is noise.

ایبستریکشن (Abstraction) کا مطلب ہے غیر ضروری تفصیلات کو چھپا کر صرف ضروری باتوں پر توجہ دینا۔

2.4 Sorting and Searching Algorithms

Sorting and searching are both fundamental techniques of computer science.

temp = A (temp = 8) → A = B (A = 4) → B = temp (B = 8)

- They work on lists of the same data type (for example, a list of all integers or a list of all names).
- **Sorting** Arranging a collection of items in an order.
- The order may be ascending (0–9, A–Z) or descending (9–0, Z–A).

Sorting	Searching
Arranging a collection of items in an order. The order may be ascending (0–9, A–Z) or descending (9–0, Z–A). Used to organise data.	Locating a specific element within a list to find its position or check its existence (is it there or not?).
Examples: Bubble Sort, Selection Sort	Examples: Linear Search, Binary Search
Real life: merit list arranged from highest to lowest marks.	Real life: finding your roll number in the merit list.

Trap: Position 1 in the textbook algorithm is index 0 in Python.

سارٹنگ (Sorting) کا مطلب ہے ڈیٹا کو صعودی (چھوٹے سے بڑے) یا نزولی (بڑے سے چھوٹے) ترتیب میں رکھنا۔

2.4.1 Sorting Algorithms: Bubble Sort ★ Golden

Bubble Sort is an elementary sorting algorithm that works by repeatedly comparing adjacent (side-by-side) elements and swapping them if they are in the wrong order.

Compare DATA[j] and DATA[j+1] → Left > Right? → Yes: swap → Move one step right → End of pass: largest is at the end → Repeat N-1 passes

- In each pass, the largest remaining value "bubbles up" to the end of the list, like an air bubble rising to the top of water.
- So after pass 1 the largest value is in the last place, after pass 2 the second largest is in second-last place, and so on.
- A list of N items needs N - 1 passes, and pass K makes N - K comparisons.

Comparison	Pair	Left > Right?	Action	List after
1st vs 2nd	[8, 4], 1, 9, 3	8 > 4 Yes	Swap	4, 8, 1, 9, 3
2nd vs 3rd	4, [8, 1], 9, 3	8 > 1 Yes	Swap	4, 1, 8, 9, 3
3rd vs 4th	4, 1, [8, 9], 3	8 > 9 No	No change	4, 1, 8, 9, 3
4th vs 5th	4, 1, 8, [9, 3]	9 > 3 Yes	Swap	4, 1, 8, 3, 9

Remember: Bubble = neighbours. Bubble Sort only ever compares neighbours (j and j+1). Each pass pushes the biggest bubble to the end. For descending order, just change >...

Exam: The most common question: "Sort the list ... using Bubble Sort. Show each pass." Draw one table per pass with columns Pair, Compare, Action, List after. Mark the element...

ببل سارٹ (Bubble Sort) میں ساتھ ساتھ موجود دو عناصر کا موازنہ کیا جاتا ہے اور اگر وہ غلط ترتیب میں ہوں تو ان کی جگہ بدل دی جاتی ہے۔

2.4.1 Sorting Algorithms: Selection Sort ★ Golden

Selection Sort is a simple sorting algorithm that works by repeatedly selecting the smallest element from the unsorted portion of the list and moving it to its correct position at the front.

Fix position $K \rightarrow$ Compare $DATA[K]$ with $DATA[PTR]$, $PTR = K+1 \dots N \rightarrow DATA[K] > DATA[PTR]? \text{ swap} \rightarrow$ End of pass: smallest is at $K \rightarrow K = K + 1$

- Think of it as filling seats from the front: first find who belongs in position 1, then position 2, and so on.
- In the textbook version, the element at position K is compared with every element after it; whenever a smaller one is found, the two are...
- So at the end of pass K , the K -th smallest value sits in position K .

Compare	List	Pos1 > PosX?	Action	List after
Pos1 vs Pos2	[8], [4], 1, 9, 3	$8 > 4$ Yes	Swap	4, 8, 1, 9, 3
Pos1 vs Pos3	[4], 8, [1], 9, 3	$4 > 1$ Yes	Swap	1, 8, 4, 9, 3
Pos1 vs Pos4	[1], 8, 4, [9], 3	$1 > 9$ No	No change	1, 8, 4, 9, 3
Pos1 vs Pos5	[1], 8, 4, 9, [3]	$1 > 3$ No	No change	1, 8, 4, 9, 3

Remember: Teacher's tip — spot the difference: in Selection Sort the inner loop starts at $outer + 1$ and compares with the fixed position $DATA[outer]$. In Bubble Sort the...

Exam: Typical questions: "Sort ... using Selection Sort, showing all steps", "Differentiate between Bubble Sort and Selection Sort" and "Why is Selection Sort not stable?" In...

سلیکشن سارٹ (Selection Sort) میں فہرست کے غیر ترتیب شدہ حصے سے سب سے چھوٹا عدد چن کر اسے سامنے اس کی درست جگہ پر رکھا جاتا ہے۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Abstraction:** Do not confuse abstraction with decomposition.
- **Sorting and Searching Algorithms:** Position 1 in the textbook algorithm is index 0 in Python.
- **Sorting Algorithms: Bubble Sort:** The slides say the time grows "exponentially".

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Sorting Algorithms: Bubble Sort**. Write: Bubble Sort is an elementary sorting algorithm that works by repeatedly comparing adjacent (side-by-side) elements and swapping them if they are in the wrong order.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) In each pass, the largest remaining value "bubbles up" to the end of the list, like an air bubble rising to the top of water. (2) So after pass 1 the largest value is in the last place, after pass 2 the second largest is in second-last place, and so on. — + example.

5 marks: 3-mark structure + table: Comparison / Pair / Left > Right? / Action / List after + the worked example from this topic.

8–10 marks: 5-mark structure + working code with correct indentation and a short trace of one example + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Sorting Algorithms: Selection Sort**. Write: Selection Sort is a simple sorting algorithm that works by repeatedly selecting the smallest element from the unsorted portion of the list and moving it to its correct position at the front.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Think of it as filling seats from the front: first find who belongs in position 1, then position 2, and so on. (2) In the textbook version, the element at position K is compared with every element after it; whenever a smaller one is found, the two are... — + example.

5 marks: 3-mark structure + table: Compare / List / Pos1 > PosX? / Action / List after + the worked example from this topic.

8–10 marks: 5-mark structure + working code with correct indentation and a short trace of one example + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. How many passes and comparisons does Bubble Sort need for 6 elements?

3 marks. Explain Sorting Algorithms: Bubble Sort with two clear points and one example.

5 marks. The most common question: "Sort the list ... using Bubble Sort. Show each pass." Draw one table per pass with columns Pair, Compare, Action, List after. Mark the element fixed at the end after every...

8 marks. Sorting Algorithms: Bubble Sort: 5-mark answer plus working code with correct indentation and a short trace of one example.

Answers: 1. 5 passes; $5 + 4 + 3 + 2 + 1 = 15$ comparisons 3. In each pass, the largest remaining value "bubbles up" to the end of the list, like an air bubble rising to the top of water.; So after pass 1 the largest value is in the last place, after pass 2 the second largest is in second-last place,... 5. Use the recipe in the blue box. Model opening: Bubble Sort is an elementary sorting algorithm that works by repeatedly comparing adjacent (side-by-side) elements and swapping them if they are in the wrong... 8. Same 5-mark body + working code with correct indentation and a short trace of one example + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Sorting Algorithms: Bubble Sort** using the recipe in this topic. Include working code with correct indentation and a short trace of one example. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 2: Computational Thinking & Algorithm

Computational Thinking & Algorithm (3/3) · Chapter exam

TODAY'S 90 MINUTES

90 minutes. 0–10 min recap yesterday ★ on the board · 10–85 min sit the 75-mark chapter exam (1 h 15 min, hall conditions) · 85–90 min collect scripts. Do not mark in front of the batch.

Board today: Searching Algorithms: Linear Search · Searching Algorithms: Binary Search · Evaluation Criteria of Algorithm · How to Decide Which Algorithm to Use

★ **Must lock:** Searching Algorithms: Binary Search

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Hall paper today. Recap the cards for ten minutes, then scripts out. Re-teach any Golden below 60% in tomorrow's starter.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. How many passes and comparisons does Bubble Sort need for 6 elements?
2. In Selection Sort, where does the inner loop start?
3. What is "noise" in abstraction?

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

2.4.2 Searching Algorithms: Linear Search

Linear Search, also known as sequential search, works by examining each element in the list one by one, from the start, until the target is found (or the list ends).

PTR = first → DATA[PTR] = FindValue? → Yes: ValueFound = 1, exit → No: PTR = PTR + 1 → List ended? print "Not Found"

- It works on both sorted and unsorted data.
- A flag variable (here ValueFound) starts at 0 and becomes 1 when the item is found.
- After the loop, the flag tells us whether to print the item or "Not Found".

Step	Element checked	Condition (== 7)	Result
1	8	8 == 7 ?	Not found
2	3	3 == 7 ?	Not found
3	6	6 == 7 ?	Not found
4	1	1 == 7 ?	Not found
5	7	7 == 7 ?	Found! (Exit loop)

Remember: Linear Search is like looking for your friend in a queue by checking every face from the front. Best case: 1 check (first person). Worst case: N checks (last...)

لینیئر سرچ (Linear Search) کو ترتیب وار تلاش (Sequential Search) بھی کہتے ہیں۔

2.4.2 Searching Algorithms: Binary Search ★ Golden

Binary Search is a highly efficient search algorithm that works by repeatedly dividing the search interval in half.

BEG = first, END = last → MID = $\text{INT}((\text{BEG} + \text{END}) / 2)$ → Item = DATA[MID]? Found → Item < DATA[MID]: END = MID-1 →
Else: BEG = MID+1 → Repeat while BEG ≤ END

- BEG (low) — first position of the current part.
- END (high) — last position of the current part.
- MID = $\text{INT}((\text{BEG} + \text{END}) / 2)$ — the middle position. INT means "drop the decimal part".
- Equal → found, stop.

Step	BEG (low)	END (high)	MID	DATA[MID]	Compare	Action
1	1	9	$\text{INT}(10/2) = 5$	5	$7 > 5$	BEG = 6 (go right)
2	6	9	$\text{INT}(15/2) = 7$	7	$7 = 7$	Found, LOC = 7

Remember: Binary Search is how you find a word in a dictionary: open the middle, decide "before or after?", and throw away half the book each time. "Sorted first, then..."

Exam: You will usually be given a sorted list and an item and asked to "search using Binary Search, showing all steps". Always draw a table with BEG, END, MID, DATA[MID],...

بائری سرچ (Binary Search) ایک تیز رفتار طریقہ ہے جو ہر قدم پر تلاش کے حصے کو آدھا کر دیتا ہے۔

2.5 Evaluation Criteria of Algorithm

Many algorithms can solve the same problem.

- To judge which is better, we evaluate them on three criteria: correctness, efficiency and clarity.

Input	Operation (divide by 2, find remainder)	Remainder == 0?	Result
3	3 / 2 (Quotient = 1, Remainder = 1)	No	Odd
6	6 / 2 (Quotient = 3, Remainder = 0)	Yes	Even
15	15 / 2 (Quotient = 7, Remainder = 1)	No	Odd

```
for number in [3, 6, 15]:
    remainder = number % 2
    if remainder == 0:
        print(number, "remainder", remainder, "→ Even")
    else:
        print(number, "remainder", remainder, "→ Odd")
```

Remember: Remember C-E-C: Correct (right answer, always finishes), Efficient (fast and light), Clear (easy to read).

کسی الگورتھم کو تین معیارات پر پرکھا جاتا ہے: درستگی (Correctness) یعنی درست نتیجہ دے، غیر معمولی صورتوں کو سنبھالے اور ہمیشہ ختم...

2.6 How to Decide Which Algorithm to Use

Before choosing an algorithm, ask three questions: What is the input size?

What is the task? → Need order? → Sorting algorithm → Need to find? Is data sorted? → No → Linear Search → Yes → Binary Search

- What is the input size? (10 items or 10 million?)
- Is the data sorted?
- Do we need speed or memory efficiency?
- If searching on unsorted data → use Linear Search.

Algorithm	Type	Best case	Worst case	Extra memory	Needs sorted data?	Stable?
Bubble Sort	Sorting	$O(N^2)$ as written; $O(N)$ with an early-stop check	$O(N^2)$	None — $O(1)$	—	Yes
Selection Sort	Sorting	$O(N^2)$	$O(N^2)$	None — $O(1)$	—	No
Linear Search	Searching	$O(1)$ (first item)	$O(N)$ (last item / absent)	None — $O(1)$	No	—
Binary Search	Searching	$O(1)$ (item at first MID)	$O(\log N)$	None — $O(1)$	Yes	—

Remember: Two keywords decide most exam answers: "sorted?" and "how big?". Unsorted or small → Linear. Sorted and big → Binary.

الگورتھم چننے سے پہلے دیکھیں کہ ڈیٹا کتنا بڑا ہے، ترتیب شدہ ہے یا نہیں، اور ہمیں رفتار چاہیے یا میموری کی بچت۔

Computational Thinking & Algorithm

13 single-page lessons · 6 Golden · read, close, say aloud, then sit the exam

Chapter in one look

- **Computational thinking** is systematic problem solving using decomposition, pattern recognition, abstraction and algorithms.
- An **algorithm** is a step-by-step logical procedure to solve a problem; it is conceptual and language-independent.
- Algorithms give structure, optimise time and memory, are reusable, ensure accuracy and act as a blueprint for code.
- **Pseudocode** is a human-readable, structured representation of an algorithm using keywords like IF, FOR and WHILE; it is not executable.
- An algorithm defines *what* to do; pseudocode shows *how* to implement it.
- **Decomposition** breaks a big problem into smaller sub-tasks; it reduces complexity, helps debugging and promotes reusability.
- **Pattern recognition** finds repetitions and rules (Row N = N stars) and is the foundation of loops.
- **Abstraction** hides noise and keeps only essential steps; situational details become variables.
- Sorting arranges data (ascending or descending); searching locates an element's position or existence.
- **Bubble Sort** compares and swaps adjacent elements; the largest value moves to the end each pass; N – 1 passes; stable; $O(N^2)$.
- **Selection Sort** brings the smallest remaining value to the front each pass; inner loop starts at outer + 1; fewer swaps; not stable; $O(N^2)$.
- **Linear Search** checks items one by one; works on unsorted data; $O(N)$ worst case.
- **Binary Search** halves the search range using BEG, MID and END; needs sorted data; $O(\log N)$ — about 20 steps for 1 million items.
- Algorithms are evaluated on **correctness** (checked with trace tables and edge cases), **efficiency** (time and space) and **clarity**.
- Choose by asking: input size, sorted or not, speed or memory? Unsorted → Linear; sorted → Binary; needs order → sort.

Key terms

Term	Meaning	اردو
Computational thinking	A systematic approach to solving problems using decomposition, patterns, abstraction and algorithms.	کمپیوٹیشنل سوچ
Algorithm	A step-by-step logical procedure to solve a problem, independent of any language.	الگورتھم، مرحلہ وار طریقہ حل
Pseudocode	A human-readable, structured but non-executable representation of an algorithm.	سوڈو کوڈ، فرضی کوڈ
Decomposition	Breaking a complex problem into smaller, manageable parts.	تقسیم کاری
Sub-task	One small part of a decomposed problem.	ذیلی کام
Pattern recognition	Identifying regularities, repetitions or rules in data.	نمونے کی پہچان
Abstraction	Hiding unnecessary details and focusing on essentials.	تجريد، غير ضروری تفصیل چھپانا
Noise	Details that do not affect the solution and can be ignored.	فالتو تفصیلات
Sorting	Arranging items in ascending or descending order.	ترتیب دینا
Searching	Locating an element in a list to find its position or existence.	تلاش کرنا
Ascending order	Smallest to largest (0–9, A–Z).	صعودی ترتیب
Descending order	Largest to smallest (9–0, Z–A).	نزولی ترتیب

Computational Thinking & Algorithm

75 marks • 1 hour 15 minutes • no notes • no phones

CHAPTER EXAM RULES

Instructions. Sit this exam after Lesson 13 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

Marks. $15 + 30 + 30 = 75$. A 5-mark / 10-mark answer needs Define • Explain • Example • Diagram • Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

Section A — Multiple choice (15 × 1 = 15)

1. An algorithm is best defined as:
 - a. a programming language
 - b. a step-by-step logical procedure to solve a problem
 - c. a type of computer hardware
 - d. compiled machine code
2. Pseudocode is:
 - a. executable machine code
 - b. a symbol used in flowcharts
 - c. a Python library
 - d. a human-readable representation of an algorithm using plain English and keywords
3. Breaking a complex problem into smaller, manageable parts is called:
 - a. decomposition
 - b. abstraction
 - c. pattern recognition
 - d. compilation
4. In the "making tea" example, ignoring the brand of the kettle is an example of:
 - a. decomposition
 - b. sorting
 - c. abstraction
 - d. searching
5. Pattern recognition forms the foundation of:
 - a. variables
 - b. loops
 - c. comments
 - d. input devices
6. Bubble Sort works by comparing:
 - a. the first and last elements
 - b. every element with the middle element
 - c. adjacent elements
 - d. random pairs of elements
7. After the first pass of Bubble Sort (ascending) on 8, 4, 1, 9, 3, the list is:
 - a. 1, 3, 4, 8, 9
 - b. 4, 8, 1, 9, 3
 - c. 1, 8, 4, 9, 3
 - d. 4, 1, 8, 3, 9
8. Binary Search works only on a list that is:
 - a. sorted
 - b. unsorted
 - c. stored in a linked list
 - d. of strings only
9. Linear Search is also known as:
 - a. binary search
 - b. sequential search
 - c. bubble search
 - d. random search
10. How many passes does Bubble Sort need to sort N elements?
 - a. N
 - b. N + 1
 - c. N - 1
 - d. 2N
11. In Selection Sort pseudocode, the inner loop starts at:
 - a. 0
 - b. N - 1
 - c. outer - 1
 - d. outer + 1
12. The worst-case time complexity of Binary Search is:
 - a. $O(\log N)$
 - b. $O(N)$
 - c. $O(N^2)$
 - d. $O(1)$
13. About how many steps does Binary Search need, at most, to find an item among 1 million sorted elements?
 - a. 2
 - b. 20
 - c. 1,000
 - d. 1,000,000
14. Which sorting algorithm is NOT stable?
 - a. Bubble Sort
 - b. Both are stable
 - c. Selection Sort
 - d. Neither sorts data
15. Which of the following is NOT an evaluation criterion of an algorithm?
 - a. Correctness
 - b. Efficiency
 - c. Clarity
 - d. Popularity

Section B — Short questions (10 × 3 = 30)

1. Define computational thinking and name its four components.
2. Write any four points on the importance of algorithms.
3. Give three differences between an algorithm and pseudocode.
4. What is decomposition? Explain it with the example of learning a new language.
5. What three questions should you ask during pattern recognition? Apply them to the triangle of stars.
6. Define abstraction. In the "making tea" example, name two variables and two noise items.
7. Differentiate between sorting and searching.
8. Why does Binary Search require a sorted list? State two disadvantages of Binary Search.
9. Give two differences between Bubble Sort and Selection Sort.
10. Name the three evaluation criteria of an algorithm. What is a trace table used for?

Section C — Long questions (attempt any 3 × 10 = 30)

1. Sort the list **5, 2, 7, 1, 6** in ascending order using Bubble Sort. Show every comparison of every pass in a table, and state the total number of comparisons and swaps. Then write the Python code for Bubble Sort.
2. Sort the list **6, 2, 9, 5, 1** in ascending order using Selection Sort (textbook method). Show every comparison of every pass and count the swaps. Also state why Selection Sort is called "blind" to an already sorted list.
3. The sorted list is **3, 8, 12, 17, 23, 31, 40, 52, 66** (Python indexes 0–8). Using Binary Search, show a low / mid / high table to search for (a) 40 and (b) 20. For each, state how many comparisons Linear Search would need.
4. Write the algorithm and pseudocode for Linear Search. Dry-run it on **8, 3, 6, 1, 7, 2, 4, 5** for FindValue = 2 and for FindValue = 9, and give two advantages and two disadvantages of Linear Search.
5. A school wants a program that reads the marks of 5 subjects for every student and prints each student's total, percentage and grade. Apply decomposition, pattern recognition and abstraction to this problem, and write a short algorithm.
6. A library has 50,000 book IDs stored in sorted order, and the librarian searches it hundreds of times a day. (a) Which search algorithm should be used and why? (b) What is the maximum number of comparisons for Linear and for Binary Search? (c) Explain the three evaluation criteria you would use to judge the chosen program.

Answer key. Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

Check yourself — answers

1. **2.1 Computational Thinking.** Answers: 1. Decomposition, pattern recognition, abstraction, algorithms. 2. Before coding (it is planning the solution). 3. Between human logic and machine code.
2. **2.2 Algorithm.** Answers: 1. Because it describes only the logic; it is not tied to Python, C++ or any other language. 2. Any two: structured approach, optimise time/memory, reusable, accurate and consistent, blueprint before coding. 3. Step 1: Start; Step 2: Read A and B; Step 3: SUM = A + B; Step 4: Print SUM; Step 5: End.
3. **2.2.2 Algorithm vs Pseudocode.** Answers: 1. Pseudocode. 2. Algorithm. 3. No; it is semi-formal and not executable.
4. **2.3.1 Decomposition.** Answers: 1. A sub-task (sub-problem). 2. Example: list subjects and teachers; decide periods per subject per week; place periods in days so no teacher has two classes at once. 3. Reduces complexity, helps in debugging, promotes reusability.
5. **2.3.2 Pattern Recognition.** Answers: 1. Loops. 2. 8 stars (Row N = N stars). 3. Term N = $2 \times N$ (each term adds 2).
6. **2.3.3 Abstraction.** Answers: 1. Unnecessary details that do not affect the solution. 2. A variable (it changes with the situation). 3. Decomposition breaks a problem into parts; abstraction hides details that are not important.
7. **2.4 Sorting and Searching Algorithms.** Answers: 1. D, C, B, A. 2. It temporarily holds one value so that two values can be swapped without losing either.
8. **2.4.1 Sorting Algorithms: Bubble Sort.** Answers: 1. 5 passes; $5 + 4 + 3 + 2 + 1 = 15$ comparisons. 2. (5,1) swap $\rightarrow$ 1, 5, 4; (5,4) swap $\rightarrow$ 1, 4, 5. 3. Pass 1: 9, 4, 7, 2; Pass 2: 9, 7, 4, 2; Pass 3: 9, 7, 4, 2.
9. **2.4.1 Sorting Algorithms: Selection Sort.** Answers: 1. At outer + 1 (PTR = K + 1). 2. Pos1 vs Pos2: $7 > 2$ swap $\rightarrow$ 2, 7, 5; Pos1 vs Pos3: $2 > 5$ no $\rightarrow$ 2, 7, 5. 3. Bubble Sort.
10. **2.4.2 Searching Algorithms: Linear Search.** Answers: 1. Sequential search. 2. 6 comparisons (2 is in position 6). 3. It is a flag: 0 means not found yet, 1 means found; it decides what to print after the loop.
11. **2.4.2 Searching Algorithms: Binary Search.** Answers: 1. The list must be sorted. 2. $\text{INT}(15/2) = 7$. 3. MID = 4 (value 4, go right, BEG = 5) $\rightarrow$ MID = $\text{INT}(13/2) = 6$ (value 6, go right, BEG = 7) $\rightarrow$ MID = $\text{INT}(15/2) = 7$ (value 7, go right, BEG = 8) $\rightarrow$ MID = 8 (value 8, found). Four comparisons.
12. **2.5 Evaluation Criteria of Algorithm.** Answers: 1. Correctness, efficiency, clarity. 2. The item is not in the list (or the list is empty, or the item is the last element). 3. Time (speed) and space (memory).
13. **2.6 How to Decide Which Algorithm to Use.** Answers: 1. Linear Search. 2. $O(\log N)$. 3. Selection Sort ($O(N^2)$ in both).

Answer key

Multiple choice

1. b
2. d
3. a
4. c
5. b
6. c
7. d
8. a
9. b
10. c
11. d
12. a
13. b
14. c
15. d

Short questions — model answers

1. Computational thinking is a systematic approach to solving problems. Its four components are decomposition (breaking down), pattern recognition (finding repetitions), abstraction (hiding unnecessary details) and algorithms (writing step-by-step solutions).
2. Algorithms (1) provide a structured approach that makes complex tasks manageable; (2) optimise time and memory usage, crucial for large applications; (3) can be reused for similar problems; (4) ensure accuracy and consistency, reducing human error; (5) act as a blueprint before coding. (Any four.)
3. (1) An algorithm is a step-by-step logical procedure; pseudocode is a human-readable representation of it using English and keywords. (2) An algorithm is conceptual and language-independent; pseudocode is semi-formal and structured but not executable. (3) An algorithm has no fixed format and is for anyone; pseudocode uses IF/FOR/WHILE, follows conventions and is mainly for programmers.
4. Decomposition is breaking a complex task into smaller, manageable parts. Learning a language is decomposed into: vocabulary (words like "I", "eat", "apple"), grammar rules (Subject → Verb → Object), tenses ("I eat" vs "I ate") and sentence building (combining words with rules).
5. Ask: What is repeating? What changes in a predictable way? Can I generalise this? For the star triangle: stars repeat in every row; each row has one more star than the previous; generalisation: Row N = N stars, so one loop can print it.
6. Abstraction means hiding unnecessary features and focusing on the essential ones. Variables: type of tea (black, green, milk), additions (sugar, lemon), tool (stove or kettle) — any two. Noise: brand of kettle, cup colour or design, where tea is made — any two.
7. Sorting arranges a collection of items in ascending or descending order to organise data (e.g., Bubble Sort, Selection Sort). Searching locates a specific element in a list to find its position or whether it exists (e.g., Linear Search, Binary Search).
8. Binary Search decides to go left or right by comparing with the middle value; this decision is only valid if all smaller values are on the left and larger on the right, i.e., the list is sorted. Disadvantages: needs sorted data; needs random access (arrays, not linked lists); code is more complex and prone to off-by-one errors. (Any two.)
9. (1) Bubble Sort compares adjacent elements; Selection Sort compares a fixed position with all later elements. (2) Bubble Sort moves the largest to the end each pass; Selection Sort moves the smallest to the front. Also: Bubble Sort is stable, Selection Sort is not; Selection Sort makes fewer swaps.
10. Correctness, efficiency (time and space) and clarity (readability). A trace table records the values of variables step by step during a dry run, so we can check that the output matches the expected result and so prove correctness.

Long questions — model answers

1. Bubble Sort on 5, 2, 7, 1, 6

Pass	Pair	Left > Right?	List after
1	[5, 2]	Yes, swap	2, 5, 7, 1, 6
1	[5, 7]	No	2, 5, 7, 1, 6
1	[7, 1]	Yes, swap	2, 5, 1, 7, 6
1	[7, 6]	Yes, swap	2, 5, 1, 6, 7
2	[2, 5]	No	2, 5, 1, 6, 7
2	[5, 1]	Yes, swap	2, 1, 5, 6, 7
2	[5, 6]	No	2, 1, 5, 6, 7
3	[2, 1]	Yes, swap	1, 2, 5, 6, 7
3	[2, 5]	No	1, 2, 5, 6, 7
4	[1, 2]	No	1, 2, 5, 6, 7

Total comparisons = 4 + 3 + 2 + 1 = 10; total swaps = 3 + 1 + 1 + 0 = 5. Sorted list: 1, 2, 5, 6, 7.

```
data = [5, 2, 7, 1, 6]
n = len(data)
for outer in range(n - 1):
    for inner in range(n - 1 - outer):
        if data[inner] > data[inner + 1]:
            temp = data[inner]
            data[inner] = data[inner + 1]
            data[inner + 1] = temp
print(data)
```

OUTPUT
[1, 2, 5, 6, 7]

2. Selection Sort (textbook method) on 6, 2, 9, 5, 1

Pass	Compare	Values	Action	List after
1	Pos1 vs Pos2	6 > 2	Swap	2, 6, 9, 5, 1
1	Pos1 vs Pos3	2 > 9	No	2, 6, 9, 5, 1
1	Pos1 vs Pos4	2 > 5	No	2, 6, 9, 5, 1
1	Pos1 vs Pos5	2 > 1	Swap	1, 6, 9, 5, 2
2	Pos2 vs Pos3	6 > 9	No	1, 6, 9, 5, 2
2	Pos2 vs Pos4	6 > 5	Swap	1, 5, 9, 6, 2
2	Pos2 vs Pos5	5 > 2	Swap	1, 2, 9, 6, 5
3	Pos3 vs Pos4	9 > 6	Swap	1, 2, 6, 9, 5
3	Pos3 vs Pos5	6 > 5	Swap	1, 2, 5, 9, 6
4	Pos4 vs Pos5	9 > 6	Swap	1, 2, 5, 6, 9

10 comparisons, 7 swaps. Sorted list: 1, 2, 5, 6, 9. (With the classic one-swap-per-pass method the passes give 1, 2, 9, 5, 6 → 1, 2, 9, 5, 6 → 1, 2, 5, 9, 6 → 1, 2, 5, 6, 9.)

Selection Sort is "blind" to a sorted list because its loops have fixed limits: it always makes all $N(N-1)/2$ comparisons, even if the list is already in order. It has no way to notice that no swaps were needed and stop early.

3. (a) Search for 40

Step	low	high	mid	data[mid]	Compare	Action
1	0	8	$(0+8)//2 = 4$	23	$40 > 23$	low = 5
2	5	8	$(5+8)//2 = 6$	40	$40 = 40$	Found at index 6

Binary: 2 comparisons. Linear Search: 40 is the 7th item, so 7 comparisons.

(b) Search for 20

Step	low	high	mid	data[mid]	Compare	Action
1	0	8	4	23	$20 < 23$	high = 3
2	0	3	$(0+3)//2 = 1$	8	$20 > 8$	low = 2
3	2	3	$(2+3)//2 = 2$	12	$20 > 12$	low = 3
4	3	3	3	17	$20 > 17$	low = 4
—	4	3	low > high, so the loop stops: Not Found			

Binary: 4 comparisons. Linear Search must check all 9 items before saying "Not Found".

4. Algorithm and pseudocode: as in section 2.4.2 (Start; DATA, N, FindValue, ValueFound := 0; repeat for PTR = 1 to N: if DATA[PTR] = FindValue then ValueFound = 1 and exit; after the loop print DATA[PTR] if ValueFound = 1, else "Not Found"; End).

FindValue = 2:

Step	1	2	3	4	5	6
Element	8	3	6	1	7	2
== 2?	No	No	No	No	No	Yes — exit

Found at position 6 after 6 comparisons; ValueFound = 1; prints 2.

FindValue = 9: 8, 3, 6, 1, 7, 2, 4, 5 are all checked (8 comparisons), none equals 9, ValueFound stays 0, so "Data Not Found" is printed.

Advantages: works on sorted and unsorted data; very simple code; fast for small lists; no extra space. Disadvantages: up to N checks (1 million for 1 million items); time grows linearly; inefficient if the same list is searched thousands of times. (Any two of each.)

5. **Decomposition:** (1) input a student's 5 marks; (2) calculate total; (3) calculate percentage = total ÷ 500 × 100; (4) decide grade; (5) print result; (6) repeat for the next student.

Pattern recognition: the same steps repeat for every student, and the same "add a mark" step repeats for every subject — so use a loop over students and a loop over subjects.

Abstraction: keep only roll number, marks, total, percentage and grade; ignore noise such as the student's address, uniform or the colour of the result card. The number of students can be a variable.

Algorithm: Step 1: Start. Step 2: Read number of students S. Step 3: Repeat Steps 4–8 for each student. Step 4: TOTAL = 0. Step 5: Repeat 5 times: read MARK, TOTAL = TOTAL + MARK. Step 6: PER = TOTAL / 500 × 100. Step 7: If PER ≥ 80 grade "A1", else if PER ≥ 70 "A", else if PER ≥ 60 "B", else if PER ≥ 50 "C", else "Fail" (school's own grading scale). Step 8: Print TOTAL, PER, GRADE. Step 9: End.

6. (a) **Binary Search**, because the data is already sorted, it is large (50,000 items) and it is searched many times a day, so the time saved per search adds up; Linear Search would be far slower.

(b) Linear Search worst case: 50,000 comparisons. Binary Search worst case: 16 comparisons, because $2^{15} = 32,768 < 50,000 \leq 2^{16} = 65,536$, so 16 halvings are enough to reduce 50,000 items to one.

(c) **Correctness:** it must return the right position for IDs that exist, say "Not Found" for missing IDs (edge cases: first ID, last ID, missing ID), and always stop. **Efficiency:** time $O(\log N)$ and no extra memory ($O(1)$). **Clarity:** meaningful names like low, mid, high, clear steps and good indentation so another programmer can maintain it.

Self-assessment: I can...

- ☐ define computational thinking and name its four parts.
- ☐ draw a table comparing algorithm and pseudocode on at least five points.
- ☐ find a pattern in data and turn it into a loop.
- ☐ dry-run Bubble Sort pass by pass and count comparisons and swaps.
- ☐ trace Linear Search with a step table, including the "not found" case.
- ☐ state best and worst time complexity for all four algorithms.
- ☐ define an algorithm and give five reasons it is important.
- ☐ decompose a real-life problem into sub-tasks and explain three benefits.
- ☐ separate necessary steps, variables and noise using abstraction.
- ☐ dry-run Selection Sort pass by pass and explain why it is not stable.
- ☐ trace Binary Search with a BEG / MID / END table and explain why it needs sorted data.
- ☐ evaluate an algorithm for correctness, efficiency and clarity, and choose the right algorithm for a situation.

Chapter 3: Programming Fundamentals

Programming Fundamentals (1/5) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Programming · Programming Language Types · Career Growth with Python · Integrated Development Environment (IDE) · Visual Studio Code (VS Code)

★ **Must lock:** none — still copy every table

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. What is the crucial condition for Binary Search to work?
2. What is another name for Linear Search?
3. Name the three evaluation criteria of an algorithm.

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

3.1 Programming

Computers are "dumb" machines. They are very fast, but they cannot think or do anything on their own.

- A set of instructions given to a computer is called a program.
- The language used to write these instructions is called a programming language.
- Writing programs is called programming, and the person who writes them is a programmer.

```
print("Hello")
```

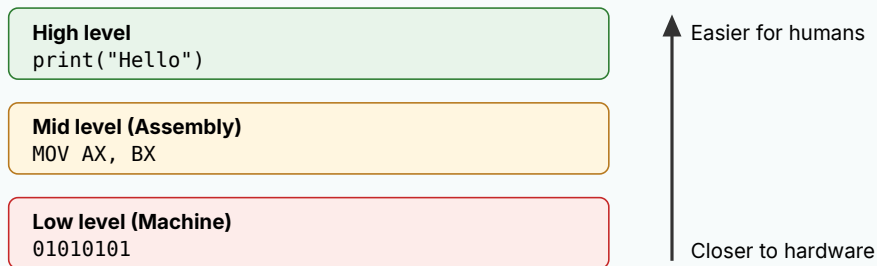
Remember: Program = recipe, programmer = recipe writer, computer = cook who follows every line exactly. If the recipe is wrong, the food is wrong. The cook does not fix...

کمپیوٹر خود سے کچھ نہیں سوچ سکتا۔

3.1.1 Programming Language Types

Programming languages are classified into three main types, depending on how close they are to the computer's own language (binary 0s and 1s) or to human language.

Type	Description	Examples
Low level	Closest to the native language of the computer (binary 0s and 1s). Very difficult for humans to read and write.	Machine language (binary)
Mid level	Bridges the gap. It can access hardware but is easier to read than binary because it uses short words called mnemonics (e.g. MOV , ADD).	Assembly language
High level	Designed to be easy for humans to read and write. Uses English-like words.	C++, Java, Python



The three levels of programming languages, with a sample line from each.

Remember: "Low = Like the machine, High = Human." Mid level sits in the Middle and uses Mnemonics (both start with M).

پروگرامنگ زبانوں کی تین اقسام ہیں: لو لیول (Low Level) جو بائری 0 اور 1 میں ہوتی ہے، مڈ لیول (Mid Level) یعنی اسمبلی جو مختصر الف...

3.2 Career Growth with Python

Python is one of the most in-demand languages in the world.

- Software Development: build desktop and web applications.
- Data Science and Data Analysis: analyse data, create reports and make predictions.
- Artificial Intelligence and Machine Learning: develop intelligent systems, chatbots and recommendation algorithms (like "videos you may...").
- Web Development: use frameworks such as Django and Flask to build websites.

```
# Python
print("Hello")
```

Remember: Six careers — remember "S-D-A-W-A-C": Software, Data, AI, Web, Automation, Cybersecurity.

پائنتھن سیکھ کر آپ سافٹ ویئر ڈیولپمنٹ، ڈیٹا سائنس، مصنوعی ذہانت (AI)، ویب ڈیولپمنٹ، آٹومیشن اور سائبر سیکیورٹی جیسے شعبوں میں کیریئر بنا سکتے ہیں۔

3.3 Integrated Development Environment (IDE)

An IDE (Integrated Development Environment) is a software tool that helps programmers write, run and manage code in one place.

- Code Editor: where you write and edit code.
- Syntax Highlighting: colours different parts of the code (keywords, strings, numbers) to make it easier to read.
- Auto-completion: suggests code while you type (type `print` and it offers `print`).
- Run/Execute Tool: runs the program with one click.

Remember: IDE features — "E-S-A-R-D": Editor, Syntax highlighting, Auto-completion, Run, Debugger.

آئی ڈی ای (IDE) ایک ایسا سافٹ ویئر ہے جس میں کوڈ لکھنا، چلانا اور غلطیاں درست کرنا ایک ہی جگہ ممکن ہوتا ہے۔

3.3.1 Visual Studio Code (VS Code)

VS Code is a free, lightweight and powerful code editor made by Microsoft and widely used for Python.

Install Python → Install VS Code → Extensions: install "Python" (Microsoft) → Explorer: new file hello.py → Type code → Click ► Run

- By default it is not a full IDE, but it becomes one when you install extensions (add-ons), such as the Python extension by Microsoft.

Element	Description
1. Top Menu Bar	Control over files, editing, navigation and workspace (File, Edit, Selection, View, Go, Run).
2. Side Bar	Contains the core tools: Explorer, Search, Source Control, Run & Debug and Extensions.
3. Explorer	Shows project files and folders. Helps you create, open and manage your Python files.
4. Search	Finds and replaces text in the open file or across the whole project.
5. Source Control	Connects with Git (a version control system) to track changes, undo mistakes and collaborate.
6. Run & Debug	Runs code, finds errors, pauses at specific lines and lets you inspect variables.
7. Extensions	Like an App Store for VS Code. Add features such as the Python extension, Prettier or Python Debugger.



A simplified VS Code window: menu bar, side bar, Explorer and the code editor.

```
print("Hello, World!")
```

Remember: Extensions are the "App Store" of VS Code: just as apps give your phone new abilities, extensions give VS Code new abilities.

وی ایس کوڈ (VS Code) ایک مفت اور ہلکا پھلکا کوڈ ایڈیٹر ہے۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Programming:** Students often think the computer "understands" what they want.
- **Programming Language Types:** "Low level" does not mean "bad" or "weak".
- **Career Growth with Python:** Django and Flask are web frameworks (for websites), not AI tools.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

ACADEMY DRILL

Timed 10 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. What is a program?

3 marks. Explain Programming with two clear points and one example.

5 marks. Write a 5-mark answer on Programming: definition, two points, a labelled diagram or table, and one example.

Answers: 1. A set of instructions given to a computer to perform a task 3. The language used to write these instructions is called a programming language.; Writing programs is called programming, and the person who writes them is a programmer. 5. Use the recipe in the blue box. Model opening: Computers are "dumb" machines.

BATCH HOMEWORK

Write a 5-mark answer on **Programming** using Define · Explain · Example · Diagram. Learn the Check yourself questions so you can answer them in 60 seconds at the start of next class.

How you will mark it tomorrow: Collect next class. Stamp full / half / zero against the 5-mark recipe.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 3: Programming Fundamentals

Programming Fundamentals (2/5) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Fundamentals of Python Programming · Variables · Data Types in Python · Input and Output Handling · Operators and Operands: Arithmetic and Assignment

★ **Must lock:** Fundamentals of Python Programming, Variables, Data Types in Python, Operators and Operands: Arithmetic and Assignment

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. What is a program?
2. Which level is MOV AX, BX?
3. Name two Python web frameworks.

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

3.4 Fundamentals of Python Programming ★ Golden

The basic structure of a Python program is simple and clear.

- A program takes data, processes it using statements, and finally shows output.
- Python runs the statements top to bottom, one after another (sequentially).
- A statement is one complete instruction, usually written on one line, such as `print("Hi")` or `x = 5`.

```
# This is a comment. It explains the code below.  
print("Hello, World!")  
if 5 > 2:  
    # The space before 'print' is the indentation  
    print("Five is greater than two!")
```

Remember: Think of indentation like the points and sub-points in your notebook. Sub-points are written a little to the right to show they "belong" to the main point. A...

Exam: Common questions: "What is the purpose of comments?", "Why is indentation important in Python?", "How is Python different from C++/Java in defining blocks?" Write:...

پائنتھن پروگرام اوپر سے نیچے ترتیب سے چلتا ہے۔

3.4.1 Variables ★ Golden

A variable is a named space in computer memory used to store data.

- Can contain letters, numbers and underscores (_) only — no spaces or symbols like -, @, \$.
- Must start with a letter or underscore, NOT a number (1name is invalid).
- Are case-sensitive: age and Age are two completely different variables.
- Cannot be a Python keyword (a reserved word such as if, for, while, else, True).

```
name = "Ali"  
age = 16  
my_height = 5.7  
score1 = 100
```

Remember: Choose meaningful names: total_marks is far clearer than tm or x. Use lowercase words joined by underscores (this style is called snake_case).

Exam: A favourite question: "Which of these variable names are valid? Give reasons." Check each name against all four rules and state the rule broken, e.g. "2nd_place —...

ویری ایبل (Variable) میموری میں ایک نام والی جگہ ہے جس میں ڈیٹا رکھا جاتا ہے، بالکل ایک لیبل لگے ڈیے کی طرح۔

3.4.1 Data Types in Python ★ Golden

A data type specifies the kind of data a variable holds — a whole number, a decimal, text, True/False, or a collection of items.

- The type decides what you can do with the data (you can divide numbers, but not names).
- Python has several built-in data types:

Name	Type	Description	Example
Integer	<code>int</code>	Whole numbers without decimals (positive, negative or zero).	<code>age = 16</code>
Floating point	<code>float</code>	Numbers with a decimal point.	<code>height = 5.7</code>
String	<code>str</code>	Text data, enclosed in quotes (" " or ' ').	<code>name = "Sammy" or year = "2026"</code>
Boolean	<code>bool</code>	Logical value: True or False (capital T / F).	<code>is_student = True</code>
List	<code>list</code>	Ordered, changeable collection in square brackets [] .	<code>marks = [85, 90, 75]</code>
Tuple	<code>tuple</code>	Ordered, unchangeable sequence in parentheses () .	<code>coords = (10, 20)</code>
Dictionary	<code>dict</code>	Key : value pairs in curly braces { } .	<code>student = {"name": "Ahmed", "age": 16}</code>
Set	<code>set</code>	Unordered collection of unique items (no duplicates).	<code>vowels = {'a', 'e', 'i'}</code>

Remember: Brackets decide the collection: [] List (changeable), () Tuple (locked), { } with key:value = Dictionary, { } with single items = Set (unique). Think: a list...

Exam: Expect: "Name any four data types with examples" or "Differentiate list and tuple". For a difference question, use a two-column table: list — [], changeable, e.g. marks...

ڈیٹا ٹائپ (Data Type) بتاتی ہے کہ ویری ایبل میں کس قسم کا ڈیٹا ہے۔

3.4.2 Input and Output Handling

Output — the `print()` function. A function is a named, ready-made piece of code that does a job when you call it with brackets.

- Items separated by commas are printed with one space between them.
- An f-string (a string with `f` before the quote) lets you put variables directly inside `{ }`. It is the fastest and most readable way to...

Remember: "`input()` always gives a string." If you plan to do maths, wrap it: `int(input(...))` for whole numbers, `float(input(...))` for decimals. Read it from the inside...

`print()` اسکرین پر معلومات دکھاتا ہے اور `input()` صارف سے معلومات لیتا ہے۔

3.4.3 Operators and Operands: Arithmetic and Assignment ★ Golden

An operand is the value or variable being worked on.

- An operator is the symbol that performs the action.
- Together they form an expression, which Python evaluates to a single value.
- In $10 + 20$: 10 is an operand, + is the operator, 20 is an operand, and the result is 30.

Operator	Purpose	Example	Result
+	Addition	$5 + 3$	8
-	Subtraction	$10 - 4$	6
*	Multiplication	$4 * 2$	8
/	Division (result is always a float)	$9 / 2$	4.5
//	Floor division (drops the decimal part)	$9 // 2$	4
%	Modulus (remainder)	$9 \% 2$	1
**	Exponentiation (power)	$2 ** 3$	8

```
print(5 + 3)
print(10 - 4)
print(4 * 2)
print(9 / 2)
print(9 // 2)
print(9 % 2)
print(2 ** 3)
print(8 / 2)
```

Remember: Three kinds of division: / = exact answer (float), // = quotient only ("floor"), % = remainder only. A number is even when $n \% 2 == 0$.

Exam: You will be asked to evaluate expressions such as $17 // 5$, $17 \% 5$, $2 ** 4$ or to trace compound assignment. Show each step and write the final value with its type (e.g....

آپرینڈ (Operand) وہ ویلیو ہے جس پر عمل ہوتا ہے اور آپریٹر (Operator) وہ علامت ہے جو عمل کرتی ہے۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Fundamentals of Python Programming:** IndentationError?
- **Variables:** = is not "equals" in the maths sense.
- **Data Types in Python:** True and False must start with a capital letter.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Fundamentals of Python Programming**. Write: The basic structure of a Python program is simple and clear.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) A program takes data, processes it using statements, and finally shows output. (2) Python runs the statements top to bottom, one after another (sequentially). — + example.

5 marks: 3-mark structure + four-point explanation + the worked example from this topic.

8–10 marks: 5-mark structure + a complete truth table and a labelled gate / logic diagram + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Variables**. Write: A variable is a named space in computer memory used to store data.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Must start with a letter or underscore, NOT a number (1name is invalid). (2) Are case-sensitive: age and Age are two completely different variables. — + example.

5 marks: 3-mark structure + four-point explanation + the worked example from this topic.

8–10 marks: 5-mark structure + working code with correct indentation and a short trace of one example + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Data Types in Python**. Write: A data type specifies the kind of data a variable holds — a whole number, a decimal, text, True/False, or a collection of items.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) The type decides what you can do with the data (you can divide numbers, but not names). (2) Python has several built-in data types: — + example.

5 marks: 3-mark structure + table: Name / Type / Description / Example + the worked example from this topic.

8–10 marks: 5-mark structure + working code with correct indentation and a short trace of one example + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. Predict the output:

3 marks. Explain Fundamentals of Python Programming with two clear points and one example.

5 marks. Common questions: "What is the purpose of comments?", "Why is indentation important in Python?", "How is Python different from C++/Java in defining blocks?" Write: comments start with # and are...

8 marks. Fundamentals of Python Programming: 5-mark answer plus a complete truth table and a labelled gate / logic diagram.

Answers: 1. B then D on separate lines (line 1 is a comment; $3 > 7$ is false so C is skipped) 3. A program takes data, processes it using statements, and finally shows output.; Python runs the statements top to bottom, one after another (sequentially). 5. Use the recipe in the blue box. Model opening: The basic structure of a Python program is simple and clear. 8. Same 5-mark body + a complete truth table and a labelled gate / logic diagram + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Fundamentals of Python Programming** using the recipe in this topic. Include a complete truth table and a labelled gate / logic diagram. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 3: Programming Fundamentals

Programming Fundamentals (3/5) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Relational, Logical and Membership Operators · Bitwise Operators · Control Structures · Sequence · Selection Statements: if and if-else

★ **Must lock:** Relational, Logical and Membership Operators, Bitwise Operators, Control Structures, Selection Statements: if and if-else

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Predict the output:
2. Valid or invalid? `_total`, `roll-no`, `class`, `marks2026`, `2marks`
3. What is the data type of each? `3.0`, `"3"`, `3`, `False`, `[3]`

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

3.4.3 Relational, Logical and Membership Operators ★ Golden

3. Relational (comparison) operators compare two values and return a Boolean (True or False).

- Relational (comparison) operators compare two values and return a Boolean (True or False).
- They are used in conditions.

Op	Description	Example (a = 10, b = 5)	Result
=	Equal to	a == b	False
≠	Not equal to	a != b	True
>	Greater than	a > b	True
<	Less than	a < b	False
≥	Greater than or equal to	a ≥ 10	True
≤	Less than or equal to	a ≤ b	False

```
a = 10
b = 5
print(a == b)
print(a != b)
print(a > b)
print(a < b)
print(a ≥ 10)
print(a ≤ b)
```

Remember: AND is strict, OR is generous. AND is like a teacher who needs both homework and attendance; OR is like a shop that accepts cash or card. Order of evaluation:...

Exam: Typical: "Evaluate a > b and b > 3 for a = 10, b = 5" or "Write the truth table of and/or". Write each sub-condition's value first, then combine. Remember: == compares,...

ریلیشنل آپریٹرز (Relational Operators) دو ویلیوز کا موازنہ کر کے True یا False دیتے ہیں۔

3.4.3 Bitwise Operators ★ Golden

6. Bitwise operators treat numbers as binary digits (bits: 0s and 1s) and work on them bit by bit, column by column.

- AND (&) — gives 1 only if both bits are 1, otherwise 0.
- OR (|) — gives 1 if at least one bit is 1.
- XOR (^) — gives 1 if the bits are different, 0 if they are the same.
- NOT (~) — inverts all bits (0 → 1, 1 → 0). In Python the result is the number made negative, minus 1: $\sim x = -(x + 1)$. So ~ 10 gives -11.

Bit A	Bit B	A & B	A B	A ^ B	~A
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

```
a = 10
b = 6
print(bin(a), bin(b))
print(a & b)
print(a | b)
print(a ^ b)
print(~a)
```

Remember: Shortcuts: $x > n = x // 2^n$, $\sim x = -x - 1$. For AND/OR/XOR always write both numbers in binary, one under the other, with the same number of bits.

Exam: Exams ask you to "evaluate $12 \& 10$, $12 | 10$, $12 \wedge 10$, ~ 5 , $7 >> 1$ showing binary working". Marks are given for the binary conversion and column working, not just the...

بٹ وائز آپریٹرز (Bitwise Operators) اعداد کو بائری میں بدل کر ہر بٹ پر الگ الگ کام کرتے ہیں۔

3.5 Control Structures ★ Golden

Control structures decide the order in which statements are executed in a program (the "flow of control").

- Sequence — do the steps one after another, top to bottom.
- Selection — choose a path depending on a condition (if, else).
- Repetition — repeat steps (loops: for, while).

Remember: Remember "S-S-R": Sequence, Selection, Repetition — "Step, Select, Repeat".

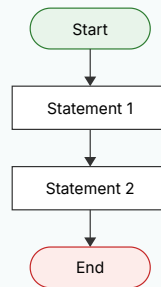
Exam: "Name and explain the three basic control structures" is a very common short question. Give a one-line definition and a two-line Python example for each.

کنٹرول اسٹرکچرز (Control Structures) طے کرتے ہیں کہ پروگرام کی ہدایات کس ترتیب سے چلیں گی۔

3.5.1 Sequence

Sequence means statements are executed one after another, strictly from top to bottom, in the exact order they are written.

- This is the default behaviour of Python.
- No statement is skipped or repeated.



Sequence: control flows straight down from one statement to the next.

```
print("Enter two numbers")
a = int(input("Enter first number: ")) # Step 1: Input
b = int(input("Enter second number: ")) # Step 2: Input
total = a + b # Step 3: Process
print("Sum =", total) # Step 4: Output
```

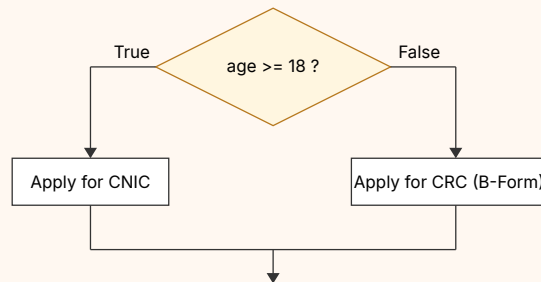
Remember: Sequence is like reading a page: line 1, then line 2, then line 3. Most programs follow the pattern Input → Process → Output.

سیکوئنس (Sequence) کا مطلب ہے کہ ہدایات بالکل اسی ترتیب سے ایک کے بعد ایک اوپر سے نیچے چلتی ہیں جس ترتیب سے لکھی گئی ہیں۔

3.5.2 Selection Statements: if and if-else ★ Golden

Selection changes the flow of a program based on a condition — an expression that is either True or False.

- It helps a program respond intelligently to different situations.
- The if statement executes a block of code only if the condition is True.
- If the condition is False, the block is completely skipped.



An if-else statement: one diamond (condition), two paths, then the paths join again.

```
if condition:
    statement(s)    # runs only when condition is True
```

Remember: Formatting rule: put a colon (:) after if and else, and indent the code inside the blocks. else never has a condition of its own.

Exam: Asked as: "Write a program that inputs a number and prints whether it is even or odd / positive or negative / pass or fail." Write the full program with `int(input())`,...

سلیکشن (Selection) میں پروگرام شرط کی بنیاد پر فیصلہ کرتا ہے۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Relational, Logical and Membership Operators:** Do not confuse = (assignment, "store") with == (comparison, "is it equal?").
- **Bitwise Operators:** Do not mix up the logical and bitwise operators: and/or/not work on True/False conditions, while &/|/~ work on the bits of integers.
- **Control Structures:** Selection and repetition both use conditions, but they are different: selection runs a block at most once; repetition can run a block many...

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Relational, Logical and Membership Operators**. Write: 3.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Relational (comparison) operators compare two values and return a Boolean (True or False). (2) They are used in conditions. — + example.

5 marks: 3-mark structure + table: Op / Description / Example (a = 10, b = 5) / Result + the worked example from this topic.

8–10 marks: 5-mark structure + a complete truth table and a labelled gate / logic diagram + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Bitwise Operators**. Write: 6.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) OR (|) — gives 1 if at least one bit is 1. (2) XOR (^) — gives 1 if the bits are different, 0 if they are the same. — + example.

5 marks: 3-mark structure + table: Bit A / Bit B / A & B / A | B / A ^ B / ~A + the worked example from this topic.

8–10 marks: 5-mark structure + a complete truth table and a labelled gate / logic diagram + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Control Structures**. Write: Control structures decide the order in which statements are executed in a program (the "flow of control").

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Selection — choose a path depending on a condition (if, else). (2) Repetition — repeat steps (loops: for, while). — + example.

5 marks: 3-mark structure + four-point explanation + the worked example from this topic.

8–10 marks: 5-mark structure + working code with correct indentation and a short trace of one example + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. Predict the output:

3 marks. Explain Relational, Logical and Membership Operators with two clear points and one example.

5 marks. Typical: "Evaluate $a > b$ and $b > 3$ for $a = 10$, $b = 5$ " or "Write the truth table of and/or". Write each sub-condition's value first, then combine. Remember: $==$ compares, $=$ assigns.

8 marks. Relational, Logical and Membership Operators: 5-mark answer plus a complete truth table and a labelled gate / logic diagram.

Answers: 1. True and True. (True and False = False; False or True = True. not True = False; $2 == 3$. Relational (comparison) operators compare two values and return a Boolean (True or False).; They are used in conditions. 5. Use the recipe in the blue box. Model opening: 3. 8. Same 5-mark body + a complete truth table and a labelled gate / logic diagram + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Relational, Logical and Membership Operators** using the recipe in this topic. Include a complete truth table and a labelled gate / logic diagram. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 3: Programming Fundamentals

Programming Fundamentals (4/5) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Selection Statements: if-elif-else and Nested if · Repetition Statements: the for Loop and range() · The while Loop, break and continue · Nested Loops and for vs while · Practical Activity: Shopping Bill Calculator

★ **Must lock:** Selection Statements: if-elif-else and Nested if, Repetition Statements: the for Loop and range(), The while Loop, break and continue, Nested Loops and for vs while

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Predict the output:
2. Predict the output (show the binary working):
3. Name the three basic control structures.

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

3.5.2 Selection Statements: if-elif-else and Nested if ★ Golden

3. The if-elif-else statement handles many conditions.

- The if-elif-else statement handles many conditions.
- (elif is short for "else if".) Python checks the conditions from top to bottom.
- As soon as it finds a True condition, it runs that block and ignores all the rest.
- If none is True, the else block runs.

marks	>= 80?	>= 70?	>= 60?	Output
92	True	(not checked)	(not checked)	A-1 Grade
75	False	True	(not checked)	A Grade
60	False	False	True	B Grade
41	False	False	False	Fail or below B

Remember: In an if-elif-else ladder, put the strictest condition first (highest marks first). Think of a ladder: you step off at the first rung that fits.

Exam: Grade-calculator programs (if-elif-else) and "largest of three numbers" are very common long questions. Check conditions in the right order, indent nested blocks...

if-elif-else میں کئی شرطیں اوپر سے نیچے ترتیب سے دیکھی جاتی ہیں، اور جو شرط سب سے پہلے درست ہو اسی کا بلاک چلتا ہے، باقی سب چھوڑ دی جاتی ہیں۔

3.5.3 Repetition Statements: the for Loop and range() ★ Golden

A repetition statement (loop) lets a block of code run again and again.

- This greatly reduces the need to write the same code many times.
- Each single run of the loop body is called an iteration.
- The for loop is used when the exact number of iterations is known.

Syntax	How it works	Example	Values produced
<code>range(stop)</code>	From 0 up to, but not including , stop.	<code>range(3)</code>	0 1 2
<code>range(start, stop)</code>	From start up to, but not including, stop.	<code>range(2, 5)</code>	2 3 4
<code>range(start, stop, step)</code>	From start towards stop, jumping by step.	<code>range(1, 6, 2)</code>	1 3 5

```
for i in range(3):  
    print(i)
```

Remember: range stops one short: range(1, 11) gives 1 to 10. To include a number n, write range(..., n + 1).

Number of values in range(a, b) = b - a.

Exam: Expect "predict the output" questions on range() with a step, and programs such as "print a table", "sum of first n numbers", "even numbers from 1 to 20". Always state...

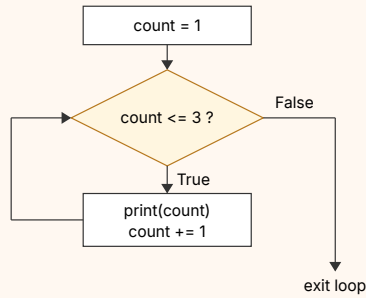
لوپ (Loop) کوڈ کے ایک بلاک کو بار بار چلاتا ہے۔

3.5.3 The while Loop, break and continue ★ Golden

2. The while loop repeats a block of code as long as a condition remains True.

- break — exits (stops) the loop entirely and immediately.
- continue — skips the rest of the current iteration and jumps to the next one.

count	count <= 3?	Printed	count after update
1	True	1	2
2	True	2	3
3	True	3	4
4	False	—	loop ends



A while loop: test the condition, run the body, go back and test again.

```
count = 1
while count ≤ 3:
    print(count)
    count += 1 # VERY IMPORTANT: update the condition!
```

Remember: break = "exit the room"; continue = "skip this turn". For every while loop, check the I-T-U rule: Initialise, Test, Update.

Exam: "Differentiate break and continue with examples" and "What is an infinite loop?" are frequent. Use the two side-by-side programs above with their outputs — examiners...

while لوپ اس وقت تک چلتا رہتا ہے جب تک شرط درست (True) رہے، اور یہ تب استعمال ہوتا ہے جب پہلے سے معلوم نہ ہو کہ کتنی بار دہرانا ہے۔

3.5.3 Nested Loops and for vs while ★ Golden

A nested loop is a loop inside another loop.

- The inner loop runs completely, from start to finish, for every single iteration of the outer loop.
- If the outer loop runs 2 times and the inner loop runs 3 times, the inner body runs $2 \times 3 = 6$ times.

for loop	while loop
Used when the number of iterations is known .	Used when the number of iterations is unknown (condition-based).
Automatically updates the loop variable.	Requires a manual update of the counter (risk of an infinite loop).
Cleaner and shorter for counting tasks.	Better for dynamic situations (like "keep playing until the user says Stop").
<code>for i in range(1, 4): print(i)</code>	<code>i = 1 / while i ≤ 3: / print(i) / i += 1</code>

```
for outer in range(1, 3):      # runs 2 times (1, 2)
    for inner in range(1, 4):  # runs 3 times per outer (1, 2, 3)
        print(f"Outer: {outer}, Inner: {inner}")
```

Remember: Ask yourself: "Do I know how many times?" Yes → for. "Do I only know when to stop?" → while. Nested loops are like a clock: the minute hand (inner loop) goes...

Exam: "Differentiate between for and while loops" (give three points and an example of each) and "How many times will the inner statement run?" (multiply the counts) are...

نیسٹڈ لوپ (Nested Loop) میں ایک لوپ دوسرے لوپ کے اندر ہوتا ہے، اور باہر والے لوپ کے ہر چکر میں اندر والا لوپ شروع سے آخر تک مکمل چلتا ہے۔

Practical Activity: Shopping Bill Calculator

This program combines all three control structures: sequence (steps in order), selection (checking "done", checking the price, checking for a discount) and repetition (a while loop that keeps adding items).

- It uses a flag — a Boolean variable (shopping) that controls when the loop stops.
- `item.lower()` converts the text to lowercase so that "done", "Done" and "DONE" all work.

Step	item	price	total	count
1	Rice	3000.0	3000.0	1
2	Soap	-50.0 (invalid, not added)	3000.0	1
3	Oil	2500.0	5500.0	2
4	DONE → shopping = False	—	5500.0	2

Remember: Label the parts of any program with comments, as here: # REPETITION, # SELECTION. It helps you, your teacher and the examiner see the structure.

یہ شاپنگ بل پروگرام تینوں کنٹرول اسٹرکچرز کو ملا کر بنایا گیا ہے: while لوپ سے اشیاء بار بار شامل ہوتی ہیں، if-else سے غلط قیمت

رو...

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Selection Statements: if-elif-else and Nested if:** Wrong order gives wrong answers.
- **Repetition Statements: the for Loop and range():** Writing `range(1, 10)` to print 1 to 10 misses 10 — an "off-by-one" logical error.
- **The while Loop, break and continue:** Infinite loop — find the error: `count = 1 while count <= 5: print(count) count` is never updated, so `count <= 5` stays True and the program...

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Selection Statements: if-elif-else and Nested if**. Write: 3.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) (elif is short for "else if".) Python checks the conditions from top to bottom. (2) As soon as it finds a True condition, it runs that block and ignores all the rest. — + example.

5 marks: 3-mark structure + table: marks / `>= 80?` / `>= 70?` / `>= 60?` / Output + the worked example from this topic.

8–10 marks: 5-mark structure + a complete truth table and a labelled gate / logic diagram + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Repetition Statements: the for Loop and range()**. Write: A repetition statement (loop) lets a block of code run again and again.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) This greatly reduces the need to write the same code many times. (2) Each single run of the loop body is called an iteration. — + example.

5 marks: 3-mark structure + table: Syntax / How it works / Example / Values produced + the worked example from this topic.

8–10 marks: 5-mark structure + working code with correct indentation and a short trace of one example + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **The while Loop, break and continue**. Write: 2.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) break — exits (stops) the loop entirely and immediately. (2) continue — skips the rest of the current iteration and jumps to the next one. — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + working code with correct indentation and a short trace of one example + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. Predict the output:

3 marks. Explain Selection Statements: if-elif-else and Nested if with two clear points and one example.

5 marks. Grade-calculator programs (if-elif-else) and "largest of three numbers" are very common long questions. Check conditions in the right order, indent nested blocks carefully (8 spaces for the inner...

8 marks. Selection Statements: if-elif-else and Nested if: 5-mark answer plus a complete truth table and a labelled gate / logic diagram.

Answers: 1. Warm ($30 > 35$ is False; $30 > 25$ is True, so the rest are skipped even though $30 > 15$ is also True) 3. (elif is short for "else if".) Python checks the conditions from top to bottom.; As soon as it finds a True condition, it runs that block and ignores all the rest. 5. Use the recipe in the blue box. Model opening: 3. 8. Same 5-mark body + a complete truth table and a labelled gate / logic diagram + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Selection Statements: if-elif-else and Nested if** using the recipe in this topic. Include a complete truth table and a labelled gate / logic diagram. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 3: Programming Fundamentals

Programming Fundamentals (5/5) · Chapter exam

TODAY'S 90 MINUTES

90 minutes. 0–10 min recap yesterday ★ on the board · 10–85 min sit the 75-mark chapter exam (1 h 15 min, hall conditions) · 85–90 min collect scripts. Do not mark in front of the batch.

Board today: Libraries in Python · Built-in Libraries · Third-Party Libraries · Debugging Code in Python

★ **Must lock:** none — still copy every table

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Hall paper today. Recap the cards for ten minutes, then scripts out. Re-teach any Golden below 60% in tomorrow's starter.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Predict the output:
2. Predict the output:
3. Predict the output:

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

3.6 Libraries in Python

A library (also called a module) is a collection of pre-written code that offers ready-made functions.

- Instead of writing a square-root algorithm from scratch, you import math and call `math.sqrt()`.
- Libraries save time, reduce bugs and let you stand on other programmers' work.
- There are five common ways to import a library.

Method	Syntax	How you call the function
1. Entire module	<code>import math</code>	<code>math.sqrt(25)</code>
2. Alias (nickname)	<code>import math as m</code>	<code>m.sqrt(16)</code>
3. Specific function	<code>from math import sqrt</code>	<code>sqrt(36)</code>
4. Several functions	<code>from math import sqrt, pi</code>	<code>sqrt(9), pi</code>
5. Everything (*)	<code>from math import *</code>	<code>sqrt(64)</code> — avoid this

Trap: `from math import *` dumps every name into your program.

لائبریری پہلے سے لکھا ہوا کوڈ ہے جسے `import` سے پروگرام میں لایا جاتا ہے۔

3.6.1 Built-in Libraries

Built-in libraries come pre-installed with Python.

- You only import them; you do not run pip.
- Three that every Class XI student must know are math, random and datetime.

```
import math
print(math.pi)
print(math.ceil(4.3))
print(math.floor(4.9))
print(math.pow(2, 4))
print(math.sqrt(81))
```

Remember: ceil always rounds up (ceiling); floor always rounds down. round() (built-in, not in math) goes to the nearest integer and is a different function.

math حساب کے لیے، random بے ترتیب عدد کے لیے اور datetime تاریخ وقت کے لیے بنی ہوئی لائبریریاں ہیں۔

3.6.2 Third-Party Libraries

Third-party libraries are written by the community.

- They are not bundled with Python, so you install them from a terminal with `pip install library_name` and then import them as usual.

Library	Primary use
NumPy	Heavy numerical computing and large arrays
Pandas	Data analysis; table-like data (CSV / Excel)
Matplotlib	Data visualisation — graphs and charts
TensorFlow / PyTorch	Artificial intelligence and machine learning

```
pip install pandas
import pandas as pd
```

Trap: `pip` is a terminal command, not a Python statement.

تھرڈ پارٹی لائبریریاں پائتھن کے ساتھ نہیں آتیں، انہیں `pip install` سے انسٹال کیا جاتا ہے۔

3.7 Debugging Code in Python

Debugging is the process of identifying, tracing and fixing bugs so a program works correctly.

- Every program you write will have bugs.
- The skill is not "never make a mistake"; it is "find the mistake quickly".
- Python errors fall into three families:

Type	When it appears	What you notice	Example
Syntax error	Before the program even starts	Python refuses to run	<code>print("Hello" — missing)</code>
Runtime error (exception)	While the program is running	It crashes with a traceback	<code>x = 10 / 0 → ZeroDivisionError</code>
Logical error	Program finishes "successfully"	The output is wrong	<code>area = length + width</code> instead of <code>*</code>

```
x = int("abc")
```

Remember: Fix syntax errors first (the program cannot start). Then run it. Then check the output against a hand-worked example to catch logical errors.

Exam: Examiners love "identify the error type" questions. Ask: (1) Does it refuse to start? → syntax. (2) Does it crash mid-run? → runtime. (3) Does it run but give a wrong...

ڈیباگنگ یعنی غلطی ڈھونڈ کر ٹھیک کرنا۔

Programming Fundamentals

24 single-page lessons · 12 Golden · read, close, say aloud, then sit the exam

Chapter in one look

- A program is a set of instructions; programming languages are low-level (close to hardware) or high-level (close to English). Python is a high-level, interpreted, readable language.
- Python is widely used in web development, data science, AI and automation — skills that grow a career in Pakistan and abroad.
- An IDE (such as VS Code plus the Python extension) combines an editor, a terminal, a debugger and syntax highlighting.
- Variables store values; Python types are dynamic. Core types: `int`, `float`, `str`, `bool`, `list`, `dict`.
- `print()` displays output; `input()` always returns a **string** — convert with `int()` or `float()` before arithmetic.
- Operators: arithmetic (`+`, `-`, `*`, `/`, `//`, `%`, `**`), assignment (`=`, `+=`, ...), relational, logical (`and`, `or`, `not`), membership (`in`), bitwise (`&`, `|`, `^`, `~`, `<<`, `>>`).
- Control structures: sequence (top to bottom), selection (`if` / `if-else` / `if-elif-else` / nested), repetition (`for` / `while`).
- `range(stop)` goes 0 ... stop-1; `range(start, stop, step)` jumps by step. `break` leaves a loop; `continue` skips one iteration.
- Use `for` when the count is known; use `while` when you wait for a condition (and always update that condition).
- Libraries extend Python. Built-in: `math`, `random`, `datetime`. Third-party (need `pip`): NumPy, Pandas, Matplotlib, TensorFlow.
- Errors: syntax (grammar, will not run), runtime (crash, traceback), logical (runs, wrong answer).

Key terms

Term	Meaning	اردو
Program	A set of instructions a computer can follow.	پروگرام
High-level language	A language close to English (Python, Java, C++).	اعلیٰ سطحی زبان
Interpreter	Runs a program line by line without a separate compile step.	انٹرپریٹر
IDE	Software that combines editor, terminal and debugger.	انٹیگریٹڈ ڈویلپمنٹ انوائرنمنٹ
Variable	A named box that holds a value.	متغیر
Data type	The kind of value: int, float, str, bool, list, dict.	ڈیٹا کی قسم
Operator	A symbol that performs an operation on operands.	عامل
Operand	A value an operator works on.	معمول
Sequence	Instructions executed from top to bottom.	تسلسل
Selection	Choosing a path with <code>if</code> / <code>else</code> .	انتخاب
Repetition / loop	Repeating a block with <code>for</code> or <code>while</code> .	تکرار / لوپ
Nested	A structure placed inside another of the same kind.	تو در تو

Programming Fundamentals

75 marks · 1 hour 15 minutes · no notes · no phones

CHAPTER EXAM RULES

Instructions. Sit this exam after Lesson 24 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

Marks. $15 + 30 + 30 = 75$. A 5-mark / 10-mark answer needs Define · Explain · Example · Diagram · Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

Section A — Multiple choice (15 × 1 = 15)

1. Python is best described as a:
 - a. low-level compiled language
 - b. high-level interpreted language
 - c. machine language
 - d. hardware description language
2. `input("Age: ")` always returns a:
 - a. int
 - b. float
 - c. str
 - d. bool
3. The value of `17 // 5` is:
 - a. 3.4
 - b. 2
 - c. 3
 - d. 3.0
4. The value of `17 % 5` is:
 - a. 2
 - b. 3
 - c. 3.4
 - d. 0
5. `2 ** 3` equals:
 - a. 6
 - b. 8
 - c. 9
 - d. 5
6. Which operator is bitwise AND?
 - a. `and`
 - b. `&&`
 - c. `&`
 - d. `^`
7. `list(range(2, 5))` is:
 - a. `[2, 3, 4, 5]`
 - b. `[2, 3, 4]`
 - c. `[2, 5]`
 - d. `[0, 1, 2, 3, 4]`
8. A `while` loop is preferred when:
 - a. you know the exact number of repeats
 - b. you must wait for a condition
 - c. you never need a counter
 - d. the body must run exactly once
9. `break` inside a loop:
 - a. skips one iteration
 - b. restarts the loop
 - c. leaves the loop immediately
 - d. pauses for input
10. `from math import sqrt` lets you write:
 - a. `math.sqrt(9)` only
 - b. `sqrt(9)`
 - c. `import.sqrt(9)`
 - d. `m.sqrt(9)`
11. Which library is built-in?
 - a. Pandas
 - b. NumPy
 - c. random
 - d. Matplotlib
12. `print("Hi")` is a:
 - a. syntax error
 - b. runtime error
 - c. logical error
 - d. warning only
13. `10 / 0` is a:
 - a. syntax error
 - b. runtime error
 - c. logical error
 - d. style error
14. `area = length + width` when you meant to multiply is a:
 - a. syntax error
 - b. runtime error
 - c. logical error
 - d. import error
15. `math.ceil(4.3)` is:
 - a. 4
 - b. 4.0
 - c. 5
 - d. 4.3

Section B — Short questions (10 × 3 = 30)

1. Differentiate between a compiler and an interpreter.
2. Why must the result of `input()` often be converted with `int()` ?
3. State the difference between `/` and `//` with one example each.
4. Write the truth table of the logical operator `and` for two Boolean values.
5. Explain `if-elif-else` . Why is the order of conditions important?
6. Differentiate between a `for` loop and a `while` loop.
7. What do `break` and `continue` do?
8. Name two built-in and two third-party Python libraries and their uses.
9. Differentiate between syntax, runtime and logical errors with one example each.
10. What three pieces of information does a Python traceback give you?

Section C — Long questions (attempt any 3 × 10 = 30)

1. Write a complete Python program that reads a student's marks (0–100) and prints the grade: 80+ A-1, 70–79 A, 60–69 B, otherwise Fail. Include input conversion and a sample run.
2. Write a program that prints all even numbers from 0 to 10 using `range` , then rewrite it with a `while` loop. Dry-run both.
3. Trace this nested loop and write every line of output:

```
for o in range(1, 3):
    for i in range(1, 4):
        print(o, i)
```
4. A shop gives 20% discount if the bill is 5000 or more. Write a program that keeps reading item prices until the user types 0, prints the total, the discount and the amount to pay. Show a sample run with prices 3000, 2500, 0.
5. Classify each bug and write the corrected line:
 - (a) `print("Total is", total`
 - (b) `avg = total / 0` when there are no items
 - (c) `discount = total + 0.20` when 20% off was intended.
6. Explain how you would use `math` , `random` and `datetime` in a small "lucky student of the day" program for a class of 40. Write the code.

Answer key. Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

Check yourself — answers

- 3.1 Programming.** Answers: 1. A set of instructions given to a computer to perform a task. 2. Because they cannot think or do anything without instructions; they only follow exactly what they are told.
- 3.1.1 Programming Language Types.** Answers: 1. Mid level (assembly language). 2. Java. 3. Any one: system software, operating systems, high-performance games.
- 3.2 Career Growth with Python.** Answers: 1. Django and Flask. 2. Artificial Intelligence and Machine Learning.
- 3.3 Integrated Development Environment (IDE).** Answers: 1. Many programming tools (editor, runner, debugger) are combined in one program. 2. Auto-completion.
- 3.3.1 Visual Studio Code (VS Code).** Answers: 1. Source Control. 2. No; it becomes an IDE by installing extensions (e.g. the Python extension). 3. Explorer.
- 3.4 Fundamentals of Python Programming.** Answers: 1. B then D on separate lines (line 1 is a comment; $3 > 7$ is false so C is skipped). 2. 4 spaces (or 1 tab).
- 3.4.1 Variables.** Answers: 1. `_total` valid; `roll-no` invalid (hyphen); `class` invalid (keyword); `marks2026` valid; `2marks` invalid (starts with a digit). 2. `Karachi` — `city` and `City` are different variables.
- 3.4.1 Data Types in Python.** Answers: 1. float, str, int, bool, list. 2. 2 (the set keeps only 5 and 6). 3. str, because it contains hyphens and you never do arithmetic on it.
- 3.4.2 Input and Output Handling.** Answers: 1. 3333 then 12 (a string times a number repeats the string). 2. `input()` returns the string "21", and a string times 2 repeats it. Fix: `num = int(input("Enter a number: "))`.
- 3.4.3 Operators and Operands: Arithmetic and Assignment.** Answers: 1. 3 1 3.5 then 49. 2. 3 ($10 + 5 = 15$; $15 // 4 = 3$).
- 3.4.3 Relational, Logical and Membership Operators.** Answers: 1. True and True. (True and False = False; False or True = True. not True = False; $2 == 2.0$ is True; False or True = True.) 2. = is assignment; a comparison needs ==. Python reports `SyntaxError: invalid syntax`. Maybe you meant '=' or ':' instead of '='? Fix: `if x == 5:`
- 3.4.3 Bitwise Operators.** Answers: 1. 8 14 6 and -6 3 24. $12 = 1100$, $10 = 1010$: AND = $1000 = 8$, OR = $1110 = 14$, XOR = $0110 = 6$. $\sim 5 = -(5 + 1) = -6$. $7 = 111 \rightarrow 11 = 3$. $3 = 11 \rightarrow 11000 = 24$ (3×8). 2. `0b1101` ($8 + 4 + 1$).
- 3.5 Control Structures.** Answers: 1. Sequence, selection, repetition. 2. Repetition (a while loop).
- 3.5.1 Sequence.** Answers: 1. 10 8 — y was calculated when x was still 4; changing x later does not change y. 2. Sequence.
- 3.5.2 Selection Statements: if and if-else.** Answers: 1. Fail then Done. 2. Missing colon after `if n > 0`; else must not have a condition — write `else:`.
- 3.5.2 Selection Statements: if-elif-else and Nested if.** Answers: 1. Warm ($30 > 35$ is False; $30 > 25$ is True, so the rest are skipped even though $30 > 15$ is also True). 2. B (outer check $15 > 10$ passes; inner check $15 > 20$ fails).
- 3.5.3 Repetition Statements: the for Loop and range().** Answers: 1. 10 7 4 1. 2. 30 ($2 + 4 + 6 + 8 + 10$). 3. The body is not indented: `IndentationError`. Indent `print(i)` by 4 spaces.
- 3.5.3 The while Loop, break and continue.** Answers: 1. 27 ($n: 1 \rightarrow 3 \rightarrow 9 \rightarrow 27$; $27 < 20$ is False, so the loop stops). 2. 1 2 4 5 (3 and 6 are skipped by `continue`; at 7 the loop breaks).
- 3.5.3 Nested Loops and for vs while.** Answers: 1. 6 (3×2). 2. A while loop, because the number of inputs is not known in advance.
- Practical Activity: Shopping Bill Calculator.** Answers: 1. Final to Pay: 4000.0 (4000 is less than 5000, so no discount). 2. It is a flag that keeps the while loop running; setting it to False stops the loop.
- 3.6 Libraries in Python.** Answers: 1. `from math import ceil`. 2. The library name is long; a short nickname keeps later lines short and is a community convention.
- 3.6.1 Built-in Libraries.** Answers: 1. 4 (already an integer). 2. `print(random.randint(1, 6))` after `import random`.
- 3.6.2 Third-Party Libraries.** Answers: 1. Matplotlib. 2. No — math is built-in.
- 3.7 Debugging Code in Python.** Answers: 1. Syntax error. 2. Logical error. 3. `ValueError` (runtime).

Answer key

Multiple choice

1. b
2. c
3. c
4. a
5. b
6. c
7. b
8. b
9. c
10. b
11. c
12. a
13. b
14. c
15. c

Short questions — model answers

1. A compiler translates the whole program into machine code before it runs (C, C++). An interpreter translates and runs line by line (Python). Compiled programs often run faster; interpreted ones are easier to test.
2. `input()` always returns a string. Adding strings concatenates them (`"2"+"3"` is `"23"`), so numeric work needs `int()` or `float()`.
3. `/` is true division: `17/5` is `3.4`. `//` is integer (floor) division: `17//5` is `3`.
4. `F` and `F` $\rightarrow$ `F`; `F` and `T` $\rightarrow$ `F`; `T` and `F` $\rightarrow$ `F`; `T` and `T` $\rightarrow$ `T`. Output is `True` only when both operands are `True`.
5. Python tests conditions from the top. The first `True` block runs and the rest are skipped. If you put `marks >= 60` above `marks >= 80`, an 85 would match the weaker condition first and get the wrong grade.
6. `for`: known count, counter updates automatically, good with `range`. `while`: unknown count, you must update the condition yourself or you get an infinite loop.
7. `break` leaves the loop immediately. `continue` skips the rest of this iteration and starts the next one.
8. Built-in: `math` (roots, ceil/floor), `random` (random integers). Third-party: Pandas (tables), Matplotlib (charts). (NumPy and TensorFlow also accepted.)
9. Syntax: missing bracket, will not run. Runtime: `10/0`, crashes. Logical: `area = l + w`, runs but answer is wrong.
10. The error type (e.g. `ValueError`), the file name, and the line number that caused the crash.

Long questions — model answers

```
1. marks = int(input("Enter marks: "))
if marks ≥ 80:
    print("A-1 Grade")
elif marks ≥ 70:
    print("A Grade")
elif marks ≥ 60:
    print("B Grade")
else:
    print("Fail or below B")
```

Sample: input 85 → A-1 Grade; input 72 → A Grade; input 40 → Fail or below B.

```
2. for i in range(0, 11, 2):
    print(i)
n = 0
while n ≤ 10:
    print(n)
    n += 2
```

Both print 0, 2, 4, 6, 8, 10.

3. Output (six lines): 1 1 / 1 2 / 1 3 / 2 1 / 2 2 / 2 3. The inner loop runs fully for each value of the outer loop.

```
4. total = 0.0
while True:
    p = float(input("Price (0 to stop): "))
    if p == 0:
        break
    if p > 0:
        total += p
    discount = total * 0.20 if total ≥ 5000 else 0
    print("Total", total)
    print("Discount", discount)
    print("Pay", total - discount)
```

Sample: 3000 + 2500 = 5500; discount 1100; pay 4400.

5. (a) Syntax — add `)`: `print("Total is", total)`. (b) Runtime `ZeroDivisionError` — only divide when the count of items is not 0. (c) Logical — 20% of total is `total * 0.20`, not `total + 0.20`.

```
6. import random, datetime, math
n = random.randint(1, 40)
today = datetime.datetime.now().strftime("%d-%m-%Y")
print("Lucky roll", n, "on", today)
print("Root of roll", math.sqrt(n))
```

`random` picks the student number, `datetime` stamps the date, `math` shows a sample extra calculation.

Self-assessment: I can...

- | | |
|--|--|
| ☐ classify a language as high-level or low-level and explain why Python is popular. | ☐ declare variables and identify int, float, str, bool, list and dict values. |
| ☐ read input, convert types, and print formatted output. | ☐ evaluate arithmetic, relational, logical and bitwise expressions by hand. |
| ☐ write if, if-else, if-elif-else and nested selections. | ☐ write for and while loops, including range, break and continue. |
| ☐ import built-in libraries (math, random, datetime) in more than one style. | ☐ name pip and at least two third-party libraries. |
| ☐ classify a bug as syntax, runtime or logical and read a traceback. | ☐ combine sequence, selection and repetition in a small complete program. |

Chapter 4: Data and Analysis

Data and Analysis (1/3) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Data, Information and Database · Database Management System (DBMS) · Database Components (Tables, Records, Fields) · Keys and Integrity Constraints · Relational Database Management System (RDBMS)

★ **Must lock:** Keys and Integrity Constraints

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Write the line that lets you call ceil without writing math. in front.
2. What does math.ceil(4.0) return?
3. Which library would you choose to draw a bar chart of class marks?

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

4.1.1 Data, Information and Database

Data are unprocessed facts, figures or observations.

Data → Processing → Information → Decision

- They can be quantitative (numbers: 80, 75, 90) or qualitative (descriptions: the colours of cars in a parking lot).
- Data on its own is not very useful — a list of marks does not yet tell you whether the class is strong.
- When you organise, process and interpret data, it becomes information.

Aspect	Data	Information
Definition	Raw facts and figures	Processed, meaningful data
Example	85, 90, 75	Average marks = 83.3%
Usefulness	Little, on its own	Useful for decisions

Remember: Exam line: Data is raw; information is processed data used for decision-making. Always add one pair of examples.

ڈیٹا خام حقائق ہیں، معلومات وہی حقائق ہیں جب انہیں ترتیب دے کر معنی دے دیا جائے۔

4.1.2 Database Management System (DBMS)

A DBMS is software that lets you create, manage and interact with databases.

- Less redundancy. The department name is stored once, not in every student row.
- Consistency. Change a phone number once; every screen that uses it shows the new number.
- Security. Only authorised users see sensitive fields (an admin sees marks; office staff see names).
- Integrity. Rules stop duplicate roll numbers and empty primary keys.

ڈی بی ایم ایس وہ سافٹ ویئر ہے جو صارف اور ڈیٹا بیس کے درمیان پل کا کام کرتا ہے۔

4.2 Database Components (Tables, Records, Fields)

The basic structure is a table (also called a relation): rows and columns, like a spreadsheet, holding facts about one kind of thing (Students, Teachers, Books).

- A record (row, tuple) is one complete instance — everything about Ali.
- A field (column, attribute) is one kind of fact — the whole Name column, or the whole Grade column.

Roll No	Name	Course	Grade
101	Ali	CS	A
102	Sara	IT	B+
103	Ahmed	CS	A

Remember: Memory: row = record = one person. column = field = one question on the admission form.

ٹیبل قطاروں اور کالموں میں ڈیٹا رکھتا ہے۔

4.3.1 Keys and Integrity Constraints ★ Golden

A key is a field (or a set of fields) that uniquely identifies records and links tables.

- Integrity constraints are the rules that keep those keys honest.
- Any field (or combination) that could uniquely identify a row.
- A table may have several.

Table	Field	Role
Student	RollNo	Primary key
Student	BFormNo	Alternate key
Result	RollNo	Foreign key → Student.RollNo
Result	ExamID + RollNo	Could be a composite primary key

Trap: A phone number is a bad PK: people change numbers, two siblings may share one, and some students have none (NULL).

Exam: Board questions: "Differentiate primary and foreign key" (table of 4 points) and "Why can a PK not be NULL?" Because a NULL would mean "this row has no identity", so the...

کینڈیڈیٹ کی وہ فیلڈ ہے جو ریکارڈ کی شناخت کر سکے۔

4.3 Relational Database Management System (RDBMS)

An RDBMS stores data in several tables and uses primary and foreign keys to keep relationships.

- Reduces redundancy by splitting data into related tables.
- Supports many users at once, with permissions.
- Is queried with SQL (Structured Query Language) — a standard language of SELECT, INSERT, UPDATE, DELETE.

آر ڈی بی ایم ایس کئی ٹیبلز میں ڈیٹا رکھتا ہے اور کیڑ سے ان کے تعلقات برقرار رکھتا ہے۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Data, Information and Database:** Do not call a jumble of WhatsApp numbers a database.
- **Database Components (Tables, Records, Fields):** Do not mix two entities in one table (students and library books together).
- **Keys and Integrity Constraints:** A phone number is a bad PK: people change numbers, two siblings may share one, and some students have none (NULL).

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Keys and Integrity Constraints**. Write: A key is a field (or a set of fields) that uniquely identifies records and links tables.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Integrity constraints are the rules that keep those keys honest. (2) Any field (or combination) that could uniquely identify a row. — + example.

5 marks: 3-mark structure + table: Table / Field / Role + the worked example from this topic.

8–10 marks: 5-mark structure + working code with correct indentation and a short trace of one example + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. Can a table have two primary keys?

3 marks. Explain Keys and Integrity Constraints with two clear points and one example.

5 marks. Board questions: “Differentiate primary and foreign key” (table of 4 points) and “Why can a PK not be NULL?” Because a NULL would mean “this row has no identity”, so the DBMS could not tell rows...

8 marks. Keys and Integrity Constraints: 5-mark answer plus working code with correct indentation and a short trace of one example.

Answers: 1. No — only one PK (it may be composite) 3. Integrity constraints are the rules that keep those keys honest.; Any field (or combination) that could uniquely identify a row. 5. Use the recipe in the blue box. Model opening: A key is a field (or a set of fields) that uniquely identifies records and links tables. 8. Same 5-mark body + working code with correct indentation and a short trace of one example + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Keys and Integrity Constraints** using the recipe in this topic. Include working code with correct indentation and a short trace of one example. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 4: Data and Analysis

Data and Analysis (2/3) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Entity Relationship Model (ER-Model) · Referential Integrity · Relational Schema Development · Case Study — Library Management System · Database Objects (Tables, Forms, Queries, Reports)

★ **Must lock:** Entity Relationship Model (ER-Model), Referential Integrity

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Can a table have two primary keys?
2. Is "Nadia, Class 11, 72 marks" data or information?
3. Name two DBMS programs.

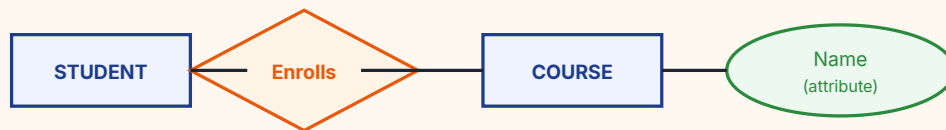
کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

4.4 Entity Relationship Model (ER-Model)

★ Golden

The ER model is the architect's blueprint: you draw what will be stored and how it connects before you create tables in Access.

Component	Meaning	Symbol	Example
Entity	A real-world object we store data about	Rectangle	STUDENT, TEACHER, COURSE
Attribute	A property of an entity. PK is underlined	Oval	StudentID, Name, Class
Relationship	How two entities are connected	Diamond	Enrolls, Teaches, Has



Rectangle = entity, diamond = relationship, oval = attribute. Underline the primary-key attribute.

Trap: The M:N rule.

Exam: Always state cardinality on the diagram (1 and M at the ends of the line). For M:N, the long question is "how do you resolve it?" — draw the junction entity with two...

ای آر ماڈل ڈیٹا بیس کا خاکہ ہے۔

4.5 Referential Integrity ★ Golden

Referential integrity means every foreign-key value in a child table must match an existing primary-key value in the parent — or be NULL if the design allows it.

- If Result.StudentID is a FK to Student.StudentID, you cannot type Result for student 999 if 999 is not in Student.
- You also cannot delete student 101 while results for 101 still exist — unless you first handle those child rows.

ID	Name
101	Ali
102	Sara

Trap: Cascade Delete is powerful and dangerous.

Exam: Write: definition, why it matters (no orphans), then one numbered example of Cascade Update and one of Cascade Delete. Mention parent vs child.

ریفرنشل انٹیگریٹی کا مطلب ہے بچہ ٹیبل کی فارن کی ہمیشہ والدین کی موجود پرائمری کی طرف اشارہ کرے۔

4.6 Relational Schema Development

A relational schema is the formal written design of the database: table names, fields, primary keys, foreign keys and how tables relate.

- You write it before clicking around in Access.
- It defines (1) how data is split into tables, (2) which columns each table has, (3) which fields are PK/FK, (4) the relationships.
- Benefits: less redundancy, integrity planned in advance, easier to grow later.

```
Department (DeptID PK, DeptName)
Student    (RollNo PK, Name, Class, DeptID FK → Department)
Course     (CourseID PK, Title, Credit, DeptID FK → Department)
```

Remember: Skipping the schema is like building a house without a drawing. You get messy, broken structures that are painful to repair.

ریلیشنل اسکیم ڈیٹا بیس کا لکھا ہوا نقشہ ہے: کون سے ٹیبل، کون سے فیلڈ، کون سی کیز، کون سے تعلقات۔

4.7 Case Study — Library Management System

A college library needs a database. Walk the seven steps examiners expect.

- A college library needs a database.
- Walk the seven steps examiners expect.
- A book has ID, title, author, publisher.
- A member (student/teacher) has ID, name, contact.

BorrowID	MemberID	BookID	IssueDate	ReturnDate
5001	102	B-17	2026-10-05	



Final library ER: two 1:M relationships through the BorrowRecord junction table.

```
Member      (MemberID PK, Name, ContactInfo)
Book        (BookID PK, Title, Author, Publisher)
BorrowRecord (BorrowID PK, MemberID FK, BookID FK, IssueDate, ReturnDate)
```

لائبریری میں ممبر اور کتاب کا تعلق بہت-سے-بہت-سے ہے، اس لیے بورو ریکارڈ نامی پل ٹیبل بناتے ہیں جس میں دونوں کی فارن کیز اور اجراء...

4.8 Database Objects (Tables, Forms, Queries, Reports)

Four objects you will meet in Access. Learn the one-line jobs cold:

- Learn the one-line jobs cold:

Object	Job	You ...
Table	Stores the data (foundation; every other object depends on it)	keep the actual records here
Form	Friendly screen for enter / edit / view (text boxes, drop-downs)	type data in; fewer mistakes than a raw grid
Query	Ask a question: filter, sort, calculate	e.g. "students above 80%"
Report	Formatted output for printing or sharing	read only — never type into a report

Remember: Store → Enter → Search → Print. Tables, Forms, Queries, Reports.

ٹیبل ڈیٹا رکھتی ہے، فارم سے اندراج ہوتا ہے، کوئری سوال پوچھ کر چھانتی ہے، رپورٹ چھپائی کے لیے سنواری ہوئی پیشکش ہے — اس میں ڈیٹا بھرا نہیں جاتا۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Entity Relationship Model (ER-Model):** The M:N rule.
- **Referential Integrity:** Cascade Delete is powerful and dangerous.
- **Database Objects (Tables, Forms, Queries, Reports):** A query does not copy the table into a second storage place.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Entity Relationship Model (ER-Model)**. Write: The ER model is the architect's blueprint: you draw what will be stored and how it connects before you create tables in Access.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) The ER model is the architect's blueprint: you draw what will be stored and how it connects before you create tables in Access. (2) Second distinct point (not a repeat of the definition) — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + a labelled ER diagram (rectangle / oval / diamond) or the relational schema + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Referential Integrity**. Write: Referential integrity means every foreign-key value in a child table must match an existing primary-key value in the parent — or be NULL if the design allows it.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) If Result.StudentID is a FK to Student.StudentID, you cannot type Result for student 999 if 999 is not in Student. (2) You also cannot delete student 101 while results for 101 still exist — unless you first handle those child rows. — + example.

5 marks: 3-mark structure + table: ID / Name + the worked example from this topic.

8–10 marks: 5-mark structure + a labelled ER diagram (rectangle / oval / diamond) or the relational schema + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. What symbol is used for a relationship?

3 marks. Explain Entity Relationship Model (ER-Model) with two clear points and one example.

5 marks. Always state cardinality on the diagram (1 and M at the ends of the line). For M:N, the long question is “how do you resolve it?” — draw the junction entity with two foreign keys.

8 marks. Entity Relationship Model (ER-Model): 5-mark answer plus a labelled ER diagram (rectangle / oval / diamond) or the relational schema.

Answers: 1. Diamond 3. The ER model is the architect's blueprint: you draw what will be stored and how it connects before you create tables in Access. 5. Use the recipe in the blue box. Model opening: The ER model is the architect's blueprint: you draw what will be stored and how it connects before you create tables in Access. 8. Same 5-mark body + a labelled ER diagram (rectangle / oval / diamond) or the relational schema + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Entity Relationship Model (ER-Model)** using the recipe in this topic. Include a labelled ER diagram (rectangle / oval / diamond) or the relational schema. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 4: Data and Analysis

Data and Analysis (3/3) · Chapter exam

TODAY'S 90 MINUTES

90 minutes. 0–10 min recap yesterday ★ on the board · 10–85 min sit the 75-mark chapter exam (1 h 15 min, hall conditions) · 85–90 min collect scripts. Do not mark in front of the batch.

Board today: Creation of Tables in Microsoft Access · Designing Forms for Data Manipulation · Creating Queries in MS Access · Data Summarization and Statistical Analysis · Data Visualization in MS Access (Charts)

★ **Must lock:** Creating Queries in MS Access

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Hall paper today. Recap the cards for ten minutes, then scripts out. Re-teach any Golden below 60% in tomorrow's starter.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. What symbol is used for a relationship?
2. What is an orphan record?
3. Name two things a schema must show.

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

4.9 Creation of Tables in Microsoft Access

Plan first (4.9.1): purpose, field list, data type of each field, which field is the PK.

- Create tab → Table.
- A blank grid appears.
- Double-click Field1, Field2, ... to rename.
- Type data; Access guesses the data type.

Data type	Use it for
Short Text	Names, addresses, phone numbers (you do not add phone numbers). Up to 255 characters.
Number	Values you calculate with: Age, Marks, Quantity.
Date/Time	Date of birth, admission date, appointment time.
Currency	Fees, salary.
AutoNumber	Access fills a unique sequential number — excellent PK.

Trap: Phone numbers look numeric but are Short Text.

پہلے ٹیبل کا منصوبہ بنائیں۔

4.10 Designing Forms for Data Manipulation

A form is the screen people actually type into.

- Create tab → Form Wizard.
- Select the table (e.g. Student).
- Move fields with > (one) or >> (all).
- Choose layout. Columnar = one record at a time, best for data entry.

فارم اندراج کی سادہ اسکرین ہے۔

4.11 Creating Queries in MS Access ★ Golden

A query retrieves, filters and analyses data without changing the original tables.

- Create tab → Query Design.
- Show Table → add Student → Close.
- Double-click the fields you want; they drop into the grid.
- Red Run (!) button executes it.

Goal	Where to type	Example
Single criterion	Criteria row, one column	Department → "Computer Science"
AND (both must match)	Same Criteria row, two columns	Name "Ali" and Department "Computer Science"
OR (either may match)	Criteria row and the or: row	Department "IT" or "Computer Science"

Trap: Text criteria need quotes: "IT".

Exam: Remember: AND = same row in the grid; OR = different rows. Examiners draw a blank grid and ask you to fill Criteria and or:.

کوئری ٹیبل کو بدلے بغیر سوال پوچھتی ہے۔

4.11.4 Data Summarization and Statistical Analysis

Grouping combines rows that share a value (all CS students together) and then an aggregate calculates a number for the group.

- In Access click Totals (Σ) to reveal the Total row in the grid.

Function	What it calculates
COUNT()	How many records (students in IT)
SUM()	Total of a number field (fees collected)
AVG()	Mean
MAX() / MIN()	Highest / lowest in the group

```
SELECT Department, AVG(Marks) AS AvgOfMarks
FROM Student
GROUP BY Department;
```

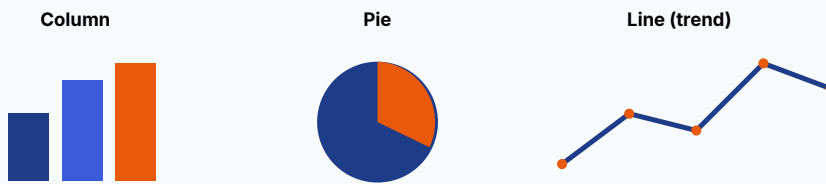
گروپ بائی ایک جیسے ریکارڈ جوڑتا ہے، پھر COUNT, SUM, AVG, MAX, MIN سے خلاصہ نکلتا ہے۔

4.11.6 Data Visualization in MS Access (Charts)

Charts turn numbers into a picture so a pattern jumps out.

- Pick the chart that matches the question:

Chart	Shape	Best for
Bar (horizontal)	Bars sideways	Categories with long names (English, Mathematics, ...)
Column (vertical)	Bars standing	Side-by-side comparison (2024 / 2025 / 2026 admissions)
Pie	Slices of a circle	Parts of one whole (percent of CS vs IT vs Eng). Few slices.
Line	Dots joined	A trend over time (visitors per month)



Column = compare categories. Pie = share of a total. Line = change over time.

Exam: "Which chart?" — line for time, pie for composition, bar/column for comparing categories. Mention one reason.

بار لمبے ناموں والی اقسام، کالم سادہ موازنہ، پائی حصوں کا تناسب، لائن وقت کے ساتھ رجحان۔

Data and Analysis

15 single-page lessons · 4 Golden · read, close, say aloud, then sit the exam

Chapter in one look

- Data = raw facts; information = processed data used for decisions; a database stores related data in a structured, updatable way.
- A DBMS is the software layer that provides less redundancy, consistency, security and integrity.
- Table = rows (records) and columns (fields). One table, one entity.
- Candidate keys could identify a row; the primary key is the one you choose (unique, not NULL); leftover candidates are alternate keys; a foreign key points at another table's PK.
- An RDBMS keeps several related tables and is queried with SQL.
- ER symbols: rectangle entity, oval attribute, diamond relationship. Cardinality 1:1, 1:M, M:N. M:N needs a junction table.
- Referential integrity forbids orphan FKs. Cascade Update copies a changed PK into children; Cascade Delete removes children with the parent.
- Write a relational schema before building. Library case: Member and Book are M:N, resolved by BorrowRecord.
- Objects: tables store, forms enter, queries search, reports print.
- Access Design View sets field names, data types and the PK. Phone numbers are Short Text; money is Currency; auto IDs are AutoNumber.
- Forms need a table/query source; Wizard + columnar layout for data entry.
- Query criteria: one value; AND on the same row; OR on two rows. Totals Σ for Group By / Count / Sum / Avg / Max / Min.
- Bar/column compare categories, pie shows parts of a whole, line shows a trend over time.

Key terms

Term	Meaning	اردو
Data	Raw facts and figures.	ڈیٹا / خام حقائق
Information	Processed data that supports a decision.	معلومات
Database	Organised collection of related data.	ڈیٹا بیس
DBMS	Software that creates and manages databases.	ڈیٹا بیس مینجمنٹ سسٹم
Table	Rows and columns about one entity.	ٹیبل
Record	One row — one instance.	ریکارڈ
Field	One column — one attribute.	فیلڈ
Primary key	Chosen unique, non-empty identifier.	پرائمری کی
Foreign key	Field that refers to another table's PK.	فارن کی
Candidate key	Any field(s) that could be the PK.	کینڈیڈیٹ کی
Alternate key	A candidate not chosen as PK.	آلٹرنیٹ کی
RDBMS	DBMS that stores related tables linked by keys.	ریلیشنل ڈی بی ایم ایس

Data and Analysis

75 marks · 1 hour 15 minutes · no notes · no phones

CHAPTER EXAM RULES

Instructions. Sit this exam after Lesson 15 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

Marks. $15 + 30 + 30 = 75$. A 5-mark / 10-mark answer needs Define · Explain · Example · Diagram · Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

Section A — Multiple choice (15 × 1 = 15)

1. Processed data used for a decision is called:
 - a. a field
 - b. information
 - c. a query
 - d. redundancy
2. A complete row about one student is a:
 - a. field
 - b. report
 - c. record
 - d. schema
3. A primary key cannot contain:
 - a. numbers
 - b. text
 - c. NULL
 - d. an AutoNumber
4. EmpLicense# after EmpID is chosen as PK is an:
 - a. foreign key
 - b. alternate key
 - c. orphan
 - d. report
5. A foreign key:
 - a. must be unique in the child table
 - b. may contain duplicates
 - c. cannot exist in an RDBMS
 - d. is always AutoNumber
6. The ER symbol for a relationship is a:
 - a. rectangle
 - b. oval
 - c. diamond
 - d. circle
7. The most common cardinality is:
 - a. 1:1
 - b. 1:M
 - c. M:N stored as-is
 - d. 0:0
8. M:N is stored in an RDBMS by:
 - a. duplicating both tables
 - b. a junction table
 - c. deleting one entity
 - d. using a pie chart
9. A child row whose StudentID is not in Student is:
 - a. a candidate key
 - b. an orphan record
 - c. a form
 - d. a primary key
10. Changing parent ID 101 → 201 and seeing the child follow is:
 - a. Cascade Delete
 - b. Cascade Update
 - c. a SELECT query
 - d. a pie chart
11. Which Access object is for printing a report card?
 - a. Table
 - b. Form
 - c. Query
 - d. Report
12. Phone numbers should be stored as:
 - a. Number
 - b. Currency
 - c. Short Text
 - d. Date/Time
13. AND criteria go:
 - a. on two different rows of the grid
 - b. on the same Criteria row
 - c. in a report header
 - d. only in SQL Server
14. To average marks per department you set Department to:
 - a. Sum
 - b. Where
 - c. Group By
 - d. Expression
15. The best chart for admissions in 2024, 2025, 2026 is a:
 - a. pie chart
 - b. column chart
 - c. junction table
 - d. foreign key

Section B — Short questions (10 × 3 = 30)

1. Differentiate data and information with one example each.
2. List four advantages of a DBMS.
3. Define record and field with reference to a Student table.
4. Differentiate primary key and foreign key (four points).
5. Why can a relational database not store M:N directly?
6. What is referential integrity? Give one rejected-insert example.
7. Differentiate Cascade Update and Cascade Delete.
8. State the four Access objects and the job of each in one line.
9. Why are phone numbers Short Text, not Number?
10. How do you express OR in Query Design?

Section C — Long questions (attempt any 3 × 10 = 30)

1. Draw a complete ER diagram for Student–Course enrolment (M:N). Resolve it with a junction entity. Write the relational schema with keys marked.
2. A Result table uses StudentID as a foreign key. Explain referential integrity, then show before/after tables for (a) Cascade Update of ID 101 → 201 and (b) Cascade Delete of student 102.
3. Work through the library case study: requirements, entities, attributes, relationship, cardinality, junction table, final schema. Include two sample BorrowRecord rows.
4. Write step-by-step how to create a Student table in Access Design View (fields RollNo Number PK, Name Short Text, Marks Number, Fee Currency, AdmDate Date/Time) and then a columnar form with the Wizard.
5. Write the Query Design settings and the SQL to list CS students with Marks ≥ 80, sorted by Name. Then write a totals query that shows COUNT of students and AVG(Marks) for each Department.
6. A principal wants to see (i) fee collected per class, (ii) the share of each class in total enrolment, (iii) enrolment over five years. Which chart for each, and why?

Answer key. Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

Check yourself — answers

1. **4.1.1 Data, Information and Database.** Answers: 1. Data (raw facts about one student). 2. It is not organised in a structured, searchable, updatable collection of related records.
2. **4.1.2 Database Management System (DBMS).** Answers: 1. MS Access, MySQL (Oracle, SQLite also). 2. Integrity (unique / primary-key constraint).
3. **4.2 Database Components (Tables, Records, Fields).** Answers: 1. 3 records, 4 fields. 2. A value stored in the Course field.
4. **4.3.1 Keys and Integrity Constraints.** Answers: 1. No — only one PK (it may be composite). 2. Yes — many child rows may point to the same parent.
5. **4.3 Relational Database Management System (RDBMS).** Answers: 1. SQL. 2. To store the department name once (less redundancy) and update it in one place.
6. **4.4 Entity Relationship Model (ER-Model).** Answers: 1. Diamond. 2. 1:M (Patient 1 — has — M Appointment).
7. **4.5 Referential Integrity.** Answers: 1. A child row whose FK value does not exist in the parent. 2. They are deleted automatically (if Cascade Delete is on).
8. **4.6 Relational Schema Development.** Answers: 1. Tables/fields and keys/relationships. 2. It reduces redundancy and mistakes (any valid benefit).
9. **4.7 Case Study — Library Management System.** Answers: 1. To break M:N into two 1:M links. 2. On BorrowRecord — they belong to a *transaction*, not to the book itself (a book is issued many times).
10. **4.8 Database Objects (Tables, Forms, Queries, Reports).** Answers: 1. Report. 2. Table.
11. **4.9 Creation of Tables in Microsoft Access.** Answers: 1. AutoNumber. 2. Currency.
12. **4.10 Designing Forms for Data Manipulation.** Answers: 1. No — it needs a data source. 2. Columnar.
13. **4.11 Creating Queries in MS Access.** Answers: 1. ≥ 80 under Marks. 2. Two rows (OR).
14. **4.11.4 Data Summarization and Statistical Analysis.** Answers: 1. Group By. 2. COUNT (of BorrowID or BookID) grouped by MemberID.
15. **4.11.6 Data Visualization in MS Access (Charts).** Answers: 1. Line. 2. Pie (two parts of a whole).

Answer key

Multiple choice

1. b
2. c
3. c
4. b
5. b
6. c
7. b
8. b
9. b
10. b
11. d
12. c
13. b
14. c
15. b

Short questions — model answers

1. Data = raw facts (80, 75, 90). Information = processed meaning ("average is 81.7%, the class is ready").
2. Less redundancy, consistency, security (authorised access), integrity (valid, unique keys). (Sharing / backup also accepted.)
3. Record = one student's full row (101, Ali, CS, A). Field = one column such as Name.
4. PK uniquely identifies a row, cannot be NULL, one per table, lives in the parent. FK refers to a PK, may duplicate, may be NULL, lives in the child, creates the relationship.
5. It would duplicate attributes of both entities. Split into two 1:M links with a junction table that holds both FKs plus extra attributes (dates).
6. Every FK must match an existing parent PK. Inserting Result.StudentID = 999 is rejected if 999 is not in Student — that would have been an orphan.
7. Cascade Update copies a changed parent PK into child FKs. Cascade Delete removes child rows when the parent is deleted.
8. Table stores; form enters; query searches/filters; report prints.
9. They are identifiers, not quantities; leading zeros would vanish; you never add two numbers of a phone.
10. Put one value on the Criteria row and the other on the or: row of the same field (or of different fields if that is the question).

Long questions — model answers

1. Entities STUDENT and COURSE, relationship Enrolls (M:N). Junction ENROLLMENT (EnrollID PK, StudentID FK, CourseID FK, Date). Schema: Student(StudentID PK, Name, Class); Course(CourseID PK, Title); Enrollment as above. Diagram: 1 — M — M — 1 through the junction.
2. (a) Parent 101 Ali → 201 Ali; child Result rows that said 101 now say 201. (b) Delete parent 102 Sara; all Result and Enrolment rows with 102 disappear. Integrity definition: FK must match a live PK; this stops orphans and keeps reports true.
3. Follow steps 1–7 in the chapter. Sample: (5001, 102, B-17, 2026-10-05, NULL); (5002, 101, B-17, 2026-10-12, 2026-10-20) — same book, two members, no duplicated titles.
4. Create → Table Design; enter the five fields and types; right-click RollNo → Primary Key; Save as Student. Create → Form Wizard → Student → >> all fields → Columnar → name Student Registration → Finish.
5. Grid: Name, Department, Marks. Criteria under Department: "CS" (or "Computer Science"); Criteria under Marks: ≥ 80 (same row = AND); Sort under Name: Ascending. SQL: `SELECT Name, Department, Marks FROM Student WHERE Department="CS" AND Marks  $\geq 80$  ORDER BY Name`; Totals query: Department Group By, StudentID Count, Marks Avg.
6. (i) Column or bar — compare categories (classes). (ii) Pie — each class is a part of one whole enrolment. (iii) Line — a trend over years.

Self-assessment: I can...

- | | |
|--|---|
| ☐ explain data vs information vs database vs DBMS. | ☐ point at records and fields in a table. |
| ☐ choose PK / FK / candidate / alternate keys correctly. | ☐ draw ER symbols and label 1:1, 1:M, M:N. |
| ☐ resolve M:N with a junction table. | ☐ explain referential integrity and both cascade operations with examples. |
| ☐ write a relational schema including the library case study. | ☐ match tables, forms, queries and reports to a task. |
| ☐ create a table in Design View and pick data types. | ☐ build a Wizard form and a criteria query (AND, OR, totals). |
| ☐ choose bar, column, pie or line for a given question. | |

Chapter 5: Application and Impacts of Computing

Application and Impacts of Computing (1/3) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Introduction to Computing Impacts · Artificial Intelligence (AI) · Internet of Things (IoT) & Components · Data Analytics & Components · Comparison: IoT vs AI vs Data Analytics

★ **Must lock:** Artificial Intelligence (AI), Internet of Things (IoT) & Components

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Criteria for marks at least 80?
2. Best data type for StudentID if you want Access to generate it?
3. Can a form exist without a table or query?

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

5.1 Introduction to Computing Impacts

Computing means using computers and computer-based devices to store, process and share information.

- Schools and colleges — online admissions, e-books, online results of the Sindh boards.
- Hospitals — computerised patient records, digital X-rays, online appointment systems.
- Banks — ATMs, mobile banking apps, Raast instant payments, JazzCash and Easypaisa wallets.
- Farms — weather forecasts on mobile phones, soil sensors, tube-well control.

کمپیوٹنگ آج ہماری روزمرہ زندگی کا اہم حصہ بن چکی ہے۔

5.1.1 Artificial Intelligence (AI) ★ Golden

Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding language and making decisions.

IoT (Eyes) — collects data → Analytics — sorts data → AI (Brain) — decides → Action

- Artificial — it is created by humans; it is not natural.
- Intelligence — it can perform tasks that require thinking.
- Face recognition on a mobile phone to unlock the screen.
- YouTube suggesting videos based on your interests.

Remember: Eyes – Sort – Brain – Do. IoT is the eyes, Analytics sorts, AI is the brain, and the actuator does the action. Think of a cricket fielder: his eyes see the...

Exam: Common questions: "Define AI and give two examples" (2–3 marks), and "Explain how AI, IoT and Data Analytics work together" (4–5 marks). To score full marks: (1) write...

مصنوعی ذہانت (AI) کو "مصنوعی" اس لیے کہتے ہیں کہ اسے انسانوں نے بنایا ہے، اور "ذہانت" اس لیے کہ یہ ایسے کام کر سکتی ہے جن کے لیے سوچنا پڑتا ہے۔

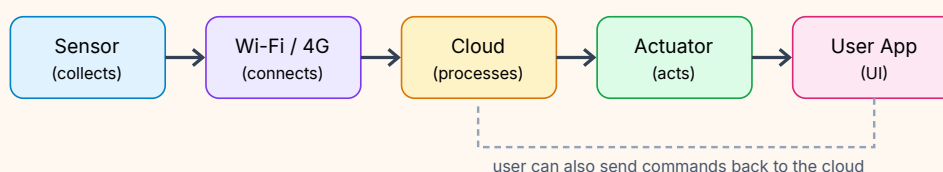
5.1.2 Internet of Things (IoT) & Components ★ Golden

The Internet of Things (IoT) is a system in which physical devices are connected to the Internet and can collect and share data to perform tasks automatically.

Sensor (Collects) → Wi-Fi (Connectivity) → Cloud (Processes) → Actuator (Acts) → User App (UI)

- The "things" are ordinary objects — watches, cameras, cars, fans, water pumps — that have been given sensors and an internet connection so...
- Source example: A smartwatch collects your heart-rate data and sends it to your mobile phone, where it is displayed.

#	Component	What it does	Examples
1	Sensors	Devices that collect information from the physical world. They are the "eyes and ears" of the system.	Temperature sensor, motion sensor, fingerprint sensor, smoke sensor, soil moisture sensor
2	Actuators	Devices that perform physical actions in response to system commands. They convert digital signals into physical movement.	Electric motor, switch, automatic door opener, water valve, fan controller
3	Connectivity	The connection that allows devices to send and receive data . Without it, sensors can't talk to the system.	Wi-Fi, Bluetooth, ZigBee, 4G/5G
4	Data Processing	Changing raw data into useful information . Steps: collecting, organising, analysing and deciding.	Checking if the room temperature is too hot to trigger the AC
5	Cloud Computing	Storing data and using services over the internet instead of local hard drives. Handles massive IoT data.	Google Drive, iCloud, online LMS
6	User Interface (UI)	The component that enables interaction between the user and the device . Helps users see information and give commands.	Mobile apps, touch screens, voice commands



The six IoT components working as one system (data processing happens inside the cloud).

Remember: Sensor senses, Actuator acts. Both words give you the answer: sensor = see/sense (input), actuator = act (output). Remember the six components with "Some Ants..."

Exam: "Define IoT and explain its components" is one of the most frequent long questions. Write the definition with the three key ideas — physical devices, connected to the...

انٹرنیٹ آف تھنگز (IoT) ایک ایسا نظام ہے جس میں مختلف آلات (جیسے گھڑیاں، کیمرے، گاڑیاں) انٹرنیٹ سے جڑے ہوتے ہیں اور خودکار طریقے سے...

5.1.3 Data Analytics & Components

Data Analytics is the process of collecting, organising, cleaning, studying and presenting data to find useful information.

1. Collection → 2. Storage → 3. Cleaning → 4. Analysis → 5. Visualization

- It helps people make better decisions.
- Data means raw facts and figures (for example, "72, 45, 88").
- Information is data that has been processed so it has meaning ("Ali scored 72%; he passed").

Roll No	Name	Marks (raw)
1	Ayesha	42
2	Bilal	18
3	Hina	35
3	Hina	35
4	Kashif	250
5	Sana	(missing)

Remember: "Cows Sit Calmly And Vote" — Collection, Storage, Cleaning, Analysis, Visualization. Or think of cooking daal: buy the lentils (collect), keep them in a jar...

ڈیٹا اینالٹکس ڈیٹا کو جمع کرنے، ترتیب دینے، صاف کرنے، اس کا مطالعہ کرنے اور پیش کرنے کا عمل ہے تاکہ مفید معلومات حاصل ہوں اور بہتر فیصلے کیے جا سکیں۔

Comparison: IoT vs AI vs Data Analytics

These three technologies are often used together, so students mix them up.

- The table below, from the source, compares them on four aspects.

Aspect	IoT	Artificial Intelligence (AI)	Data Analytics
Definition	Network of physical devices connected to the internet.	Technology enabling machines to think and decide.	Process of examining data to find patterns.
Main purpose	Data collection and automation.	Intelligent decision making and predictions.	Extracting meaningful insights.
Type of work	Works with physical devices (hardware).	Works with smart software systems.	Works with data (numbers, records).
Output	Raw data from the environment.	Smart decisions or actions.	Reports, graphs, insights.
Role in a smart system	Eyes and hands	Brain	Organiser / reporter
Pakistani example	Safe City cameras in Lahore/Karachi capturing traffic	Software recognising a car's number plate for an e-challan	Monthly report showing which roads have the most violations

Remember: Ask "What is the output?" Raw readings → IoT. Charts and reports → Data Analytics. A decision or prediction → AI.

IoT جسمانی آلات (ہارڈویئر) کا نیٹ ورک ہے جو ماحول سے خام ڈیٹا جمع کرتا ہے۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Artificial Intelligence (AI):** Mistake: "AI means robots." Fix: Most AI is just software — YouTube suggestions, spam filters and face unlock have no robot body.
- **Internet of Things (IoT) & Components:** Mistake: Calling a fan controller or water valve a "sensor".
- **Data Analytics & Components:** Mistake: Skipping cleaning and going straight to analysis.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Artificial Intelligence (AI)**. Write: Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding language and making decisions.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Intelligence — it can perform tasks that require thinking. (2) Face recognition on a mobile phone to unlock the screen. — + example.

5 marks: 3-mark structure + four-point explanation + the worked example from this topic.

8–10 marks: 5-mark structure + a complete truth table and a labelled gate / logic diagram + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Internet of Things (IoT) & Components**. Write: The Internet of Things (IoT) is a system in which physical devices are connected to the Internet and can collect and share data to perform tasks automatically.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) The "things" are ordinary objects — watches, cameras, cars, fans, water pumps — that have been given sensors and an internet connection so... (2) Source example: A smartwatch collects your heart-rate data and sends it to your mobile phone, where it is displayed. — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + a fully labelled diagram copied from the board + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. Why is AI called "artificial"?

3 marks. Explain Artificial Intelligence (AI) with two clear points and one example.

5 marks. Common questions: "Define AI and give two examples" (2–3 marks), and "Explain how AI, IoT and Data Analytics work together" (4–5 marks). To score full marks: (1) write the two-part meaning of the...

8 marks. Artificial Intelligence (AI): 5-mark answer plus a complete truth table and a labelled gate / logic diagram.

Answers: 1. Because it is created by humans 3. Intelligence — it can perform tasks that require thinking.; Face recognition on a mobile phone to unlock the screen. 5. Use the recipe in the blue box. Model opening: Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding... 8. Same 5-mark body + a complete truth table and a labelled gate / logic diagram + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Artificial Intelligence (AI)** using the recipe in this topic. Include a complete truth table and a labelled gate / logic diagram. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 5: Application and Impacts of Computing

Application and Impacts of Computing (2/3) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Uses of IoT in Different Areas · Scope of AI in Education in Pakistan · Career Connection in IoT and AI · Information Sources (Primary, Secondary, Tertiary)

★ **Must lock:** Information Sources (Primary, Secondary, Tertiary)

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Why is AI called "artificial"?
2. Classify as sensor or actuator: (a) fingerprint scanner, (b) automatic door opener, (c) motion detector, (d) electric motor.
3. Name four places in Pakistan where computing has become vital.

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

5.2 Uses of IoT in Different Areas

IoT improves efficiency (doing more with less effort), saves time, and enhances the overall quality of life.

Soil Sensor → Cloud AI (Needs water?) → Water Pump

- It plays a significant role in developing modern infrastructure (the basic systems of a country — roads, power, water, hospitals).
- The source lists six areas.

Remember: Remember the six areas with "CHIEF H": Cities, Health, Industry, Education, Farming, Homes.

IoT کارکردگی بہتر بناتا ہے، وقت بچاتا ہے اور زندگی کا معیار بلند کرتا ہے۔

5.2.1 Scope of AI in Education in Pakistan

Artificial Intelligence is gradually becoming an important part of Pakistan's education system, improving teaching, learning and administration (managing the school: admissions, records, timetables).

- Learning Management Systems (LMS) — online platforms (e.g. Google Classroom, Moodle) where teachers post lessons and assignments; AI...
- Automated grading and digital testing — MCQ tests marked instantly by computer.
- AI-based career counselling — software that studies a student's marks and interests and suggests suitable fields.
- Helps students learn at their own speed.

Practical uses	Benefits	Challenges
LMS	Learn at own speed	Limited rural internet
Automated grading & digital testing	Identifies students needing help	Lack of computers/devices
AI career counselling	Supports students with disabilities	Teacher training needed
—	—	Data privacy & security

Trap: Mistake: Writing that AI will replace teachers.

پاکستان کے تعلیمی نظام میں مصنوعی ذہانت آہستہ آہستہ اہم کردار ادا کر رہی ہے، جیسے لرننگ مینجمنٹ سسٹم (LMS)، خودکار جانچ اور کیریئر کونسلنگ۔

5.2.2 Career Connection in IoT and AI

Learning computing skills opens doors to many employers in Pakistan: software houses (e.g.

- IoT Engineer: designs smart systems and connected devices (e.g. a smart irrigation kit).
- Data Analyst: analyses the data generated by IoT systems and produces reports.
- Programming — Python, C++.
- Networking — Wi-Fi, protocols (rules for how devices communicate).

Remember: Skills = "PNEP": Programming, Networking, Electronics, Problem-solving. You start building the first one this year when you learn Python!

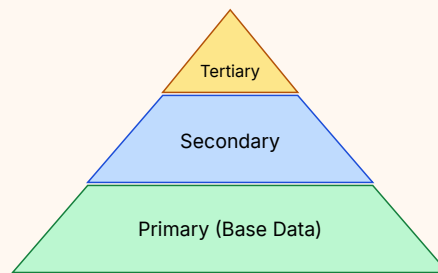
کمپیوٹنگ کی مہارتیں سیکھ کر آپ سافٹ ویئر ہاؤسز، ٹیلی کام کمپنیوں (جیسے جاز، ٹیلی نار)، بینکوں اور سرکاری آئی ٹی محکموں میں کام کر سکتے ہیں۔

5.3 Information Sources (Primary, Secondary, Tertiary) ★ Golden

Information sources are ways to find knowledge and data.

- The internet has vast information, but not all of it is accurate.
- Sources are divided into three main categories.
- Primary sources give original, firsthand information straight from people, events or official records.

Primary (firsthand)	Secondary (explained by others)	Tertiary (summarised, quick reference)
Interview with a flood victim in Dadu	Newspaper report on the 2022 floods	Encyclopedia entry "Floods in Pakistan"
Photograph taken at an event	Textbook chapter explaining the event	Wikipedia article
Personal diary or letter	Biography written by a historian	Dictionary
NADRA official records (CNIC data)	Pakistan Bureau of Statistics report analysing census data	Handbook / almanac of facts
Original survey responses	Research paper analysing the survey	Index or list of research papers
Sensor readings from an IoT weather station	TV news analysis of the weather	A school "fact sheet" on climate
Quaid-e-Azam's original speech recording	Article discussing the speech	Encyclopedia summary of his life



The information pyramid: primary data is the base; each higher level is built from the levels below it and is more summarised.

Remember: Rule of thumb (Author – Date – References): If an online article does not state the author's name, the date of publication, or list its references, it should...

Exam: Typical questions: "Differentiate between primary, secondary and tertiary sources with two examples each" (long/short), "Classify the following sources..." (MCQ/short),...

معلومات کے ذرائع علم اور ڈیٹا تلاش کرنے کے طریقے ہیں، جنہیں تین اقسام میں تقسیم کیا جاتا ہے۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Uses of IoT in Different Areas:** Mistake: Giving only the name of the area ("smart cities") in an exam answer.
- **Scope of AI in Education in Pakistan:** Mistake: Writing that AI will replace teachers.
- **Information Sources (Primary, Secondary, Tertiary):** Mistake: Thinking "primary" means "most important" or "tertiary" means "useless".

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Information Sources (Primary, Secondary, Tertiary)**. Write: Information sources are ways to find knowledge and data.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) The internet has vast information, but not all of it is accurate. (2) Sources are divided into three main categories. — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + a four-row comparison table and one rewritten survey question + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. A video interview with a cricketer after a match.

3 marks. Explain Information Sources (Primary, Secondary, Tertiary) with two clear points and one example.

5 marks. Typical questions: "Differentiate between primary, secondary and tertiary sources with two examples each" (long/short), "Classify the following sources..." (MCQ/short), and "When does a source become..."

8 marks. Information Sources (Primary, Secondary, Tertiary): 5-mark answer plus a four-row comparison table and one rewritten survey question.

Answers: 1. P 3. The internet has vast information, but not all of it is accurate.; Sources are divided into three main categories. 5. Use the recipe in the blue box. Model opening: Information sources are ways to find knowledge and data. 8. Same 5-mark body + a four-row comparison table and one rewritten survey question + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Information Sources (Primary, Secondary, Tertiary)** using the recipe in this topic. Include a four-row comparison table and one rewritten survey question. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 5: Application and Impacts of Computing

Application and Impacts of Computing (3/3) · Chapter exam

TODAY'S 90 MINUTES

90 minutes. 0–10 min recap yesterday ★ on the board · 10–85 min sit the 75-mark chapter exam (1 h 15 min, hall conditions) · 85–90 min collect scripts. Do not mark in front of the batch.

Board today: Impacts of Computing in Various Fields · Assistive Technologies · Importance of Assistive Technologies · Career Connection in Assistive Tech

★ **Must lock:** Assistive Technologies

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Hall paper today. Recap the cards for ten minutes, then scripts out. Re-teach any Golden below 60% in tomorrow's starter.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. A video interview with a cricketer after a match.
2. How do smart bins help waste management?
3. Give three practical uses of AI in education.

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

5.4 Impacts of Computing in Various Fields

Computing has drastically changed society.

- It makes life faster and more connected, but it also brings challenges like privacy risks and technology addiction.
- For every field, think of both sides.

Field	Positive impact (+)	Negative impact (-)
Education	Online classes, e-books and learning apps make studying highly flexible and accessible.	Excessive screen time reduces attention spans and causes physical health issues (eye strain, back pain).
Communication	Emails and video calls allow instant, global communication at almost zero cost (e.g. a WhatsApp video call to a relative in Dubai).	People interact less in person, leading to social isolation and reduced face-to-face skills.
Business	Online shopping, digital banking and automation dramatically improve speed and efficiency.	Traditional manual jobs are lost to machines; risk of severe cyber-attacks on banks.
Healthcare	Digital patient records and telemedicine allow doctors to treat remote patients accurately.	Patient data privacy is at risk if hospital servers are hacked.
Agriculture	Smart irrigation and weather forecasting massively increase crop yields.	Poor internet access in rural areas means many farmers cannot use these tools.

Remember: For the five fields, remember "E-C-B-H-A" — Education, Communication, Business, Healthcare, Agriculture — and always write one + and one – for each.

کمپیوٹنگ نے معاشرے کو بہت بدل دیا ہے۔

5.5 Assistive Technologies ★ Golden

Assistive technologies are devices, software and systems that help people with disabilities or functional limitations (difficulty in seeing, hearing, moving, speaking, remembering, etc.).

Mobility → Sensory → Communication → Cognitive

- They allow more people to participate equally in school, work and society.
- There are four categories:

Category	Purpose	Examples
1. Mobility aids	Help people move from one place to another.	Smart wheelchairs, stair lifts
2. Sensory aids	Support vision and hearing to improve interaction.	Hearing aids, screen readers (VoiceOver), Braille displays
3. Communication aids	Assist individuals with speech difficulties to express themselves.	Speech-generating devices, text-to-speech apps
4. Cognitive / learning aids	Help people with memory, reading and understanding.	Audiobooks, spell-checkers, digital reminder apps

Remember: Four categories = "My Small Cat Cries": Mobility, Sensory, Communication, Cognitive. Link each to a body function: legs (mobility), eyes/ears (sensory), mouth...

Exam: Expect: "What are assistive technologies? Explain their categories with examples" (long), "Name two sensory aids" (short), or "A screen reader is an example of ___"...

معاون ٹیکنالوجیز (Assistive Technologies) وہ آلات، سافٹ ویئر اور نظام ہیں جو معذور افراد کی مدد کرتے ہیں تاکہ وہ اسکول، کام اور

مع...

5.5.1 Importance of Assistive Technologies

Assistive technologies are important for three main reasons: Equal access to education: Screen readers allow blind students to read digital textbooks equally with their peers (classmates).

Equal Access → Independence → Inclusion

- Equal access to education: Screen readers allow blind students to read digital textbooks equally with their peers (classmates).
- Independence in daily life: Mobility aids reduce the need for constant human assistance — a person in a smart wheelchair can go to the...
- Social inclusion: They break down barriers, allowing individuals to work, study and socialise confidently.

Remember: Remember "All": Access, Independence, Inclusion.

معاون ٹیکنالوجیز کی اہمیت تین باتوں میں ہے: تعلیم تک برابر رسائی (ناہینا طلبہ اسکرین ریڈر سے کتابیں پڑھ سکتے ہیں)، روزمرہ زندگی می...

5.5.2 Career Connection in Assistive Tech

Designing assistive technology is a highly rewarding career path — you earn a living while improving people's lives.

- CareerWhat they do Assistive Software DeveloperCreates tools such as speech-to-text software (e.g. Urdu voice typing for people who cannot use a keyboard).
- Biomedical EngineerDesigns artificial limbs and hearing aids.

Career	What they do
Assistive Software Developer	Creates tools such as speech-to-text software (e.g. Urdu voice typing for people who cannot use a keyboard).
Biomedical Engineer	Designs artificial limbs and hearing aids.
UX/UI Accessibility Expert	Ensures websites are easy to navigate for visually impaired users (e.g. adding text descriptions to images so screen readers can read them).

معاون ٹیکنالوجی بنانا ایک قابلِ قدر پیشہ ہے۔

Application and Impacts of Computing

13 single-page lessons · 4 Golden · read, close, say aloud, then sit the exam

Chapter in one look

- Computing is vital in Pakistan's schools, hospitals, banks, farms and industries; it has both positive and negative impacts.
- **AI** is "artificial" (made by humans) and "intelligent" (does tasks needing thinking); it uses data to learn, recognise patterns and make decisions.
- In smart systems, IoT is the "eyes" that gather data, Analytics sorts it, and AI is the "brain" that decides the action.
- **IoT** is a system of physical devices connected to the internet that collect and share data to perform tasks automatically.
- The six IoT components are sensors (collect), actuators (act), connectivity, data processing, cloud computing and user interface.
- **Data Analytics** is a pipeline: collection → storage → cleaning → analysis → visualization.
- Data cleaning removes duplicates, incorrect values, missing records and improper formats.
- IoT outputs raw data (hardware); AI outputs smart decisions (software); Data Analytics outputs reports, graphs and insights (data).
- IoT is used in smart cities, industrial automation, healthcare, education, home automation and smart farming.
- Smart farming (soil sensor → cloud AI → water pump) saves water and improves yields in Sindh and Punjab.
- AI in Pakistani education: LMS, automated grading, career counselling; challenges are rural internet, lack of devices, teacher training and data privacy.
- IoT/AI careers (IoT engineer, data analyst) need programming, networking, electronics and problem-solving.
- Information sources: primary (firsthand/original), secondary (collected and explained by others), tertiary (summarised for quick reference).
- Do not trust an online article that lacks an author, a publication date or references.
- Computing helps education, communication, business, healthcare and agriculture but risks screen addiction, isolation, job loss, cyber-attacks and privacy loss.
- **Assistive technologies** — mobility, sensory, communication and cognitive aids — give equal access, independence and social inclusion to people with disabilities.

Key terms

Term	Meaning	اردو
Computing	Using computers and digital devices to store, process and share information.	کمپیوٹنگ
Artificial Intelligence (AI)	Human-made technology that lets machines think, learn and decide using data.	مصنوعی ذہانت
Internet of Things (IoT)	Physical devices connected to the internet that collect and share data to work automatically.	انٹرنیٹ آف تھنگز، اشیاء کا انٹرنیٹ
Sensor	Device that collects information from the physical world.	حساس آلہ (سینسر)
Actuator	Device that performs a physical action in response to a command.	عمل کار آلہ (ایکچوایٹر)
Connectivity	The link (Wi-Fi, Bluetooth, ZigBee, 4G/5G) that lets devices send and receive data.	رابطہ کاری
Data processing	Changing raw data into useful information.	ڈیٹا کی پروسیسنگ
Cloud computing	Storing data and using services over the internet instead of local drives.	کلاؤڈ کمپیوٹنگ
User interface (UI)	The part that lets a user see information and give commands to a device.	صارف انٹرفیس
Data Analytics	Collecting, storing, cleaning, analysing and presenting data to find useful information.	ڈیٹا کا تجزیہ
Data cleaning	Finding and correcting errors such as duplicates and missing values.	ڈیٹا کی صفائی
Data visualization	Showing data as charts, plots or graphs.	ڈیٹا کی بصری پیشکش

Application and Impacts of Computing

75 marks • 1 hour 15 minutes • no notes • no phones

CHAPTER EXAM RULES

Instructions. Sit this exam after Lesson 13 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

Marks. $15 + 30 + 30 = 75$. A 5-mark / 10-mark answer needs Define • Explain • Example • Diagram • Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

Section A — Multiple choice (15 × 1 = 15)

1. AI is called "artificial" because it:
 - a. runs only on electricity
 - b. is created by humans
 - c. is always wrong
 - d. works without data
2. Which of the following is an actuator?
 - a. Temperature sensor
 - b. Motion sensor
 - c. Water valve
 - d. Wi-Fi router
3. ZigBee is an example of which IoT component?
 - a. Connectivity
 - b. Sensor
 - c. User interface
 - d. Actuator
4. Google Drive and iCloud are examples of:
 - a. Sensors
 - b. Data cleaning
 - c. Actuators
 - d. Cloud computing
5. Removing duplicate student records belongs to which component of Data Analytics?
 - a. Data collection
 - b. Data storage
 - c. Data cleaning
 - d. Data visualization
6. A pie chart showing pass vs fail percentage is an example of:
 - a. Data collection
 - b. Data storage
 - c. Data cleaning
 - d. Data visualization
7. According to the comparison table, the output of IoT is:
 - a. Raw data from the environment
 - b. Smart decisions
 - c. Reports and graphs
 - d. Printed documents
8. In a smart system, AI acts as the:
 - a. eyes
 - b. brain
 - c. hands
 - d. wires
9. NADRA's official records are a:
 - a. tertiary source
 - b. secondary source
 - c. primary source
 - d. unreliable source
10. Wikipedia is classified as a:
 - a. primary source
 - b. tertiary source
 - c. secondary source
 - d. firsthand record
11. Smart bins that send alerts when full are a use of IoT in:
 - a. smart farming
 - b. healthcare
 - c. home automation
 - d. smart cities
12. A system in which a soil sensor tells the cloud AI to switch on a water pump is used in:
 - a. smart farming
 - b. industrial automation
 - c. smart classrooms
 - d. traffic management
13. A screen reader such as VoiceOver is a:
 - a. mobility aid
 - b. communication aid
 - c. sensory aid
 - d. cognitive aid
14. A speech-generating device is a:
 - a. mobility aid
 - b. cognitive aid
 - c. sensory aid
 - d. communication aid
15. An online article should NOT be trusted for academic work if it lacks:
 - a. colourful pictures
 - b. the author's name, publication date or references
 - c. a long title
 - d. advertisements

Section B — Short questions (10 × 3 = 30)

1. Define Artificial Intelligence and give two examples from daily life.
2. Define the Internet of Things (IoT) with one example.
3. Differentiate between a sensor and an actuator with two examples of each.
4. What is the role of cloud computing in an IoT system?
5. List the five components of Data Analytics. What kinds of errors are removed in data cleaning?
6. Give two uses of IoT in smart cities and two in healthcare.
7. State two benefits and two challenges of AI in education in Pakistan.
8. Differentiate between primary and secondary sources with two examples each.
9. When does a secondary source become unreliable?
10. Give one positive and one negative impact of computing on (a) communication and (b) business.

Section C — Long questions (attempt any 3 × 10 = 30)

1. Define IoT. Explain its six components with an example of each. Draw a diagram showing how they connect.
2. Compare IoT, AI and Data Analytics in a table (definition, purpose, type of work, output). Then explain, with an example, how the three work together in a smart system.
3. A college in Mirpurkhas wants to find out why many students are failing Mathematics. Explain how it would use each of the five components of Data Analytics to solve this problem.
4. Explain the three types of information sources with examples. Classify the following: (i) a school textbook, (ii) a photograph of an event, (iii) an encyclopedia, (iv) an interview, (v) a Bureau of Statistics report, (vi) a dictionary. When does each type become unreliable?
5. What are assistive technologies? Explain their four categories with examples, state their importance, and name two related careers.
6. "Computing has both positive and negative impacts on society." Discuss with reference to education, healthcare and agriculture, and suggest three rules for smart and responsible use.

Answer key. Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

Check yourself — answers

1. **5.1 Introduction to Computing Impacts.** Answers: 1. Any four of: schools, hospitals, banks, farms, industries. 2. The effect computing has on people and society — it can be positive or negative.
2. **5.1.1 Artificial Intelligence (AI).** Answers: 1. Because it is created by humans. 2. AI is the brain; IoT is the eyes. 3. Yes — it uses data (past spam emails) to decide whether a new email is spam, a task that needs "thinking".
3. **5.1.2 Internet of Things (IoT) & Components.** Answers: 1. (a) sensor, (b) actuator, (c) sensor, (d) actuator. 2. Wi-Fi, Bluetooth, ZigBee, 4G/5G. 3. Because IoT devices produce massive amounts of data; the cloud stores and processes it over the internet instead of on small local hard drives.
4. **5.1.3 Data Analytics & Components.** Answers: 1. Collection, Storage, Cleaning, Analysis, Visualization. 2. Duplicate records, incorrect values (typos), missing records, improper formats. 3. Data Visualization.
5. **Comparison: IoT vs AI vs Data Analytics.** Answers: 1. IoT. 2. Reports, graphs and insights.
6. **5.2 Uses of IoT in Different Areas.** Answers: 1. They send alerts when full, so only full bins are emptied. 2. Rural areas lack doctors; devices send patient data directly to doctors far away. 3. The water pump (its switch).
7. **5.2.1 Scope of AI in Education in Pakistan.** Answers: 1. LMS, automated grading and digital testing, AI-based career counselling. 2. Any two: limited rural internet, lack of devices, need for teacher training, data privacy and security.
8. **5.2.2 Career Connection in IoT and AI.** Answers: 1. Designs smart systems and connected devices. 2. Programming (Python, C++), networking, electronics, problem-solving.
9. **5.3 Information Sources (Primary, Secondary, Tertiary).** Answers: 1. P 2. S 3. T 4. P 5. S 6. T 7. P 8. S
10. **5.4 Impacts of Computing in Various Fields.** Answers: 1. + Digital records and telemedicine help treat remote patients; – patient data privacy is at risk if servers are hacked. 2. Poor internet access in rural areas. 3. Protect personal information/use strong passwords; do not share fake news; balance screen time with physical activity.
11. **5.5 Assistive Technologies.** Answers: 1. Sensory aids. 2. Any two: audiobooks, spell-checkers, digital reminder apps. 3. Devices, software and systems that help people with disabilities or functional limitations participate equally in school, work and society.
12. **5.5.1 Importance of Assistive Technologies.** Answers: 1. Equal access to education, independence in daily life, social inclusion. 2. They reduce the need for constant human assistance.
13. **5.5.2 Career Connection in Assistive Tech.** Answers: 1. A biomedical engineer. 2. That websites are easy to navigate for visually impaired users.

Answer key

Multiple choice

1. b
2. c
3. a
4. d
5. c
6. d
7. a
8. b
9. c
10. b
11. d
12. a
13. c
14. d
15. b

Short questions — model answers

1. AI is human-made ("artificial") technology that can perform tasks requiring thinking ("intelligence"); it uses data to make decisions. Examples: face recognition to unlock a phone; YouTube suggesting videos; chatbots answering questions.
2. IoT is a system in which physical devices are connected to the internet and can collect and share data to perform tasks automatically. Example: a smartwatch collects heart-rate data and sends it to a mobile phone for display.
3. A sensor collects information from the physical world (input; the "eyes and ears"), e.g. temperature sensor, motion sensor. An actuator performs a physical action in response to a command, converting digital signals into movement (output), e.g. electric motor, automatic door opener.
4. Cloud computing stores IoT data and provides processing services over the internet instead of local hard drives, so it can handle the massive amounts of data produced by many devices. Examples: Google Drive, iCloud, online LMS.
5. Collection, storage, cleaning, analysis, visualization. Cleaning removes duplicated records, incorrect values (typos), missing files/records, and improper formats.
6. Smart cities: sensors adjust traffic signals to reduce congestion; smart bins send alerts when full (also street lighting, parking). Healthcare: smartwatches measure heart rate and vitals; remote monitoring devices send patient data to doctors (also emergency alerts).
7. Benefits: students learn at their own speed; AI identifies students who need extra help (also supports students with disabilities). Challenges: limited internet access in rural areas; lack of computers/devices (also teacher training, data privacy).
8. Primary sources give original, firsthand information from people, events or official records (e.g. interviews, NADRA records). Secondary sources contain information already collected and explained by someone else, summarising or analysing primary data (e.g. textbooks, news reports).
9. When the information is outdated, lacks proper references/citations, or has not been verified by an expert.
10. (a) Communication: + emails and video calls give instant, global communication at almost no cost; – less in-person interaction causes social isolation. (b) Business: + online shopping, digital banking and automation improve speed and efficiency; – manual jobs are lost to machines and banks face cyber-attacks.

Long questions — model answers

1. **Definition:** IoT is a system in which physical devices are connected to the internet and collect and share data to perform tasks automatically. **Components:** (1) Sensors — collect information from the physical world, e.g. temperature, motion, fingerprint, smoke, soil moisture sensors. (2) Actuators — perform physical actions in response to commands, converting digital signals to movement, e.g. motor, switch, door opener, water valve, fan controller. (3) Connectivity — lets devices send/receive data, e.g. Wi-Fi, Bluetooth, ZigBee, 4G/5G. (4) Data processing — changes raw data into useful information through collecting, organising, analysing and deciding, e.g. checking if a room is too hot to switch on the AC. (5) Cloud computing — stores data and provides services over the internet, handles massive data, e.g. Google Drive, iCloud, LMS. (6) User interface — lets users see information and give commands, e.g. mobile apps, touch screens, voice commands. **Diagram:** Sensor (collects) → Wi-Fi → Cloud (processes) → Actuator (acts) → User App (UI).
2. **Table:** IoT — network of physical devices connected to the internet; purpose: data collection and automation; works with hardware; output: raw data from environment. AI — technology enabling machines to think and decide; purpose: intelligent decisions and predictions; works with smart software; output: smart decisions or actions. Data Analytics — process of examining data to find patterns; purpose: extracting meaningful insights; works with data (numbers, records); output: reports, graphs, insights. **Working together:** IoT is the eyes that gather data, Analytics sorts it, AI is the brain that decides. Example: a temperature sensor reads 36 °C (IoT) → recent readings are organised showing the room is heating up (Analytics) → AI decides the room is too hot (AI) → a switch turns on the AC (action).
3. (1) **Collection:** gather marks of all Maths tests, attendance and survey responses from students. (2) **Storage:** keep them safely in a spreadsheet or database, e.g. on the cloud. (3) **Cleaning:** remove duplicate records, correct typing errors (e.g. 120 out of 100), fill or mark missing records, and make formats (dates, roll numbers) consistent. (4) **Analysis:** study the clean data to find patterns — e.g. which chapters most students got wrong, whether low attendance links to failing. (5) **Visualization:** present results in bar charts or a pie chart of pass vs fail so the principal sees the problem at a glance and plans extra classes for weak chapters.
4. **Primary:** original, firsthand, highly authentic (interviews, photographs, diaries, NADRA records). **Secondary:** collected and explained by someone else, summarises/analyses primary data (textbooks, news reports, research papers, Bureau of Statistics reports). **Tertiary:** highly summarised and organised from primary and secondary sources for quick reference (encyclopedias, Wikipedia, dictionaries, handbooks). **Classification:** (i) secondary, (ii) primary, (iii) tertiary, (iv) primary, (v) secondary, (vi) tertiary. **Unreliability:** primary — incomplete, recorded incorrectly, or deliberately false; secondary — outdated, no references/citations, not verified by an expert; tertiary — unknown/unverified website, or oversimplified and missing key facts. Rule: do not trust an article without author, date or references.
5. **Definition:** devices, software and systems that help people with disabilities or functional limitations participate equally in school, work and society. **Categories:** (1) Mobility aids — help people move; smart wheelchairs, stair lifts. (2) Sensory aids — support vision and hearing; hearing aids, screen readers (VoiceOver), Braille displays. (3) Communication aids — help people with speech difficulties express themselves; speech-generating devices, text-to-speech apps. (4) Cognitive/learning aids — help with memory, reading, understanding; audiobooks, spell-checkers, reminder apps. **Importance:** equal access to education (screen readers let blind students read digital textbooks), independence in daily life (mobility aids reduce need for constant help), social inclusion (people work, study and socialise confidently). **Careers:** any two of assistive software developer (speech-to-text tools), biomedical engineer (artificial limbs, hearing aids), UX/UI accessibility expert (websites usable by visually impaired users).
6. **Introduction:** computing makes life faster and more connected but brings privacy risks and addiction. **Education:** + online classes, e-books and learning apps make study flexible and accessible; – excessive screen time reduces attention span and harms physical health. **Healthcare:** + digital patient records and telemedicine let doctors treat remote patients accurately (e.g. rural Sindh); – patient data privacy is at risk if hospital servers are hacked. **Agriculture:** + smart irrigation and weather forecasting greatly increase yields; – poor rural internet means many farmers cannot use these tools. **Responsible use:** protect personal information and use strong passwords; do not share fake news or unverified forwards; balance screen time with sports and physical activity. **Conclusion:** benefits are large if technology is used wisely.

Self-assessment: I can...

- ☐ define AI and explain why it is called "artificial intelligence".
- ☐ define IoT and name its six components with examples.
- ☐ list the five components of Data Analytics in order and describe data cleaning.
- ☐ describe IoT uses in smart cities, industry, healthcare, education, homes and farms.
- ☐ classify any source as primary, secondary or tertiary and judge its reliability.
- ☐ explain the four categories of assistive technologies and their importance.
- ☐ explain how IoT, Data Analytics and AI work together (eyes → sort → brain → action).
- ☐ tell the difference between a sensor and an actuator.
- ☐ compare IoT, AI and Data Analytics in a table.
- ☐ state uses, benefits and challenges of AI in education in Pakistan.
- ☐ discuss positive and negative impacts of computing in five fields.
- ☐ name careers linked to IoT, AI and assistive technology and the skills they need.

Chapter 6: Digital Literacy

Digital Literacy (1/4) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Introduction to Digital Literacy · Digital Literacy in the Information Age · Types of Data · Data-Collection Strategies · Interviews and Surveys

★ **Must lock:** Types of Data, Data-Collection Strategies, Interviews and Surveys

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Which category does a hearing aid belong to?
2. Give one positive and one negative impact of computing in healthcare.
3. State three reasons why assistive technologies are important.

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

6.1 Introduction to Digital Literacy

Digital literacy is the ability to understand and use digital devices, the internet, and online tools in a safe, effective, and responsible manner.

Locate information → Organise it properly → Turn it into something useful → Share it responsibly

- Safe – you protect yourself and others online (for example, you do not share passwords or private details).
- Effective – you get the job done well (for example, you find the right information quickly).
- Responsible – you use information honestly and fairly (for example, you do not copy someone else's work or spread false news).
- find information,

Remember: Remember the four verbs of a digitally literate student: Locate – Organise – Create – Share ("LOCS": think of a lock that keeps your digital life safe).

ڈیجیٹل خواندگی (Digital Literacy) سے مراد ڈیجیٹل آلات، انٹرنیٹ اور آن لائن ٹولز کو محفوظ، مؤثر اور ذمہ دارانہ طریقے سے سمجھنا اور استعمال کرنا ہے۔

6.2 Digital Literacy in the Information Age

The Information Age is the period we live in now, where vast amounts of data are produced, shared and accessed instantly through the internet, mobile devices and digital platforms (websites, apps, social media).

- Simply having access to technology is not enough.
- Owning a smartphone does not make you good at using information.
- You must develop the skills to understand, manage and use information wisely.

Ability	What it means	Example
Technical	Knowing how to operate devices and software.	Creating a Google Form; using a spreadsheet.
Cognitive (thinking)	Thinking critically about information.	Checking whether a news story is true before believing it.
Social	Communicating and behaving well with others online.	Being polite in a class WhatsApp group; respecting privacy.

Remember: Three abilities = "Tech, Think, Talk": Technical, Cognitive (think), Social (talk with others).

معلوماتی دور (Information Age) میں انٹرنیٹ اور موبائل کے ذریعے بے پناہ معلومات فوراً تیار اور شیئر ہوتی ہیں۔

6.3 Types of Data ★ Golden

Data means facts, numbers, words or observations that we collect.

- The data gathered through a survey, an investigation or a classroom project can be described in two ways: qualitatively or quantitatively.
- Knowing the difference is critical, because each type is analysed in a different way.

Type of data	Nature	Examples	Best used for
Qualitative	Descriptive, non-numerical	Opinions, reasons, behaviour	Understanding why and how something happens
Quantitative	Numerical, countable or measurable	Hours, scores, percentages, totals	Comparing results and making graphs/charts

Remember: QUALitative = QUALity = words and feelings. QUANTitative = QUANTity = numbers and counts. Qualitative answers why/how; quantitative answers how many/how much.

Exam: Common questions: "Differentiate between qualitative and quantitative data" (4–5 marks) or "Classify the following data". Write the answer as a table with four rows:...

کیفیتی ڈیٹا (Qualitative Data) الفاظ پر مشتمل ہوتا ہے اور بتاتا ہے کہ لوگ کیا سوچتے، محسوس کرتے یا تجربہ کرتے ہیں، جیسے آن لائن کل...

6.4 Data-Collection Strategies ★ Golden

Digital literacy is not limited to finding information that already exists on the internet.

Interviews → Surveys → Prototypes → Observation → Simulations

- You should also learn how to gather original (new) information yourself.
- A data-collection strategy is a planned method for gathering this information.
- The five common strategies are: Interviews·Surveys·Prototypes·Observation·Simulations

Strategy	How it works (one line)	Usual data type	Best when...
Interview	Planned conversation with a person or small group	Mostly qualitative	You need deep, detailed answers from a few people
Survey	Same questions asked to many people	Quantitative and qualitative	You need answers from many people quickly
Prototype	Users test an early version and give feedback	Both	You want to improve a product or solution
Observation	Watch and record; no questions asked	Notes (qualitative) or counts (quantitative)	You want to see what people actually do
Simulation	Computer model; change variables and see what happens	Mostly quantitative	Real experiments are costly, risky or difficult

Remember: Mnemonic for the five strategies: "I S P O S" – "I Saw People On Screens": Interviews, Surveys, Prototypes, Observation, Simulations.

Exam: "List and explain the data-collection strategies" is a frequent long question. Name all five, give a one-sentence definition and one example for each, and say what type...

ڈیجیٹل خواندگی صرف موجود معلومات تلاش کرنے تک محدود نہیں، بلکہ نئی معلومات خود جمع کرنا بھی اس کا حصہ ہے۔

6.4.1 – 6.4.2 Interviews and Surveys ★ Golden

An interview is a planned conversation used to gather information directly from a person.

- It may be one-to-one (you and one person) or in a small group.
- Questions are planned in advance, so you do not forget anything important.
- Answers are usually qualitative (opinions, reasons, experiences).

Question type	What it looks like	Data it gives
Closed: Yes/No	Do you study with your phone nearby? (Yes / No)	Countable (quantitative)
Closed: Multiple choice	Which activity do you use your phone for most? (Lectures / Notes / Social media / Games)	Categories you can count
Number / scale	How many hours do you use your phone daily? ____	Quantitative
Rating scale	"My phone helps me study." 1 = Strongly disagree ... 5 = Strongly agree	Quantitative (you can average the ratings)
Open-ended	How does your phone help you in studies?	Qualitative (free text)

Remember: Interview = In-depth with few. Survey = Same questions to many. Think of a doctor talking to one patient (interview) vs a census form sent to every house...

Exam: You may be asked to "differentiate between an interview and a survey" or to "identify what is wrong with this survey question and rewrite it". For rewriting, name the...

انٹرویو ایک منصوبہ بند گفتگو ہے جس میں کسی ایک فرد یا چھوٹے گروہ سے براہ راست معلومات حاصل کی جاتی ہیں؛ اس کے جوابات عموماً کیفیتی ہوتے ہیں۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Introduction to Digital Literacy:** Mistake: "Digital literacy means knowing how to use a computer." Fix: Using a computer is only the start.
- **Digital Literacy in the Information Age:** Mistake: Thinking that "more information" automatically means "better information".
- **Types of Data:** Mistake: Thinking that any data containing a number is quantitative.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Types of Data**. Write: Data means facts, numbers, words or observations that we collect.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) The data gathered through a survey, an investigation or a classroom project can be described in two ways: qualitatively or quantitatively. (2) Knowing the difference is critical, because each type is analysed in a different way. — + example.

5 marks: 3-mark structure + table: Type of data / Nature / Examples / Best used for + the worked example from this topic.

8–10 marks: 5-mark structure + a four-row comparison table and one rewritten survey question + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Data-Collection Strategies**. Write: Digital literacy is not limited to finding information that already exists on the internet.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) You should also learn how to gather original (new) information yourself. (2) A data-collection strategy is a planned method for gathering this information. — + example.

5 marks: 3-mark structure + table: Strategy / How it works (one line) / Usual data type / Best when... + the worked example from this topic.

8–10 marks: 5-mark structure + a one-page canvas: problem, user, riskiest assumption, test + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Interviews and Surveys**. Write: An interview is a planned conversation used to gather information directly from a person.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Questions are planned in advance, so you do not forget anything important. (2) Answers are usually qualitative (opinions, reasons, experiences). — + example.

5 marks: 3-mark structure + table: Question type / What it looks like / Data it gives + the worked example from this topic.

8–10 marks: 5-mark structure + a four-row comparison table and one rewritten survey question + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. Is "time spent online in hours" qualitative or quantitative?

3 marks. Explain Types of Data with two clear points and one example.

5 marks. Common questions: "Differentiate between qualitative and quantitative data" (4–5 marks) or "Classify the following data". Write the answer as a table with four rows: nature, examples, best used for,...

8 marks. Types of Data: 5-mark answer plus a four-row comparison table and one rewritten survey question.

Answers: 1. Quantitative (measurable number) 3. The data gathered through a survey, an investigation or a classroom project can be described in two ways: qualitatively or quantitatively.; Knowing the difference is critical, because each type is analysed in a different way. 5. Use the recipe in the blue box. Model opening: Data means facts, numbers, words or observations that we collect. 8. Same 5-mark body + a four-row comparison table and one rewritten survey question + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Types of Data** using the recipe in this topic. Include a four-row comparison table and one rewritten survey question. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 6: Digital Literacy

Digital Literacy (2/4) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Prototypes, Observation and Simulations · Primary and Secondary Data · Designing a Data-Collection Approach · Presenting Data Using Digital Tools: 6.7.1 Spreadsheets and 6.7.2 Presentations

★ **Must lock:** Prototypes, Observation and Simulations, Primary and Secondary Data

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Is "time spent online in hours" qualitative or quantitative?
2. Name the five data-collection strategies.
3. Name two digital tools for creating surveys.

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

6.4.3 – 6.4.5 Prototypes, Observation and Simulations ★ Golden

A prototype is a simple early version of a product, model or solution.

- It allows an idea to be tested before time and effort are spent on the final version.
- Users try the prototype and give feedback, and this feedback becomes useful data for improvement.
- Example: Before building a full school app, you draw the screens on paper (or make a few clickable screens in a design tool) and ask five...

Remember: Watch it → Observation. Ask it → Survey/Interview. Try it → Prototype. Model it → Simulation.

Exam: Scenario questions are popular: "Which data-collection method is most suitable for ... ? Give a reason."
Always name the method and justify it using its key feature (e.g....

پروٹوٹائپ کسی پروڈکٹ یا حل کا ابتدائی سادہ نمونہ ہے جسے صارفین آزما کر رائے دیتے ہیں تاکہ اسے بہتر بنایا جا سکے۔

6.5 Primary and Secondary Data ★ Golden

Whenever you use data, it is important to know where it came from.

- The source affects how much control the researcher had over collecting it, and how confidently the results can be used.

Aspect	Primary Data	Secondary Data
Source	Collected directly by the researcher.	Already collected by others.
Examples	Survey responses, interviews, observations.	Reports, articles, websites, databases.
Purpose	Prepared for a specific study.	Reused for a new purpose.
Control	High control over questions and collection process.	Limited control because the data already exists.

Primary Data

Fresh, direct collection

(You do the work)

Surveys · Interviews · Observation

Secondary Data

Existing archives

(Someone else did it)

Books · Reports · Databases

Primary data is collected by you; secondary data was collected by others.

Remember: Primary = "1st hand" (you collect it first). Secondary = "2nd hand" (like a second-hand book – someone else used it first).

Exam: "Differentiate between primary and secondary data" is one of the most-asked questions in this chapter. Draw the four-row table: Source, Examples, Purpose, Control. Learn...

بنیادی ڈیٹا (Primary Data) محقق خود کسی خاص مقصد کے لیے براہ راست جمع کرتا ہے، مثلاً سروے، انٹرویو یا مشاہدہ۔

6.6 Designing a Data-Collection Approach

Before collecting data, a researcher must have a plan.

Question → Data type → Method → Sample → Tools → Collect → Store + backup → Ethics

- Define the research question clearly. What exactly do you want to find out?
- Decide the type of data needed: quantitative, qualitative, or both.
- Select suitable methods (survey, interview, observation).
- Choose the sample or target group – the people who will give you data.

Remember: Remember the steps with: "Quiet Teachers Make Students Think, Check, Save Ethically" → Question, Type, Method, Sample, Tools, Collect, Store, Ethics.

ڈیٹا جمع کرنے سے پہلے منصوبہ بندی ضروری ہے: تحقیقی سوال واضح کریں، ڈیٹا کی قسم طے کریں، مناسب طریقہ چنیں، نمونہ (Sample) منتخب کریں...

6.7 Presenting Data Using Digital Tools: 6.7.1 Spreadsheets and 6.7.2 Presentations

Collecting data is only one part of the work.

Spreadsheet data → Chart / Graph → Insert → Slides

- You must also present your findings so that others can understand them quickly and easily.
- Digital tools make this much easier.
- The four main tools are spreadsheets, presentations, infographics and reports.

Formula	What it does
=SUM(B2:B4)	Adds all values in B2 to B4.
=AVERAGE(B2:B4)	Mean: sum ÷ number of values.
=MAX(B2:B4) / =MIN(B2:B4)	Largest / smallest value.
=MEDIAN(B2:B4)	Middle value when sorted.
=COUNT(B2:B4)	Counts cells containing numbers.
=COUNTA(C2:C4)	Counts cells that are not empty (text or numbers).
=COUNTIF(C2:C4, "Yes")	Counts cells that match a condition.

Remember: Presentation rule of thumb: one idea per slide, and no more than about six short bullet points. If you are reading long sentences off the slide, the slide has...

ڈیٹا جمع کرنے کے بعد اسے پیش کرنا بھی ضروری ہے۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Prototypes, Observation and Simulations:** Mistake: Saying that observation involves asking people questions.
- **Primary and Secondary Data:** Mistake: Thinking data from a website is always primary because "I found it myself".
- **Designing a Data-Collection Approach:** Mistake: Starting to collect data before writing a clear research question.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Prototypes, Observation and Simulations**. Write: A prototype is a simple early version of a product, model or solution.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) It allows an idea to be tested before time and effort are spent on the final version. (2) Users try the prototype and give feedback, and this feedback becomes useful data for improvement. — + example.

5 marks: 3-mark structure + four-point explanation + the worked example from this topic.

8–10 marks: 5-mark structure + a one-page canvas: problem, user, riskiest assumption, test + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Primary and Secondary Data**. Write: Whenever you use data, it is important to know where it came from.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Whenever you use data, it is important to know where it came from. (2) The source affects how much control the researcher had over collecting it, and how confidently the results can be used. — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + a four-row comparison table and one rewritten survey question + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. Why would an engineer use a simulation instead of a real test?

3 marks. Explain Prototypes, Observation and Simulations with two clear points and one example.

5 marks. Scenario questions are popular: "Which data-collection method is most suitable for ... ? Give a reason." Always name the method and justify it using its key feature (e.g. "Simulation, because a real...").

8 marks. Prototypes, Observation and Simulations: 5-mark answer plus a one-page canvas: problem, user, riskiest assumption, test.

Answers: 1. Because the real experiment may be costly, risky or difficult 3. It allows an idea to be tested before time and effort are spent on the final version.; Users try the prototype and give feedback, and this feedback becomes useful data for improvement. 5. Use the recipe in the blue box. Model opening: A prototype is a simple early version of a product, model or solution. 8. Same 5-mark body + a one-page canvas: problem, user, riskiest assumption, test + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Prototypes, Observation and Simulations** using the recipe in this topic. Include a one-page canvas: problem, user, riskiest assumption, test. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 6: Digital Literacy

Digital Literacy (3/4) · Teach

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter · 10–35 min concept lock · 35–55 min board working · 55–75 min academy drill · 75–85 min error clinic · 85–90 min homework out.

Board today: Infographics and Reports · Case Study: Digital Inquiry Project · Steps 1 – 3: Advanced Search and Methodology · Steps 4 – 6: Survey Preparation and Primary Data Collection

★ **Must lock:** Case Study: Digital Inquiry Project, Steps 1 – 3: Advanced Search and Methodology, Steps 4 – 6: Survey Preparation and Primary Data Collection

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Two students redo the worked example on the board after you hide it.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Why would an engineer use a simulation instead of a real test?
2. Which type of data gives the researcher high control?
3. What is the first step in designing a data-collection approach?

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

6.7.3 – 6.7.4 Infographics and Reports

An infographic (information + graphic) is a visual way of presenting information, facts or data so that people can understand the main idea quickly and easily.

- Simplify complex concepts.
- Highlight the most important findings and results.
- Use clear visuals, limited text and accurate information.

Situation	Best tool	Why
You must calculate the average marks of 40 students.	Spreadsheet	Formulas calculate instantly.
You will explain results to the class in assembly.	Presentation	Designed for a live audience.
You want a one-page poster for the notice board.	Infographic	Main result understood at a glance.
The principal wants the full study with method and recommendations.	Report	Detailed, structured document.

Remember: Match tool to job: Spreadsheet = Calculate, Presentation = Tell, Infographic = Show quickly, Report = Explain fully.

انفوگرافک معلومات کو تصویری انداز میں پیش کرنے کا طریقہ ہے تاکہ اصل بات جلدی سمجھ آ جائے؛ اس کے لیے کینوا (Canva) اور پاور بی آئی...

6.8 Case Study: Digital Inquiry Project ★ Golden

This case study shows how to apply all the ideas of this chapter to solve a real-world problem from start to finish.

1 Question → 2 Search → 3 Collect → 4 Analyse → 5 Create

- An inquiry is an investigation to find the answer to a question.
- Research question: "How does daily mobile phone use affect the study habits of students?" Scenario: Students notice classmates using phones...
- Some say phones help them learn; others say phones waste time.

Workflow stage	Steps
Question & Search	1 Advanced searches · 2 Identify data types · 3 Select methods
Collect	4 Prepare the survey · 5 Choose the sample · 6 Collect primary data ethically · 7 Use secondary data
Analyse	8 Organise in a spreadsheet · 9 Create charts · 10 Analyse results
Create	11 Write the conclusion · 12 Recommendations → final digital artefact

Remember: Workflow in five words: Ask → Search → Collect → Analyse → Create. Every research project you ever do will follow this pattern.

Exam: Expect a long question: "Describe the steps of a digital inquiry project using the example of mobile phone use." Write the research question first, then the 12 steps...

اس کیس اسٹڈی میں تحقیقی سوال یہ ہے: "روزانہ موبائل فون کا استعمال طلبہ کی پڑھائی کی عادات پر کیسے اثر ڈالتا ہے؟"

6.8 Steps 1– 3: Advanced Search and Methodology ★ Golden

Students first search for background information using search engines (such as Google or Bing).

- They use advanced search operators – special symbols or words typed into the search box – to narrow down results and find reliable...
- Suggested queryOperator usedWhat it does "mobile phone use" "study habits" students" " (quotes)Finds the exact phrases "mobile phone use"...
- "academic performance" filetype:pdfFinds only PDF documents (often official reports and research papers).

Suggested query	Operator used	What it does
"mobile phone use" "study habits" students	" " (quotes)	Finds the exact phrases "mobile phone use" and "study habits", not the words scattered randomly.
"academic performance" filetype:pdf	filetype:	Finds only PDF documents (often official reports and research papers).
digital distraction site:.edu	site:	Searches only educational websites (addresses ending in .edu).

Remember: Three operators, three jobs: " " = exact, filetype: = format, site: = where. Note: there is no space after the colon (site:.edu, not site: .edu).

Exam: "Explain any three advanced search operators with examples" is a likely short question. For each, write the operator, an example query, and what it does. Also learn the...

پہلے قدم میں طلبہ سرچ انجن پر جدید سرچ آپریٹرز (Advanced Search Operators) استعمال کرتے ہیں: کوٹیشن مارکس (" ") بالکل وہی جملہ تلا...

6.8 Steps 4 – 6: Survey Preparation and Primary Data Collection ★ Golden

Design clear, unbiased questions. Each question is linked to the type of data it produces: QuestionType of data
How many hours do you use your phone daily?Quantitative Which activity do you use your phone for the most?
Quantitative...

- Design clear, unbiased questions.
- Each question is linked to the type of data it produces: QuestionType of data How many hours do you use your phone daily?Quantitative Which...

Question	Type of data
How many hours do you use your phone daily?	Quantitative
Which activity do you use your phone for the most?	Quantitative (categories)
How does your phone help you in studies?	Qualitative
Suggest one way to use phones responsibly.	Qualitative

Remember: Ethics in five words: Consent, No-Sensitive, Anonymous, Private, Educational – "C-NAPE". Think: "Cover the person's nape (neck) – protect them."

Exam: Questions often ask: "What ethical guidelines should be followed when collecting data?" List all five points. Use the words consent and anonymity – examiners look for...

چوتھے قدم میں واضح اور غیر جانبدار سوالات تیار کیے جاتے ہیں۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Infographics and Reports:** Mistake: Making a chart look more dramatic than the data – for example, starting the bar-chart axis at 18 instead of 0 so that 20 looks...
- **Steps 1–3: Advanced Search and Methodology:** Mistake: Assuming the first search result is the most reliable.
- **Steps 4–6: Survey Preparation and Primary Data Collection:** Mistake: Thinking a bigger sample is always enough.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Case Study: Digital Inquiry Project**. Write: This case study shows how to apply all the ideas of this chapter to solve a real-world problem from start to finish.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) An inquiry is an investigation to find the answer to a question. (2) Research question: "How does daily mobile phone use affect the study habits of students?" Scenario: Students notice classmates using phones... — + example.

5 marks: 3-mark structure + table: Workflow stage / Steps + the worked example from this topic.

8–10 marks: 5-mark structure + the full comparison table with every cell filled + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Steps 1–3: Advanced Search and Methodology**. Write: Students first search for background information using search engines (such as Google or Bing).

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) They use advanced search operators – special symbols or words typed into the search box – to narrow down results and find reliable... (2) Suggested query/Operator used/What it does "mobile phone use" "study habits" students " (quotes) Finds the exact phrases "mobile phone use"... — + example.

5 marks: 3-mark structure + table: Suggested query / Operator used / What it does + the worked example from this topic.

8–10 marks: 5-mark structure + a four-row comparison table and one rewritten survey question + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Steps 4–6: Survey Preparation and Primary Data Collection**. Write: Design clear, unbiased questions.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Design clear, unbiased questions. (2) Each question is linked to the type of data it produces: Question/Type of data How many hours do you use your phone daily? Quantitative Which... — + example.

5 marks: 3-mark structure + table: Question / Type of data + the worked example from this topic.

8–10 marks: 5-mark structure + a four-row comparison table and one rewritten survey question + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. What is the research question in the case study?

3 marks. Explain Case Study: Digital Inquiry Project with two clear points and one example.

5 marks. Expect a long question: "Describe the steps of a digital inquiry project using the example of mobile phone use." Write the research question first, then the 12 steps with one or two lines each, and...

8 marks. Case Study: Digital Inquiry Project: 5-mark answer plus the full comparison table with every cell filled.

Answers: 1. "How does daily mobile phone use affect the study habits of students?" 3. An inquiry is an investigation to find the answer to a question.; Research question: "How does daily mobile phone use affect the study habits of students?" Scenario: Students notice classmates using phones... 5. Use the recipe in the blue box. Model opening: This case study shows how to apply all the ideas of this chapter to solve a real-world problem from start to finish. 8. Same 5-mark body + the full comparison table with every cell filled + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Case Study: Digital Inquiry Project** using the recipe in this topic. Include the full comparison table with every cell filled. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Chapter 6: Digital Literacy

Digital Literacy (4/4) · Chapter exam

TODAY'S 90 MINUTES

90 minutes. 0–10 min recap yesterday ★ on the board · 10–85 min sit the 75-mark chapter exam (1 h 15 min, hall conditions) · 85–90 min collect scripts. Do not mark in front of the batch.

Board today: Steps 7 – 9: Secondary Data and Spreadsheet Organisation · Steps 10 – 12: Data Analysis, Conclusion and Recommendations · The Final Digital Artefact · Careers in Digital Research

★ **Must lock:** Steps 7 – 9: Secondary Data and Spreadsheet Organisation, Steps 10 – 12: Data Analysis, Conclusion and Recommendations, The Final Digital Artefact

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Hall paper today. Recap the cards for ten minutes, then scripts out. Re-teach any Golden below 60% in tomorrow's starter.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. What is the research question in the case study?
2. Write a query that finds the exact phrase "online learning" only on educational websites.
3. What is a sample?

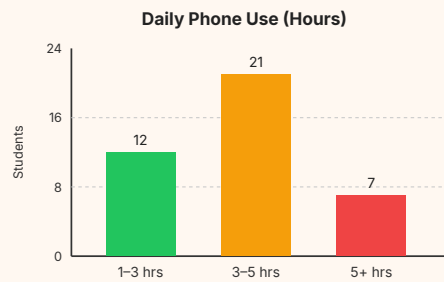
کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

6.8 Steps 7 – 9: Secondary Data and Spreadsheet Organisation ★ Golden

Students include a reliable online article stating that "excessive screen time lowers concentration." They will compare this secondary data with their own primary survey results.

- If both agree, the conclusion is stronger.
- If they disagree, the students must think about why (for example, a different age group or country).

Student	Hours	Study Use	Main Activity	Distracted?
1	3	Yes	Lectures	Sometimes
2	5	No	Social Media	Yes
3	2	Yes	Notes	No



Bar chart of daily phone use for 40 students. The counts (12, 21, 7) are estimated from the bar heights in the source chart.

Remember: Chart chooser: Bar = Compare, Pie = Parts of a whole, Line = Change over time.

Exam: You may get the spreadsheet table and be asked to "write formulas to find the total and average hours" or "count students who use the phone for study". Write the full...

ساتویں قدم میں ایک قابل اعتماد آن لائن مضمون کا ثانوی ڈیٹا ("زیادہ اسکرین ٹائم توجہ کم کرتا ہے") اپنے سروے کے بنیادی ڈیٹا سے مواز...

6.8 Steps 10 – 12: Data Analysis, Conclusion and Recommendations ★ Golden

To analyse means to study the data carefully to find patterns and trends.

- Most students (the majority) use phones for 3–5 hours daily.
- Half use it for educational purposes (lectures/notes).
- A clear trend: students who use phones mostly for entertainment report much higher distraction levels than those who use them for study.

Remember: Conclusion formula: "The data shows ... However ... Therefore ..." – state the main finding, mention the other side, then give the overall answer.

Exam: You may be asked to "write a conclusion and two recommendations" for given data. Use the "shows ... however ... therefore" pattern and make sure each recommendation follows...

دسویں قدم میں چارٹس اور اسپریڈشیٹ سے رجحانات تلاش کیے جاتے ہیں: زیادہ تر طلبہ روزانہ تین سے پانچ گھنٹے فون استعمال کرتے ہیں،
نصف ا...

6.8 The Final Digital Artefact ★ Golden

A digital artefact is a final product created using digital technologies.

- Students must create one final artefact that answers the original research question.
- It is not just a decorated file; it must present evidence and conclusions.
- Artefact option Suitable digital tools Main purpose
Digital Report MS Word, Google Docs Explain the full study in a structured text document.

Artefact option	Suitable digital tools	Main purpose
Digital Report	MS Word, Google Docs	Explain the full study in a structured text document.
Presentation	PowerPoint, Canva, Slides	Share key findings with a live audience.
Infographic	Canva, Adobe Express	Show the main results visually and quickly on one page.
Dashboard	Power BI, Tableau	Explore data through interactive charts and filters.

Remember: The structure follows the inquiry itself: What (title, question, intro) → How (method) → Found (primary, secondary, charts) → So what (conclusion,...)

Exam: Learn the four artefact options with tools and purpose (a table question), and the ten-part structure in order. "What is a digital artefact?" – stress that it must...

ڈیجیٹل آرٹیفیکٹ (Digital Artefact) ڈیجیٹل ٹیکنالوجی سے بنائی گئی آخری پروڈکٹ ہے جو تحقیقی سوال کا جواب ثبوت اور نتائج کے ساتھ پیش...

Careers in Digital Research

The skills in this chapter lead to real jobs: CareerWhat they doChapter skill used Market Research AnalystDesigns surveys to collect primary data to guide business decisions.Surveys, primary data UX (User Experience)...

Career	What they do	Chapter skill used
Market Research Analyst	Designs surveys to collect primary data to guide business decisions.	Surveys, primary data
UX (User Experience) Researcher	Gathers qualitative and quantitative data to improve app and website designs.	Prototypes, interviews, observation
Data Journalist	Uses advanced search skills to research stories and presents them via infographics.	Advanced search, infographics
Digital Marketing Specialist	Collects audience data and presents results through dashboards and reports.	Dashboards, reports

Remember: Link each career to one tool: Market Researcher → Survey; UX Researcher → Prototype; Data Journalist → Search + Infographic; Digital Marketer → Dashboard.

اس باب کی مہارتوں سے کئی کیریئر بنتے ہیں: مارکیٹ ریسرچ اینالسٹ کاروباری فیصلوں کے لیے سروے بناتا ہے، یو ایکس (UX) ریسرچر ایپس اور...

Digital Literacy

17 single-page lessons · 11 Golden · read, close, say aloud, then sit the exam

Chapter in one look

- **Digital literacy** is the ability to understand and use digital devices, the internet and online tools safely, effectively and responsibly – not just using devices, but finding, evaluating and creating information.
- In the **Information Age**, access to technology is not enough; digital literacy needs technical, cognitive and social abilities.
- **Qualitative data** is descriptive and non-numerical (opinions, reasons, behaviour) and explains why/how.
- **Quantitative data** is numerical, countable or measurable (hours, scores, percentages) and is used for comparing and charting.
- The five **data-collection strategies** are interviews, surveys, prototypes, observation and simulations.
- **Interviews** are planned conversations giving mostly qualitative data; **surveys** ask the same questions to many people, on paper or with Google/Microsoft Forms.
- Good survey questions are clear, short, unbiased and ask about one thing; avoid leading, double-barrelled, vague and overlapping questions.
- A **prototype** is an early version tested for feedback; **observation** records what happens without asking questions; a **simulation** is a computer model used when real experiments are costly, risky or difficult.
- **Primary data** is collected first-hand (high control); **secondary data** was collected by others and reused (limited control).
- Designing a data-collection approach has eight steps: question, data type, method, sample, tools, collect, store with backup, and ethics (consent is mandatory).
- **Spreadsheets** organise and calculate (e.g. `=SUM()`, `=AVERAGE()`, `=COUNTIF()`); **presentations** deliver to an audience; **infographics** give quick visual summaries; **reports** give detailed structure.
- Visuals must always be accurate and based strictly on real data, never exaggerated.
- **Advanced search operators** such as `" "`, `filetype:pdf` and `site:.edu` help find reliable sources efficiently; sources must be evaluated for author, date, reliability and bias.
- **Ethics**: get consent, avoid sensitive information, keep anonymity, keep responses private, and use data only for educational purposes.
- The **digital inquiry project** follows Question → Search → Collect → Analyse → Create in 12 steps.
- A **digital artefact** (report, presentation, infographic or dashboard) answers the research question using evidence and visuals, following a ten-part structure ending in references.

Key terms

Term	Meaning	اردو
Digital literacy	Ability to understand and use digital devices, the internet and online tools safely, effectively and responsibly.	ڈیجیٹل خواندگی
Information Age	Period in which vast amounts of data are produced, shared and accessed instantly.	معلوماتی دور
Data	Facts, numbers, words or observations collected for study.	مواد / ڈیٹا
Qualitative data	Descriptive, non-numerical data about thoughts, feelings and experiences.	کیفیتی ڈیٹا
Quantitative data	Numerical data that can be counted or measured.	مقداری ڈیٹا
Interview	Planned conversation to gather information directly from a person or small group.	انٹرویو / بالمشافہ گفتگو
Survey / Questionnaire	Same set of questions asked to many people.	سروے / سوالنامہ
Prototype	Simple early version of a product used to test an idea and get feedback.	ابتدائی نمونہ
Observation	Watching and recording what happens without asking questions.	مشاہدہ
Simulation	Computer-based model of a real situation in which variables can be changed.	نقلی ماڈل (سمیولیشن)
Variable	A factor that can be changed or measured.	متغیر
Primary data	Data collected first-hand by the researcher for a specific purpose.	بنیادی ڈیٹا

Digital Literacy

75 marks • 1 hour 15 minutes • no notes • no phones

CHAPTER EXAM RULES

Instructions. Sit this exam after Lesson 17 and the recap. **Section A** — all 15 MCQs (1 mark). **Section B** — all 10 short questions (3 marks). **Section C** — attempt any **three** long questions (10 marks). At least one Section C answer must be a ★ Golden topic. Write in your notebook. Do not open the key until time is up.

Marks. $15 + 30 + 30 = 75$. A 5-mark / 10-mark answer needs Define • Explain • Example • Diagram • Working.

سبق اور خلاصے کے بعد یہ امتحان بغیر نوٹس کے حل کریں۔ وقت پورا ہونے پر کلید کھولیں۔

Section A — Multiple choice (15 × 1 = 15)

1. Digital literacy is best described as the ability to:
 - a. type quickly on a keyboard
 - b. understand and use digital devices, the internet and online tools safely, effectively and responsibly
 - c. repair computer hardware
 - d. write computer programs
2. Which of the following is qualitative data?
 - a. Marks obtained in a test
 - b. Number of hours spent online
 - c. Students' opinions about online classes
 - d. Percentage of students using the internet
3. Which method is most suitable when the same questions must be asked to many people?
 - a. Survey
 - b. Interview
 - c. Prototype
 - d. Simulation
4. Simulations are especially useful when:
 - a. you want people's opinions
 - b. the data already exists in a report
 - c. you want to watch people without asking questions
 - d. real experiments are costly, risky or difficult
5. To find out how many cars cross an intersection at 8 AM, the best method is:
 - a. Interview
 - b. Observation
 - c. Survey
 - d. Prototype
6. Which of these is primary data?
 - a. Responses to a Google Form you created for your project
 - b. An official government report
 - c. A published research article
 - d. Web statistics from a website
7. A researcher has limited control over:
 - a. their own interviews
 - b. their own survey questions
 - c. their own observations
 - d. secondary data
8. The first step in designing a data-collection approach is to:
 - a. collect the data
 - b. choose the sample
 - c. define the research question clearly
 - d. store the data with a backup
9. Which spreadsheet formula finds the mean of cells B2 to B6?
 - a. =AVERAGE(B2:B6)
 - b. =SUM(B2:B6)
 - c. =COUNT(B2:B6)
 - d. =MAX(B2:B6)
10. The search query `digital distraction site:.edu` will:
 - a. find only PDF files
 - b. search only educational websites
 - c. find the exact phrase "digital distraction" only
 - d. exclude educational websites
11. The operator `filetype:pdf` is used to:
 - a. search for an exact phrase
 - b. search only government websites
 - c. convert a web page into PDF
 - d. find only PDF documents
12. "Do not record names unless absolutely necessary" describes the principle of:
 - a. consent
 - b. accuracy
 - c. anonymity
 - d. sampling
13. In the case study, how many Class XI students were surveyed?
 - a. 20
 - b. 40
 - c. 50
 - d. 100
14. Which tools are suggested for a dashboard artefact?
 - a. Power BI, Tableau
 - b. MS Word, Google Docs
 - c. PowerPoint, Google Slides
 - d. Google Forms, Microsoft Forms
15. A prototype is:
 - a. the final finished product
 - b. a published report
 - c. a type of survey question
 - d. a simple early version of a product used to test an idea

Section B — Short questions (10 × 3 = 30)

1. Define digital literacy. Name the three kinds of ability it involves.
2. Why is simply having access to technology not enough in the Information Age?
3. Differentiate between qualitative and quantitative data, giving two examples of each.
4. How is an interview different from a survey?
5. What is a prototype, and how does it produce useful data?
6. When is a simulation used? Give one example.
7. Compare primary and secondary data in terms of purpose and control.
8. List any four ethical guidelines to follow when collecting primary data.
9. Explain three advanced search operators with an example of each.
10. What is a digital artefact? Name its four options with one suitable tool each.

Section C — Long questions (attempt any 3 × 10 = 30)

1. Explain the eight steps of designing a data-collection approach. Apply each step to the research question: "How much water is wasted at the college taps during break time, and why?"
2. **Design a survey for** finding out how Class XI students in your college travel to college and how long the journey takes. Write a consent statement and at least six questions (mixing closed, number, rating and open-ended types), state the data type of each question, describe your sample, and explain how you will keep the data ethical.
3. Ten students recorded their daily phone use (hours) in cells B2 to B11: 2, 4, 3, 5, 6, 4, 3, 4, 1, 4. (a) Write the spreadsheet formulas and give the results for the total, mean, median, mode, maximum and minimum. (b) Write a formula to count students using the phone for more than 4 hours, and find the percentage. (c) Group the data into "1–2 hrs", "3–4 hrs" and "5+ hrs" and name a suitable chart.
4. Describe all 12 steps of the Digital Inquiry Project on the question "How does daily mobile phone use affect the study habits of students?", including the final digital artefact and its structure.
5. **Design a survey for** investigating "How do Class XI students use AI tools for homework?" Then explain which secondary sources you would search for (with advanced search queries), and which artefact you would create to present your findings to the principal, with reasons.

Answer key. Finish the Chapter Exam first, then mark. Do not peek while the clock is running.

پہلے امتحان مکمل کریں، پھر اس کلید سے نمبر لگائیں۔

Check yourself — answers

- 6.1 Introduction to Digital Literacy.** Answers: 1. The ability to understand and use digital devices, the internet and online tools in a safe, effective and responsible manner. 2. Find information, communicate, produce digital content, and solve problems through technology.
- 6.2 Digital Literacy in the Information Age.** Answers: 1. A period in which vast amounts of data are produced, shared and accessed instantly through the internet, mobile devices and digital platforms. 2. Technical, cognitive and social abilities.
- 6.3 Types of Data.** Answers: 1. Quantitative (measurable number). 2. Qualitative. 3. Qualitative (an opinion in words).
- 6.4 Data-Collection Strategies.** Answers: 1. Interviews, surveys, prototypes, observation, simulations. 2. New information that you gather yourself, rather than information that already exists.
- 6.4.1 – 6.4.2 Interviews and Surveys.** Answers: 1. Google Forms and Microsoft Forms. 2. It is double-barrelled (two subjects in one question); ask about each subject separately. 3. Usually qualitative.
- 6.4.3 – 6.4.5 Prototypes, Observation and Simulations.** Answers: 1. Because the real experiment may be costly, risky or difficult. 2. Yes – for example, counting the number of cars passing a point. 3. The feedback users give after trying the prototype.
- 6.5 Primary and Secondary Data.** Answers: 1. Primary data. 2. Secondary (someone else collected it). 3. Any two: interviews, surveys, observations, experiments.
- 6.6 Designing a Data-Collection Approach.** Answers: 1. Define the research question clearly. 2. Step 8 – follow ethical practice and respect privacy. 3. Digitally, in a spreadsheet, with a backup copy.
- 6.7 Presenting Data Using Digital Tools: 6.7.1 Spreadsheets and 6.7.2 Presentations.** Answers: 1. `=AVERAGE(D2:D20)`. 2. MS Excel, Google Sheets, Apple Numbers. 3. Any two: start with a clear title and research question; use short bullet points; include tables/charts/key findings; keep design simple and readable.
- 6.7.3 – 6.7.4 Infographics and Reports.** Answers: 1. Any two: Canva, Power BI, Tableau. 2. Review it carefully for accuracy. 3. Conclusions and recommendations based on the data.
- 6.8 Case Study: Digital Inquiry Project.** Answers: 1. "How does daily mobile phone use affect the study habits of students?" 2. Question, Search, Collect, Analyse, Create.
- 6.8 Steps 1 – 3: Advanced Search and Methodology.** Answers: 1. "online learning" site:.edu. 2. Because the same questions had to be asked to many students, and a survey is the most efficient way to do that. 3. Quantitative: hours used daily (or % who feel distracted). Qualitative: opinions on why they get distracted (or suggestions for better use).
- 6.8 Steps 4 – 6: Survey Preparation and Primary Data Collection.** Answers: 1. The group of people who provide the data. 2. Not recording the names of participants (unless absolutely necessary). 3. Qualitative.
- 6.8 Steps 7 – 9: Secondary Data and Spreadsheet Organisation.** Answers: 1. A reliable online article stating "excessive screen time lowers concentration". 2. `=COUNTIF(C2:C41, "Yes")`. 3. A bar (column) chart.
- 6.8 Steps 10 – 12: Data Analysis, Conclusion and Recommendations.** Answers: 1. Students who use phones mostly for entertainment. 2. No – they should not be banned but managed wisely. 3. Any two: turn off social media notifications while studying; set daily screen-time limits; take physical breaks from screens.
- 6.8 The Final Digital Artefact.** Answers: 1. A dashboard (Power BI, Tableau). 2. References. 3. Canva and Adobe Express.
- Careers in Digital Research.** Answers: 1. Market Research Analyst. 2. User Experience.

Answer key

Multiple choice

1. b
2. c
3. a
4. d
5. b
6. a
7. d
8. c
9. a
10. b
11. d
12. c
13. b
14. a
15. d

Short questions — model answers

1. Digital literacy is the ability to understand and use digital devices, the internet and online tools in a safe, effective and responsible manner. It involves technical, cognitive and social abilities.
2. In the Information Age vast amounts of data are produced and shared instantly. Owning a device does not mean you can judge or use this information well. Users must develop skills to understand, manage and use information wisely, and to tell reliable information from unreliable.
3. Qualitative data is descriptive and non-numerical; it explains why and how (e.g. students' opinions about online classes; observation notes). Quantitative data is numerical, countable or measurable; it is used for comparing and making charts (e.g. marks in a test; hours spent online).
4. An interview is a planned conversation with one person or a small group, giving mostly qualitative, in-depth answers. A survey asks the same set of questions to many people (on paper or via Google/Microsoft Forms) and is efficient for large groups; online surveys also collect responses and make charts automatically.
5. A prototype is a simple early version of a product, model or solution. Users try it and give feedback; this feedback is data used to improve the product before time and effort are spent on the final version.
6. A simulation is used when real experiments are costly, risky or difficult. Example: a flight simulator to train pilots, or a traffic simulation to test the effect of closing a road.
7. Purpose: primary data is prepared for a specific study; secondary data is reused for a new purpose. Control: the researcher has high control over primary data (questions and process) and limited control over secondary data because it already exists.
8. Any four: ask permission (consent); avoid sensitive personal information; do not record names unless necessary (anonymity); keep responses private; use data only for educational purposes.
9. Quotes " " find an exact phrase, e.g. "study habits" students. filetype: finds a file format, e.g. "academic performance" filetype:pdf gives only PDFs. site: limits to a type of website, e.g. digital distraction site:.edu searches only educational sites.
10. A digital artefact is a final product created with digital technology that answers the research question with evidence and conclusions. Options: digital report (MS Word/Google Docs), presentation (PowerPoint/Canva/Slides), infographic (Canva/Adobe Express), dashboard (Power BI/Tableau).

Long questions — model answers

- Eight steps applied to water wastage.** (1) **Define the question:** "How much water is wasted at the college taps during break time, and why?" (2) **Data type:** both – quantitative (number of taps left running, minutes running) and qualitative (students' reasons). (3) **Methods:** observation to count running taps; a short survey to learn habits and reasons. (4) **Sample:** all 12 taps observed over 5 break times; 30 students from different classes surveyed. (5) **Tools:** an observation sheet (Tap no., Time, Left running Y/N, Minutes) and a Google Form. (6) **Collect** carefully, at the same time each day, recording accurately. (7) **Store** data in Google Sheets and keep a backup copy (download or USB). (8) **Ethics:** get the principal's permission; the survey is voluntary and anonymous with a consent question; data used only for the project.
- Survey on travel to college (model).** *Consent:* "This anonymous survey is for a Class XI project on travel to college. Do not write your name. Answers will be kept private and used only for this project. Do you agree to take part? Yes / No." *Questions:* Q1 How do you usually travel to college? Walk / Bus / Rickshaw / Motorbike / Car / Other – quantitative (categories). Q2 How many minutes does your journey usually take? ____ – quantitative. Q3 How many days a week are you late because of travel? 0 / 1 / 2 / 3 or more – quantitative. Q4 "My journey to college is safe." 1 (strongly disagree) to 5 (strongly agree) – quantitative (rating). Q5 Do you travel alone? Yes / No – quantitative (count). Q6 What is the biggest problem you face while travelling? – qualitative. Q7 Suggest one way the college could make travel easier. – qualitative. Questions are unbiased, ask one thing each, and options do not overlap. *Sample:* about 40 Class XI students, roughly 10 from each section, to be representative. *Ethics:* consent, no sensitive information (no home address), no names (anonymity), private storage, educational use only. Data is collected through Google Forms and exported to Google Sheets for analysis.
- Spreadsheet problem.** (a) $\text{=SUM(B2:B11)} \rightarrow 2+4+3+5+6+4+3+4+1+4 = 36$. $\text{=AVERAGE(B2:B11)} \rightarrow 36 \div 10 = 3.6$ hours. Sorted: 1, 2, 3, 3, 4, 4, 4, 4, 5, 6; $\text{=MEDIAN(B2:B11)} \rightarrow (4 + 4) \div 2 = 4$. $\text{=MODE(B2:B11)} \rightarrow 4$ (appears 4 times). $\text{=MAX(B2:B11)} \rightarrow 6$; $\text{=MIN(B2:B11)} \rightarrow 1$. (b) $\text{=COUNTIF(B2:B11, ">4")}$ → values 5 and 6 → 2 students; percentage $\text{=COUNTIF(B2:B11, ">4")}/\text{COUNT(B2:B11)} \rightarrow 2 \div 10 = 20\%$. (c) 1–2 hrs: 2 students (2, 1); 3–4 hrs: 6 students (4, 3, 4, 3, 4, 4); 5+ hrs: 2 students (5, 6). Check: $2 + 6 + 2 = 10 \checkmark$. A **bar/column chart** is suitable for comparing the groups (or a pie chart to show 20%, 60%, 20%).
- Twelve steps.** (1) Advanced searches with operators such as "mobile phone use" "study habits" students, "academic performance" filetype:pdf, digital distraction site:.edu; evaluate sources (author, recent, reliable site, facts/bias). (2) Identify data types: quantitative (hours daily, % distracted) and qualitative (reasons, suggestions). (3) Select method: survey via Google Forms/paper, because the same questions go to many students. (4) Prepare the survey: hours daily (quantitative), main activity (quantitative categories), how the phone helps (qualitative), one responsible-use suggestion (qualitative). (5) Sample: 40 Class XI students from different sections. (6) Collect ethically: consent, no sensitive data, anonymity, privacy, educational use only. (7) Secondary data: an article saying excessive screen time lowers concentration, to compare with results. (8) Organise in Excel/Google Sheets (Student, Hours, Study Use, Main Activity, Distracted?), sort, total, average. (9) Charts: bar chart of daily phone use (1–3, 3–5, 5+ hrs). (10) Analyse: majority use 3–5 hours; half use for education; entertainment users report more distraction. (11) Conclusion: phones have positive and negative effects; should be managed wisely, not banned. (12) Recommendations: turn off social media notifications, set screen-time limits, take physical breaks. **Artefact:** one of report, presentation, infographic or dashboard, structured as Title, Research Question, Introduction, Method, Primary Findings, Secondary Summary, Charts, Conclusion, Recommendations, References.
- AI tools for homework (model).** *Survey:* consent statement (anonymous, voluntary, educational use only). Q1 Have you used an AI tool (e.g. ChatGPT, Gemini) for homework? Yes / No – quantitative. Q2 How many times a week? 0 / 1–2 / 3–5 / 6 or more – quantitative (non-overlapping categories). Q3 What do you use it for most? Explaining topics / Checking answers / Writing assignments / Translation / Other – quantitative (categories). Q4 "AI tools help me understand my subjects better." 1–5 rating – quantitative. Q5 Do you check AI answers against your textbook? Always / Sometimes / Never – quantitative. Q6 Describe one problem you have faced with AI answers. – qualitative. *Sample:* about 40 Class XI students from all sections. *Secondary sources:* "generative AI" students homework filetype:pdf; "AI in education" site:.edu; "artificial intelligence" policy site:.gov.pk; evaluate each for author, date, reliability and bias. *Artefact:* a **digital report** (MS Word/Google Docs) because the principal needs the full study – method, findings, charts, conclusion, recommendations and references – in a structured document; a one-page infographic could be attached as a quick summary. Charts would be made in Google Sheets from the form responses, and every visual must match the real data.

Self-assessment: I can...

- ☐ define digital literacy and explain why it matters in the Information Age.
- ☐ name and explain the five data-collection strategies and choose the right one for a scenario.
- ☐ compare primary and secondary data by source, examples, purpose and control.
- ☐ follow ethical rules: consent, anonymity, privacy and educational use.
- ☐ use advanced search operators and evaluate whether a source is reliable.
- ☐ describe the 12 steps of the digital inquiry project.
- ☐ classify data as qualitative or quantitative and explain the difference in a table.
- ☐ write clear, unbiased survey questions and fix leading, double-barrelled or vague ones.
- ☐ list the eight steps for designing a data-collection approach.
- ☐ use spreadsheet formulas such as `=SUM()` , `=AVERAGE()` and `=COUNTIF()` and calculate results correctly.
- ☐ choose a suitable tool (spreadsheet, presentation, infographic, report, dashboard) to present findings.
- ☐ plan a digital artefact using the ten-part standard structure.

Golden blitz — Chapters 1–2

★ only · Chapters 1–2 · Golden blitz

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter from yesterday's Goldens · 10–35 min lock the tables from memory · 35–55 min two students write a 5-mark Golden answer on the board · 55–75 min timed 8-mark · 75–85 min error clinic · 85–90 min homework: redraw from memory.

Board today: Analog and Digital Signals · Logic Gates (Basic Gates: AND, OR, NOT) · Logic Diagrams · Karnaugh Maps (K-Maps) · Software Development Life Cycle (SDLC) · The Waterfall Model · The Agile Model · Communication Models and the OSI 7-Layer Model

★ **Must lock:** Analog and Digital Signals, Logic Gates (Basic Gates: AND, OR, NOT), Logic Diagrams, Karnaugh Maps (K-Maps), Software Development Life Cycle (SDLC), The Waterfall Model, The Agile Model, Communication Models and the OSI 7-Layer Model, Algorithm, Algorithm vs Pseudocode, Decomposition, Sorting Algorithms: Bubble Sort, Sorting Algorithms: Selection Sort, Searching Algorithms: Binary Search

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Write every table and diagram on this day from memory on the board, then check the cards.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. What is the secondary data used in Step 7?
2. According to Step 10, which group reported higher distraction?
3. Which artefact lets viewers explore data with interactive filters?

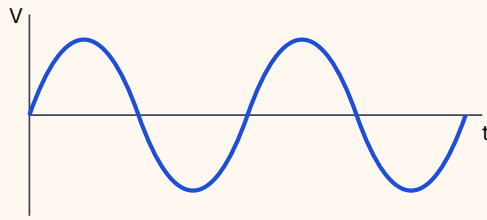
کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

1.1.3 Analog and Digital Signals ★ Golden

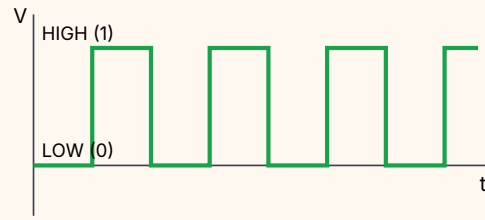
A signal is a changing voltage (or other quantity) that carries information.

- Analog signals are continuous signals that change smoothly over time, like a sine wave.
- Examples: the human voice, a traditional mercury thermometer.
- Digital signals have only two levels: HIGH (1) and LOW (0).

Feature	Analog data	Digital data
Nature	Continuous flow.	Discrete, separate values (bits).
Wave shape	Sine wave (smooth curve).	Square wave (sudden jumps).
Reliability	Low (affected by noise).	High (resistant to noise).
Examples	Human voice, traditional thermometer.	Computer data.



Analog Signal (Sine Wave)



Digital Signal (Square Wave)

An analog signal changes smoothly; a digital signal jumps between only two levels.

Remember: Analog = A smooth curve. Digital = Doorsteps. Analog looks like ocean waves; digital looks like a castle wall (square teeth).

Exam: Common questions: "Differentiate between analog and digital signals" (3–4 marks) and "Draw an analog and a digital signal." Write the comparison as a table with at least...

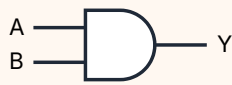
اینا لگ سگنل وقت کے ساتھ ہموار انداز میں بدلتا ہے، جیسے سائن ویو (Sine Wave)؛ انسانی آواز اس کی مثال ہے۔

1.1.7 Logic Gates (Basic Gates: AND, OR, NOT) ★ Golden

Logic gates are electronic circuits that perform Boolean operations.

- They take one or more input bits and produce one output bit.
- They are the building blocks of all digital systems — a processor contains billions of them.
- The three basic gates match the three basic operations: the AND gate, the OR gate and the NOT gate (also called an inverter).

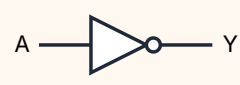
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



$$\text{AND: } Y = A \cdot B$$



$$\text{OR: } Y = A + B$$



$$\text{NOT: } Y = A'$$

Symbols of the basic gates. AND has a flat back and round front ("D" shape); OR has a curved back and pointed front; NOT is a triangle with a small circle (bubble).

Remember: Shortcut: AND output is 0 if any input is 0. OR output is 1 if any input is 1. Learn just the "special row": AND is 1 only at 11; OR is 0 only at 00.

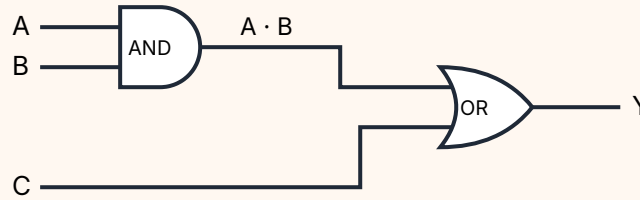
Exam: You are often asked to "draw the symbol, write the expression and truth table" of a gate (2–3 marks each). Draw neat symbols with labelled inputs A, B and output Y....

لاجک گیٹس الیکٹرانک سرکٹ ہیں جو بولین آپریشن انجام دیتے ہیں اور تمام ڈیجیٹل نظاموں کی بنیادی اینٹیں ہیں۔

1.1.11 Logic Diagrams ★ Golden

A logic diagram shows how logic gates are connected to create a circuit.

- It is a "picture" of a Boolean expression.
- To draw one, use the order of precedence: the operation with the highest priority is drawn first (nearest the inputs), and the last...



$Y = A \cdot B + C$: AND first, then OR.

Remember: Read the expression and ask "What is done last?" That gate goes on the far right, next to Y. Work backwards to the inputs. Count gates: $Y = A \cdot B + C$ needs 2...

Exam: Typical question: "Draw the logic diagram of $Y = \dots$ " and sometimes "also make its truth table" (4–5 marks). Write the precedence steps in one line, draw neat gate symbols...

لاچک ڈایاگرام دکھاتا ہے کہ گیٹس آپس میں کیسے جڑ کر سرکٹ بناتے ہیں۔

1.1.13 Karnaugh Maps (K-Maps) ★ Golden

A Karnaugh map (K-map) is a graphical tool used to simplify Boolean expressions.

- It is a grid of cells — one cell for each row of the truth table.
- The cells are arranged so that neighbouring (adjacent) cells differ by only ONE bit.
- This ordering is called Gray code.

	B = 0	B = 1
A = 0	Cell 0 ($A'B'$)	Cell 1 ($A'B$)
A = 1	Cell 2 (AB')	Cell 3 (AB)

Remember: Gray code song: "zero-zero, zero-one, one-one, one-zero" — only one bit flips at each step.

Imagine the K-map is drawn on a tube, so the left and right edges...

Exam: K-maps are a favourite long question (5–6 marks): "Simplify $F(A,B,C) = \sum m(\dots)$ using a K-map." Draw the map with labelled Gray-code headings, fill every cell (write the 0s...

کارنو میپ (K-Map) بولین ایکسپریشن کو آسان بنانے کا تصویری طریقہ ہے۔

1.2.1 – 1.2.3 Software Development Life Cycle (SDLC) ★ Golden

The Software Development Life Cycle (SDLC) is a structured process that guides software development from beginning to end.

1. Requirement Analysis → 2. System Design → 3. Implementation → 4. Testing → 5. Deployment → 6. Maintenance

- It ensures clear goals, better planning, risk reduction and high quality.
- Without it, teams forget features, go over budget and deliver buggy software.
- SDLC has 6 phases: 1.

Phase	Activities and details	Deliverable (what it produces)
1. Requirement Analysis	Meet clients, understand their needs, list the features. This is the most important phase — mistakes here spread to every later phase.	SRS Document (Software Requirements Specification)
2. System Design	Create the High-Level Design (overall architecture), the Low-Level Design (database, UI screens) and flowcharts.	Design Document
3. Implementation	Write the actual code in programming languages (Python, Java, etc.) based on the design documents.	Working Source Code

Remember: Remember the order with "Real Developers Implement, Test, Deploy, Maintain" (R-D-I-T-D-M). Black box = a closed box (you can't see inside). White box = a glass...

Exam: "Explain the phases of SDLC" is a very common long question (5–8 marks). Name all 6 phases in order, write 1–2 lines of activities for each, and state the deliverable of...

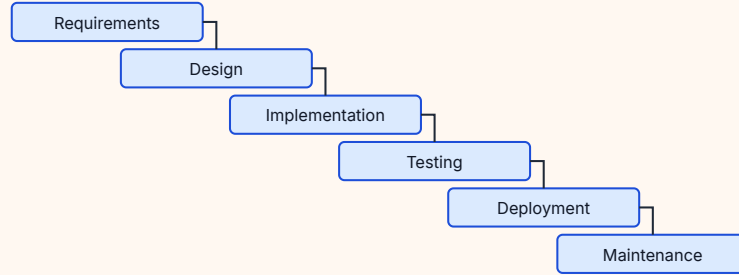
ایس ڈی ایل سی (SDLC) سافٹ ویئر بنانے کا منظم طریقہ ہے جس کے چھ مراحل ہیں: ضروریات کا تجزیہ، سسٹم ڈیزائن، کوڈنگ (Implementation)، ...

1.2.4 The Waterfall Model ★ Golden

The Waterfall model is a linear, sequential approach to the SDLC.

- Each phase must be completed before the next one begins, like water falling down steps.
- There is no easy way to go back to a previous phase.

Advantages (Pros)	Disadvantages (Cons)
Simple to understand and manage.	Not flexible to changes.
Well documented at every step.	Testing happens very late in the process.



The Waterfall model: each phase "falls" into the next, in one direction only.

Remember: Waterfall = water only flows downwards. Once water has fallen it cannot climb back up — once a phase is finished you do not go back.

Exam: Expect "Explain the Waterfall model with its advantages and disadvantages" or "When should the Waterfall model be used?" Draw the step diagram, list the three "when to..."

واثر فال (آبشار) ماڈل ایک سیدھا اور ترتیب وار طریقہ ہے؛ ہر مرحلہ مکمل ہونے کے بعد ہی اگلا شروع ہوتا ہے اور پیچھے جانا مشکل ہوتا ہے۔

1.2.4 The Agile Model ★ Golden

The Agile model is an iterative approach: the software is built in small, repeated cycles called sprints (usually 2–4 weeks long).

Plan → Design → Build → Test → Review & feedback → Next sprint ↻

- Each sprint plans, builds and tests a small part of the software, and then shows it to the customer.
- Customer feedback is gathered continuously and used in the next sprint.
- Plan → Design → Build → Test → Review & feedback → Next sprint ↻

Advantages (Pros)	Disadvantages (Cons)
Highly flexible to changes.	Less documentation.
Early delivery of working software.	Unpredictable end-time and cost.

Remember: Agile = agile athlete: quick and able to change direction. Think of making a pizza where the customer tastes each slice and you adjust the next one.

Exam: Questions ask you to define Agile, explain sprints, list pros/cons, or say when to use it. Always mention the words "iterative", "sprints (2–4 weeks)" and "continuous..."

ایجائل (Agile) ماڈل میں سافٹ ویئر چھوٹے چھوٹے ادوار میں بنتا ہے جنہیں اسپرینٹ (Sprint) کہتے ہیں، جو عموماً دو سے چار ہفتے کے ہوتے ہیں۔

1.3.1 – 1.3.4 Communication Models and the OSI 7-Layer Model ★ Golden

A communication model explains how data travels from one device to another through a network.

- Because sending data is complicated, the job is split into layers; each layer does one part of the job and passes the data to the next...
- The OSI (Open Systems Interconnection) model was developed by ISO (International Organization for Standardization) in 1984.
- It divides communication into 7 layers.

Layer	Function	Protocols / examples
7. Application	Provides network services directly to user applications (browser, email program).	HTTP, HTTPS, FTP, SMTP, DNS
6. Presentation	Data translation, encryption (security) and compression.	SSL/TLS, JPEG, MP3, ASCII
5. Session	Opens, maintains and closes communication sessions.	NetBIOS, RPC
4. Transport	Reliable end-to-end delivery, error checking, port numbering.	TCP (reliable), UDP (fast)



The 7 layers of the OSI model with the mnemonic "Please Do Not Throw Sausage Pizza Away".

Remember: Mnemonic (bottom to top, 1 → 7): "Please Do Not Throw Sausage Pizza Away" = Physical, Data Link, Network, Transport, Session, Presentation, Application. (Top...

Exam: A guaranteed long question: "Explain the seven layers of the OSI model." Draw the 7-layer stack (numbered, layer 7 on top), then write one function and one or two...

کمیونیکیشن ماڈل بتاتا ہے کہ ڈیٹا نیٹ ورک کے ذریعے ایک ڈیوائس سے دوسری تک کیسے پہنچتا ہے۔

2.2 Algorithm ★ Golden

An algorithm is a step-by-step logical procedure used to solve a problem.

- Input — the data it starts with.
- Clear steps — each step has only one meaning.
- A definite end — it must finish after a limited number of steps.
- Output — the result it produces.

Reason	What it means for you
Structured approach	Algorithms provide a structured approach to problem-solving, making complex tasks manageable.
Optimisation	They optimise time and memory usage, which is crucial for large-scale applications (for example, searching millions of records).
Reusability	They can be reused for similar problems, saving time and effort. A sorting algorithm for marks can also sort prices.
Accuracy and consistency	They ensure accuracy and consistency, reducing human error. The same steps always give the same result.
Blueprint	They act as a blueprint (a plan) before actual code implementation, just like an architect's map before building a house.

Remember: Remember the five reasons as S-O-R-A-B: Structured, Optimised, Reusable, Accurate, Blueprint.

Exam: Very common short question: "Define algorithm and write its importance." Write the definition word for word — a step-by-step logical procedure used to solve a problem;...

الگورتھم (Algorithm) کسی مسئلے کو حل کرنے کا مرحلہ وار منطقی طریقہ ہے۔

2.2.2 Algorithm vs Pseudocode ★ Golden

Both algorithms and pseudocode are used in problem-solving, but they serve different purposes in the design phase.

Problem → Algorithm (what) → Pseudocode (how) → Program (code) → Output

- The algorithm says what must be done.
- The pseudocode shows how it could be implemented, using programming-like keywords such as IF, FOR and WHILE, but it still cannot be run on...

Point	Algorithm	Pseudocode
Definition	A step-by-step logical procedure to solve a problem.	A human-readable representation of an algorithm using plain English & keywords.
Nature	Conceptual and language-independent.	Semi-formal and structured, but not executable code.
Purpose	To define what needs to be done logically.	To present how it can be implemented in a readable format.
Form	Usually described in plain steps, bullets or numbered points.	Written using structured constructs (IF, FOR, WHILE) like programming.
Focus	Focuses on the logic and steps of the problem.	Focuses on representing steps clearly before coding.
Audience	For anyone trying to understand the logic (including non-programmers).	Mainly for programmers or students preparing for coding.
Format rules	No fixed format; can be informal.	Follows standard pseudocode conventions for clarity.

Remember: Teacher's analogy: an algorithm is like a recipe ("Bake a cake at 350 degrees"). Pseudocode is like a structured kitchen manual using standard verbs (SET...

Exam: "Differentiate between algorithm and pseudocode" is a favourite 4–5 mark question. Draw a two-column table and give at least five points (definition, nature, purpose,...

الگورتھم بتاتا ہے کہ کیا کرنا ہے، جبکہ سوڈو کوڈ (Pseudocode) بتاتا ہے کہ اسے کیسے کیا جا سکتا ہے۔

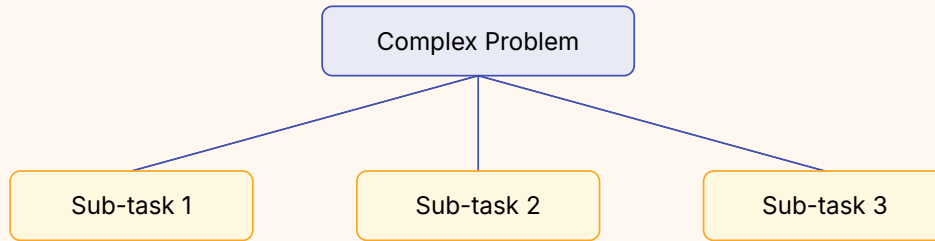
2.3.1 Decomposition ★ Golden

Decomposition is a problem-solving technique in which a complex task or system is broken down into smaller, manageable parts.

Vocabulary → Grammar rules → Tenses → Sentence building

- By dividing a big problem into simpler steps, it becomes easier to understand, develop and solve.
- Each small part is called a sub-task (or sub-problem).
- You solve each sub-task separately, then join the solutions to solve the whole problem.

Sub-task	Description
Learn vocabulary	Learn the words for people, actions and things (e.g., "I", "eat", "apple").
Grammar rules	Learn the sentence structure: Subject → Verb → Object.
Learn tenses	Present, past, future (e.g., "I eat" vs. "I ate").
Sentence building	Combine words using the rules to create valid sentences: "I ate an apple."



Decomposition: one complex problem is split into smaller sub-tasks.

Remember: Think of eating a big roti: you cannot eat it in one bite, so you break it into small pieces.
Decomposition = "break the roti".

Exam: Expect "Define decomposition with a real-life example" or "Explain the benefits of decomposition". Give the definition, draw the small tree (Complex Problem → Sub-task...)

ڈیکمپوزیشن (Decomposition) کا مطلب ہے کسی بڑے اور پیچیدہ مسئلے کو چھوٹے اور آسان حصوں میں تقسیم کرنا۔

2.4.1 Sorting Algorithms: Bubble Sort ★ Golden

Bubble Sort is an elementary sorting algorithm that works by repeatedly comparing adjacent (side-by-side) elements and swapping them if they are in the wrong order.

Compare DATA[j] and DATA[j+1] → Left > Right? → Yes: swap → Move one step right → End of pass: largest is at the end → Repeat N-1 passes

- In each pass, the largest remaining value "bubbles up" to the end of the list, like an air bubble rising to the top of water.
- So after pass 1 the largest value is in the last place, after pass 2 the second largest is in second-last place, and so on.
- A list of N items needs N - 1 passes, and pass K makes N - K comparisons.

Comparison	Pair	Left > Right?	Action	List after
1st vs 2nd	[8, 4], 1, 9, 3	8 > 4 Yes	Swap	4, 8, 1, 9, 3
2nd vs 3rd	4, [8, 1], 9, 3	8 > 1 Yes	Swap	4, 1, 8, 9, 3
3rd vs 4th	4, 1, [8, 9], 3	8 > 9 No	No change	4, 1, 8, 9, 3
4th vs 5th	4, 1, 8, [9, 3]	9 > 3 Yes	Swap	4, 1, 8, 3, 9

Remember: Bubble = neighbours. Bubble Sort only ever compares neighbours (j and j+1). Each pass pushes the biggest bubble to the end. For descending order, just change >...

Exam: The most common question: "Sort the list ... using Bubble Sort. Show each pass." Draw one table per pass with columns Pair, Compare, Action, List after. Mark the element...

ببل سارٹ (Bubble Sort) میں ساتھ ساتھ موجود دو عناصر کا موازنہ کیا جاتا ہے اور اگر وہ غلط ترتیب میں ہوں تو ان کی جگہ بدل دی جاتی ہے۔

2.4.1 Sorting Algorithms: Selection Sort ★ Golden

Selection Sort is a simple sorting algorithm that works by repeatedly selecting the smallest element from the unsorted portion of the list and moving it to its correct position at the front.

Fix position $K \rightarrow$ Compare $DATA[K]$ with $DATA[PTR]$, $PTR = K+1 \dots N \rightarrow DATA[K] > DATA[PTR]? \text{ swap} \rightarrow$ End of pass: smallest is at $K \rightarrow K = K + 1$

- Think of it as filling seats from the front: first find who belongs in position 1, then position 2, and so on.
- In the textbook version, the element at position K is compared with every element after it; whenever a smaller one is found, the two are...
- So at the end of pass K , the K -th smallest value sits in position K .

Compare	List	Pos1 > PosX?	Action	List after
Pos1 vs Pos2	[8], [4], 1, 9, 3	$8 > 4$ Yes	Swap	4, 8, 1, 9, 3
Pos1 vs Pos3	[4], 8, [1], 9, 3	$4 > 1$ Yes	Swap	1, 8, 4, 9, 3
Pos1 vs Pos4	[1], 8, 4, [9], 3	$1 > 9$ No	No change	1, 8, 4, 9, 3
Pos1 vs Pos5	[1], 8, 4, 9, [3]	$1 > 3$ No	No change	1, 8, 4, 9, 3

Remember: Teacher's tip — spot the difference: in Selection Sort the inner loop starts at $outer + 1$ and compares with the fixed position $DATA[outer]$. In Bubble Sort the...

Exam: Typical questions: "Sort ... using Selection Sort, showing all steps", "Differentiate between Bubble Sort and Selection Sort" and "Why is Selection Sort not stable?" In...

سلیکشن سارٹ (Selection Sort) میں فہرست کے غیر ترتیب شدہ حصے سے سب سے چھوٹا عدد چن کر اسے سامنے اس کی درست جگہ پر رکھا جاتا ہے۔

2.4.2 Searching Algorithms: Binary Search ★ Golden

Binary Search is a highly efficient search algorithm that works by repeatedly dividing the search interval in half.

BEG = first, END = last → MID = $\text{INT}((\text{BEG} + \text{END}) / 2)$ → Item = DATA[MID]? Found → Item < DATA[MID]: END = MID-1 →
Else: BEG = MID+1 → Repeat while BEG ≤ END

- BEG (low) — first position of the current part.
- END (high) — last position of the current part.
- MID = $\text{INT}((\text{BEG} + \text{END}) / 2)$ — the middle position. INT means "drop the decimal part".
- Equal → found, stop.

Step	BEG (low)	END (high)	MID	DATA[MID]	Compare	Action
1	1	9	$\text{INT}(10/2) = 5$	5	$7 > 5$	BEG = 6 (go right)
2	6	9	$\text{INT}(15/2) = 7$	7	$7 = 7$	Found, LOC = 7

Remember: Binary Search is how you find a word in a dictionary: open the middle, decide "before or after?", and throw away half the book each time. "Sorted first, then..."

Exam: You will usually be given a sorted list and an item and asked to "search using Binary Search, showing all steps". Always draw a table with BEG, END, MID, DATA[MID],...

بائری سرچ (Binary Search) ایک تیز رفتار طریقہ ہے جو ہر قدم پر تلاش کے حصے کو آدھا کر دیتا ہے۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Analog and Digital Signals:** Do not draw a digital signal with sloping sides.
- **Logic Gates (Basic Gates: AND, OR, NOT):** Do not mix up the AND and OR symbols.
- **Logic Diagrams:** Do not draw a separate wire for a variable each time it appears and then leave one unconnected.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Analog and Digital Signals**. Write: A signal is a changing voltage (or other quantity) that carries information.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Analog signals are continuous signals that change smoothly over time, like a sine wave. (2) Examples: the human voice, a traditional mercury thermometer. — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + a fully labelled diagram copied from the board + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Logic Gates (Basic Gates: AND, OR, NOT)**. Write: Logic gates are electronic circuits that perform Boolean operations.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) They take one or more input bits and produce one output bit. (2) They are the building blocks of all digital systems — a processor contains billions of them. — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + a complete truth table and a labelled gate / logic diagram + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Logic Diagrams**. Write: A logic diagram shows how logic gates are connected to create a circuit.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) It is a "picture" of a Boolean expression. (2) To draw one, use the order of precedence: the operation with the highest priority is drawn first (nearest the inputs), and the last... — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + a complete truth table and a labelled gate / logic diagram + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. What shape of wave does a digital signal make?

3 marks. Explain Analog and Digital Signals with two clear points and one example.

5 marks. Common questions: "Differentiate between analog and digital signals" (3–4 marks) and "Draw an analog and a digital signal." Write the comparison as a table with at least three features (nature, wave...

8 marks. Analog and Digital Signals: 5-mark answer plus a fully labelled diagram copied from the board.

Answers: 1. A square wave 3. Analog signals are continuous signals that change smoothly over time, like a sine wave.; Examples: the human voice, a traditional mercury thermometer. 5. Use the recipe in the blue box.

Model opening: A signal is a changing voltage (or other quantity) that carries information. 8. Same 5-mark body + a fully labelled diagram copied from the board + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Analog and Digital Signals** using the recipe in this topic. Include a fully labelled diagram copied from the board. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Golden blitz — Chapters 3–4

★ only · Chapters 3–4 · Golden blitz

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter from yesterday's Golden · 10–35 min lock the tables from memory · 35–55 min two students write a 5-mark Golden answer on the board · 55–75 min timed 8-mark · 75–85 min error clinic · 85–90 min homework: redraw from memory.

Board today: Fundamentals of Python Programming · Variables · Data Types in Python · Operators and Operands: Arithmetic and Assignment · Relational, Logical and Membership Operators · Bitwise Operators · Control Structures · Selection Statements: if and if-else

★ **Must lock:** Fundamentals of Python Programming, Variables, Data Types in Python, Operators and Operands: Arithmetic and Assignment, Relational, Logical and Membership Operators, Bitwise Operators, Control Structures, Selection Statements: if and if-else, Selection Statements: if-elif-else and Nested if, Repetition Statements: the for Loop and range(), The while Loop, break and continue, Nested Loops and for vs while, Keys and Integrity Constraints, Entity Relationship Model (ER-Model), Referential Integrity, Creating Queries in MS Access

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Write every table and diagram on this day from memory on the board, then check the cards.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. What shape of wave does a digital signal make?
2. What is the output of an OR gate when $A = 0$ and $B = 0$?
3. In $Y = A + B \cdot C$, which gate is drawn first?

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

3.4 Fundamentals of Python Programming ★ Golden

The basic structure of a Python program is simple and clear.

- A program takes data, processes it using statements, and finally shows output.
- Python runs the statements top to bottom, one after another (sequentially).
- A statement is one complete instruction, usually written on one line, such as `print("Hi")` or `x = 5`.

```
# This is a comment. It explains the code below.  
print("Hello, World!")  
if 5 > 2:  
    # The space before 'print' is the indentation  
    print("Five is greater than two!")
```

Remember: Think of indentation like the points and sub-points in your notebook. Sub-points are written a little to the right to show they "belong" to the main point. A...

Exam: Common questions: "What is the purpose of comments?", "Why is indentation important in Python?", "How is Python different from C++/Java in defining blocks?" Write:...

پائنتھن پروگرام اوپر سے نیچے ترتیب سے چلتا ہے۔

3.4.1 Variables ★ Golden

A variable is a named space in computer memory used to store data.

- Can contain letters, numbers and underscores (_) only — no spaces or symbols like -, @, \$.
- Must start with a letter or underscore, NOT a number (1name is invalid).
- Are case-sensitive: age and Age are two completely different variables.
- Cannot be a Python keyword (a reserved word such as if, for, while, else, True).

```
name = "Ali"  
age = 16  
my_height = 5.7  
score1 = 100
```

Remember: Choose meaningful names: total_marks is far clearer than tm or x. Use lowercase words joined by underscores (this style is called snake_case).

Exam: A favourite question: "Which of these variable names are valid? Give reasons." Check each name against all four rules and state the rule broken, e.g. "2nd_place —...

ویری ایبل (Variable) میموری میں ایک نام والی جگہ ہے جس میں ڈیٹا رکھا جاتا ہے، بالکل ایک لیبل لگے ڈیے کی طرح۔

3.4.1 Data Types in Python ★ Golden

A data type specifies the kind of data a variable holds — a whole number, a decimal, text, True/False, or a collection of items.

- The type decides what you can do with the data (you can divide numbers, but not names).
- Python has several built-in data types:

Name	Type	Description	Example
Integer	<code>int</code>	Whole numbers without decimals (positive, negative or zero).	<code>age = 16</code>
Floating point	<code>float</code>	Numbers with a decimal point.	<code>height = 5.7</code>
String	<code>str</code>	Text data, enclosed in quotes (" " or ' ').	<code>name = "Sammy" or year = "2026"</code>
Boolean	<code>bool</code>	Logical value: True or False (capital T / F).	<code>is_student = True</code>
List	<code>list</code>	Ordered, changeable collection in square brackets [] .	<code>marks = [85, 90, 75]</code>
Tuple	<code>tuple</code>	Ordered, unchangeable sequence in parentheses () .	<code>coords = (10, 20)</code>
Dictionary	<code>dict</code>	Key : value pairs in curly braces { } .	<code>student = {"name": "Ahmed", "age": 16}</code>
Set	<code>set</code>	Unordered collection of unique items (no duplicates).	<code>vowels = {'a', 'e', 'i'}</code>

Remember: Brackets decide the collection: [] List (changeable), () Tuple (locked), { } with key:value = Dictionary, { } with single items = Set (unique). Think: a list...

Exam: Expect: "Name any four data types with examples" or "Differentiate list and tuple". For a difference question, use a two-column table: list — [], changeable, e.g. marks...

ڈیٹا ٹائپ (Data Type) بتاتی ہے کہ ویری ایبل میں کس قسم کا ڈیٹا ہے۔

3.4.3 Operators and Operands: Arithmetic and Assignment ★ Golden

An operand is the value or variable being worked on.

- An operator is the symbol that performs the action.
- Together they form an expression, which Python evaluates to a single value.
- In $10 + 20$: 10 is an operand, + is the operator, 20 is an operand, and the result is 30.

Operator	Purpose	Example	Result
+	Addition	$5 + 3$	8
-	Subtraction	$10 - 4$	6
*	Multiplication	$4 * 2$	8
/	Division (result is always a float)	$9 / 2$	4.5
//	Floor division (drops the decimal part)	$9 // 2$	4
%	Modulus (remainder)	$9 \% 2$	1
**	Exponentiation (power)	$2 ** 3$	8

```
print(5 + 3)
print(10 - 4)
print(4 * 2)
print(9 / 2)
print(9 // 2)
print(9 % 2)
print(2 ** 3)
print(8 / 2)
```

Remember: Three kinds of division: / = exact answer (float), // = quotient only ("floor"), % = remainder only. A number is even when $n \% 2 == 0$.

Exam: You will be asked to evaluate expressions such as $17 // 5$, $17 \% 5$, $2 ** 4$ or to trace compound assignment. Show each step and write the final value with its type (e.g....

آپرینڈ (Operand) وہ ویلیو ہے جس پر عمل ہوتا ہے اور آپریٹر (Operator) وہ علامت ہے جو عمل کرتی ہے۔

3.4.3 Relational, Logical and Membership Operators ★ Golden

3. Relational (comparison) operators compare two values and return a Boolean (True or False).

- Relational (comparison) operators compare two values and return a Boolean (True or False).
- They are used in conditions.

Op	Description	Example (a = 10, b = 5)	Result
=	Equal to	a == b	False
≠	Not equal to	a != b	True
>	Greater than	a > b	True
<	Less than	a < b	False
≥	Greater than or equal to	a ≥ 10	True
≤	Less than or equal to	a ≤ b	False

```
a = 10
b = 5
print(a == b)
print(a != b)
print(a > b)
print(a < b)
print(a ≥ 10)
print(a ≤ b)
```

Remember: AND is strict, OR is generous. AND is like a teacher who needs both homework and attendance; OR is like a shop that accepts cash or card. Order of evaluation:...

Exam: Typical: "Evaluate a > b and b > 3 for a = 10, b = 5" or "Write the truth table of and/or". Write each sub-condition's value first, then combine. Remember: == compares,...

ریلیشنل آپریٹرز (Relational Operators) دو ویلیوز کا موازنہ کر کے True یا False دیتے ہیں۔

3.4.3 Bitwise Operators ★ Golden

6. Bitwise operators treat numbers as binary digits (bits: 0s and 1s) and work on them bit by bit, column by column.

- AND (&) — gives 1 only if both bits are 1, otherwise 0.
- OR (|) — gives 1 if at least one bit is 1.
- XOR (^) — gives 1 if the bits are different, 0 if they are the same.
- NOT (~) — inverts all bits (0 → 1, 1 → 0). In Python the result is the number made negative, minus 1: $\sim x = -(x + 1)$. So ~ 10 gives -11.

Bit A	Bit B	A & B	A B	A ^ B	~A
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

```
a = 10
b = 6
print(bin(a), bin(b))
print(a & b)
print(a | b)
print(a ^ b)
print(~a)
```

Remember: Shortcuts: $x > n = x // 2^n$, $\sim x = -x - 1$. For AND/OR/XOR always write both numbers in binary, one under the other, with the same number of bits.

Exam: Exams ask you to "evaluate $12 \& 10$, $12 | 10$, $12 \wedge 10$, ~ 5 , $7 >> 1$ showing binary working". Marks are given for the binary conversion and column working, not just the...

بٹ وائز آپریٹرز (Bitwise Operators) اعداد کو بائری میں بدل کر ہر بٹ پر الگ الگ کام کرتے ہیں۔

3.5 Control Structures ★ Golden

Control structures decide the order in which statements are executed in a program (the "flow of control").

- Sequence — do the steps one after another, top to bottom.
- Selection — choose a path depending on a condition (if, else).
- Repetition — repeat steps (loops: for, while).

Remember: Remember "S-S-R": Sequence, Selection, Repetition — "Step, Select, Repeat".

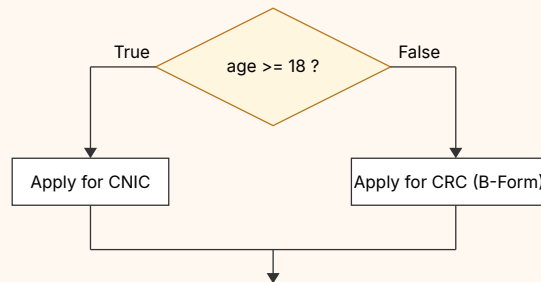
Exam: "Name and explain the three basic control structures" is a very common short question. Give a one-line definition and a two-line Python example for each.

کنٹرول اسٹرکچرز (Control Structures) طے کرتے ہیں کہ پروگرام کی ہدایات کس ترتیب سے چلیں گی۔

3.5.2 Selection Statements: if and if-else ★ Golden

Selection changes the flow of a program based on a condition — an expression that is either True or False.

- It helps a program respond intelligently to different situations.
- The if statement executes a block of code only if the condition is True.
- If the condition is False, the block is completely skipped.



An if-else statement: one diamond (condition), two paths, then the paths join again.

```
if condition:
    statement(s)    # runs only when condition is True
```

Remember: Formatting rule: put a colon (:) after if and else, and indent the code inside the blocks. else never has a condition of its own.

Exam: Asked as: "Write a program that inputs a number and prints whether it is even or odd / positive or negative / pass or fail." Write the full program with `int(input())`,...

سلیکشن (Selection) میں پروگرام شرط کی بنیاد پر فیصلہ کرتا ہے۔

3.5.2 Selection Statements: if-elif-else and Nested if ★ Golden

3. The if-elif-else statement handles many conditions.

- The if-elif-else statement handles many conditions.
- (elif is short for "else if".) Python checks the conditions from top to bottom.
- As soon as it finds a True condition, it runs that block and ignores all the rest.
- If none is True, the else block runs.

marks	>= 80?	>= 70?	>= 60?	Output
92	True	(not checked)	(not checked)	A-1 Grade
75	False	True	(not checked)	A Grade
60	False	False	True	B Grade
41	False	False	False	Fail or below B

Remember: In an if-elif-else ladder, put the strictest condition first (highest marks first). Think of a ladder: you step off at the first rung that fits.

Exam: Grade-calculator programs (if-elif-else) and "largest of three numbers" are very common long questions. Check conditions in the right order, indent nested blocks...

if-elif-else میں کئی شرطیں اوپر سے نیچے ترتیب سے دیکھی جاتی ہیں، اور جو شرط سب سے پہلے درست ہو اسی کا بلاک چلتا ہے، باقی سب چھوڑ دی جاتی ہیں۔

3.5.3 Repetition Statements: the for Loop and range() ★ Golden

A repetition statement (loop) lets a block of code run again and again.

- This greatly reduces the need to write the same code many times.
- Each single run of the loop body is called an iteration.
- The for loop is used when the exact number of iterations is known.

Syntax	How it works	Example	Values produced
<code>range(stop)</code>	From 0 up to, but not including , stop.	<code>range(3)</code>	0 1 2
<code>range(start, stop)</code>	From start up to, but not including, stop.	<code>range(2, 5)</code>	2 3 4
<code>range(start, stop, step)</code>	From start towards stop, jumping by step.	<code>range(1, 6, 2)</code>	1 3 5

```
for i in range(3):  
    print(i)
```

Remember: range stops one short: range(1, 11) gives 1 to 10. To include a number n, write range(..., n + 1).

Number of values in range(a, b) = b - a.

Exam: Expect "predict the output" questions on range() with a step, and programs such as "print a table", "sum of first n numbers", "even numbers from 1 to 20". Always state...

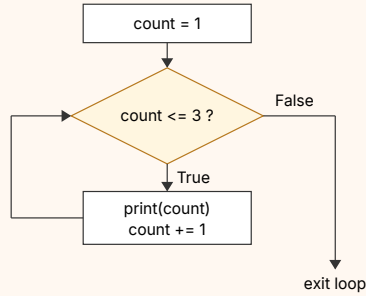
لوپ (Loop) کوڈ کے ایک بلاک کو بار بار چلاتا ہے۔

3.5.3 The while Loop, break and continue ★ Golden

2. The while loop repeats a block of code as long as a condition remains True.

- break — exits (stops) the loop entirely and immediately.
- continue — skips the rest of the current iteration and jumps to the next one.

count	count <= 3?	Printed	count after update
1	True	1	2
2	True	2	3
3	True	3	4
4	False	—	loop ends



A while loop: test the condition, run the body, go back and test again.

```
count = 1
while count ≤ 3:
    print(count)
    count += 1 # VERY IMPORTANT: update the condition!
```

Remember: break = "exit the room"; continue = "skip this turn". For every while loop, check the I-T-U rule: Initialise, Test, Update.

Exam: "Differentiate break and continue with examples" and "What is an infinite loop?" are frequent. Use the two side-by-side programs above with their outputs — examiners...

while لوپ اس وقت تک چلتا رہتا ہے جب تک شرط درست (True) رہے، اور یہ تب استعمال ہوتا ہے جب پہلے سے معلوم نہ ہو کہ کتنی بار دہرانا ہے۔

3.5.3 Nested Loops and for vs while ★ Golden

A nested loop is a loop inside another loop.

- The inner loop runs completely, from start to finish, for every single iteration of the outer loop.
- If the outer loop runs 2 times and the inner loop runs 3 times, the inner body runs $2 \times 3 = 6$ times.

for loop	while loop
Used when the number of iterations is known .	Used when the number of iterations is unknown (condition-based).
Automatically updates the loop variable.	Requires a manual update of the counter (risk of an infinite loop).
Cleaner and shorter for counting tasks.	Better for dynamic situations (like "keep playing until the user says Stop").
<code>for i in range(1, 4): print(i)</code>	<code>i = 1 / while i ≤ 3: / print(i) / i += 1</code>

```
for outer in range(1, 3):      # runs 2 times (1, 2)
    for inner in range(1, 4):  # runs 3 times per outer (1, 2, 3)
        print(f"Outer: {outer}, Inner: {inner}")
```

Remember: Ask yourself: "Do I know how many times?" Yes → for. "Do I only know when to stop?" → while. Nested loops are like a clock: the minute hand (inner loop) goes...

Exam: "Differentiate between for and while loops" (give three points and an example of each) and "How many times will the inner statement run?" (multiply the counts) are...

نیسٹڈ لوپ (Nested Loop) میں ایک لوپ دوسرے لوپ کے اندر ہوتا ہے، اور باہر والے لوپ کے ہر چکر میں اندر والا لوپ شروع سے آخر تک مکمل چلتا ہے۔

4.3.1 Keys and Integrity Constraints ★ Golden

A key is a field (or a set of fields) that uniquely identifies records and links tables.

- Integrity constraints are the rules that keep those keys honest.
- Any field (or combination) that could uniquely identify a row.
- A table may have several.

Table	Field	Role
Student	RollNo	Primary key
Student	BFormNo	Alternate key
Result	RollNo	Foreign key → Student.RollNo
Result	ExamID + RollNo	Could be a composite primary key

Trap: A phone number is a bad PK: people change numbers, two siblings may share one, and some students have none (NULL).

Exam: Board questions: "Differentiate primary and foreign key" (table of 4 points) and "Why can a PK not be NULL?" Because a NULL would mean "this row has no identity", so the...

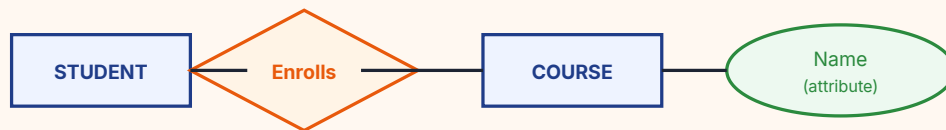
کینڈیڈیٹ کی وہ فیلڈ ہے جو ریکارڈ کی شناخت کر سکے۔

4.4 Entity Relationship Model (ER-Model)

★ Golden

The ER model is the architect's blueprint: you draw what will be stored and how it connects before you create tables in Access.

Component	Meaning	Symbol	Example
Entity	A real-world object we store data about	Rectangle	STUDENT, TEACHER, COURSE
Attribute	A property of an entity. PK is underlined	Oval	StudentID, Name, Class
Relationship	How two entities are connected	Diamond	Enrolls, Teaches, Has



Rectangle = entity, diamond = relationship, oval = attribute. Underline the primary-key attribute.

Trap: The M:N rule.

Exam: Always state cardinality on the diagram (1 and M at the ends of the line). For M:N, the long question is "how do you resolve it?" — draw the junction entity with two...

ای آر ماڈل ڈیٹا بیس کا خاکہ ہے۔

4.5 Referential Integrity ★ Golden

Referential integrity means every foreign-key value in a child table must match an existing primary-key value in the parent — or be NULL if the design allows it.

- If Result.StudentID is a FK to Student.StudentID, you cannot type Result for student 999 if 999 is not in Student.
- You also cannot delete student 101 while results for 101 still exist — unless you first handle those child rows.

ID	Name
101	Ali
102	Sara

Trap: Cascade Delete is powerful and dangerous.

Exam: Write: definition, why it matters (no orphans), then one numbered example of Cascade Update and one of Cascade Delete. Mention parent vs child.

ریفرنشل انٹیگریٹی کا مطلب ہے بچہ ٹیبل کی فارن کی ہمیشہ والدین کی موجود پرائمری کی طرف اشارہ کرے۔

4.11 Creating Queries in MS Access ★ Golden

A query retrieves, filters and analyses data without changing the original tables.

- Create tab → Query Design.
- Show Table → add Student → Close.
- Double-click the fields you want; they drop into the grid.
- Red Run (!) button executes it.

Goal	Where to type	Example
Single criterion	Criteria row, one column	Department → "Computer Science"
AND (both must match)	Same Criteria row, two columns	Name "Ali" and Department "Computer Science"
OR (either may match)	Criteria row and the or: row	Department "IT" or "Computer Science"

Trap: Text criteria need quotes: "IT".

Exam: Remember: AND = same row in the grid; OR = different rows. Examiners draw a blank grid and ask you to fill Criteria and or:.

کوئری ٹیبل کو بدلے بغیر سوال پوچھتی ہے۔

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Fundamentals of Python Programming:** IndentationError?
- **Variables:** = is not "equals" in the maths sense.
- **Data Types in Python:** True and False must start with a capital letter.

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Fundamentals of Python Programming**. Write: The basic structure of a Python program is simple and clear.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) A program takes data, processes it using statements, and finally shows output. (2) Python runs the statements top to bottom, one after another (sequentially). — + example.

5 marks: 3-mark structure + four-point explanation + the worked example from this topic.

8–10 marks: 5-mark structure + a complete truth table and a labelled gate / logic diagram + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Variables**. Write: A variable is a named space in computer memory used to store data.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Must start with a letter or underscore, NOT a number (1name is invalid). (2) Are case-sensitive: age and Age are two completely different variables. — + example.

5 marks: 3-mark structure + four-point explanation + the worked example from this topic.

8–10 marks: 5-mark structure + working code with correct indentation and a short trace of one example + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Data Types in Python**. Write: A data type specifies the kind of data a variable holds — a whole number, a decimal, text, True/False, or a collection of items.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) The type decides what you can do with the data (you can divide numbers, but not names). (2) Python has several built-in data types: — + example.

5 marks: 3-mark structure + table: Name / Type / Description / Example + the worked example from this topic.

8–10 marks: 5-mark structure + working code with correct indentation and a short trace of one example + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. Predict the output:

3 marks. Explain Fundamentals of Python Programming with two clear points and one example.

5 marks. Common questions: "What is the purpose of comments?", "Why is indentation important in Python?", "How is Python different from C++/Java in defining blocks?" Write: comments start with # and are...

8 marks. Fundamentals of Python Programming: 5-mark answer plus a complete truth table and a labelled gate / logic diagram.

Answers: 1. B then D on separate lines (line 1 is a comment; $3 > 7$ is false so C is skipped) 3. A program takes data, processes it using statements, and finally shows output.; Python runs the statements top to bottom, one after another (sequentially). 5. Use the recipe in the blue box. Model opening: The basic structure of a Python program is simple and clear. 8. Same 5-mark body + a complete truth table and a labelled gate / logic diagram + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Fundamentals of Python Programming** using the recipe in this topic. Include a complete truth table and a labelled gate / logic diagram. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Golden blitz — Chapters 5–6

★ only · Chapters 5–6 · Golden blitz

TODAY'S 90 MINUTES

90 minutes. 0–10 min starter from yesterday's Golden · 10–35 min lock the tables from memory · 35–55 min two students write a 5-mark Golden answer on the board · 55–75 min timed 8-mark · 75–85 min error clinic · 85–90 min homework: redraw from memory.

Board today: Artificial Intelligence (AI) · Internet of Things (IoT) & Components · Information Sources (Primary, Secondary, Tertiary) · Assistive Technologies · Types of Data · Data-Collection Strategies · Interviews and Surveys · Prototypes, Observation and Simulations

★ **Must lock:** Artificial Intelligence (AI), Internet of Things (IoT) & Components, Information Sources (Primary, Secondary, Tertiary), Assistive Technologies, Types of Data, Data-Collection Strategies, Interviews and Surveys, Prototypes, Observation and Simulations, Primary and Secondary Data, Case Study: Digital Inquiry Project, Steps 1 – 3: Advanced Search and Methodology, Steps 4 – 6: Survey Preparation and Primary Data Collection, Steps 7 – 9: Secondary Data and Spreadsheet Organisation, Steps 10 – 12: Data Analysis, Conclusion and Recommendations, The Final Digital Artefact

If the batch is behind, drop non-Golden cards. Never skip a Golden or an exam day.

Write every table and diagram on this day from memory on the board, then check the cards.

STARTER TEST

0–10 min. Books closed. Three 1-mark questions from yesterday. Marks go in the batch register.

1. Predict the output:
2. Valid or invalid? _total, roll-no, class, marks2026, 2marks
3. What is the data type of each? 3.0, "3", 3, False, [3]

کل کے گولڈن سے تین سوال زبانی پوچھیں؛ نمبر رجسٹر میں لکھیں۔

5.1.1 Artificial Intelligence (AI) ★ Golden

Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding language and making decisions.

IoT (Eyes) — collects data → Analytics — sorts data → AI (Brain) — decides → Action

- Artificial — it is created by humans; it is not natural.
- Intelligence — it can perform tasks that require thinking.
- Face recognition on a mobile phone to unlock the screen.
- YouTube suggesting videos based on your interests.

Remember: Eyes – Sort – Brain – Do. IoT is the eyes, Analytics sorts, AI is the brain, and the actuator does the action. Think of a cricket fielder: his eyes see the...

Exam: Common questions: "Define AI and give two examples" (2–3 marks), and "Explain how AI, IoT and Data Analytics work together" (4–5 marks). To score full marks: (1) write...

مصنوعی ذہانت (AI) کو "مصنوعی" اس لیے کہتے ہیں کہ اسے انسانوں نے بنایا ہے، اور "ذہانت" اس لیے کہ یہ ایسے کام کر سکتی ہے جن کے لیے سوچنا پڑتا ہے۔

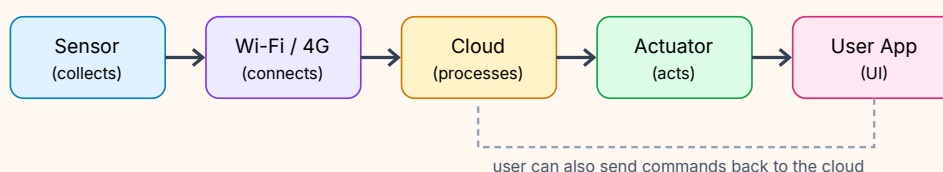
5.1.2 Internet of Things (IoT) & Components ★ Golden

The Internet of Things (IoT) is a system in which physical devices are connected to the Internet and can collect and share data to perform tasks automatically.

Sensor (Collects) → Wi-Fi (Connectivity) → Cloud (Processes) → Actuator (Acts) → User App (UI)

- The "things" are ordinary objects — watches, cameras, cars, fans, water pumps — that have been given sensors and an internet connection so...
- Source example: A smartwatch collects your heart-rate data and sends it to your mobile phone, where it is displayed.

#	Component	What it does	Examples
1	Sensors	Devices that collect information from the physical world. They are the "eyes and ears" of the system.	Temperature sensor, motion sensor, fingerprint sensor, smoke sensor, soil moisture sensor
2	Actuators	Devices that perform physical actions in response to system commands. They convert digital signals into physical movement.	Electric motor, switch, automatic door opener, water valve, fan controller
3	Connectivity	The connection that allows devices to send and receive data . Without it, sensors can't talk to the system.	Wi-Fi, Bluetooth, ZigBee, 4G/5G
4	Data Processing	Changing raw data into useful information . Steps: collecting, organising, analysing and deciding.	Checking if the room temperature is too hot to trigger the AC
5	Cloud Computing	Storing data and using services over the internet instead of local hard drives. Handles massive IoT data.	Google Drive, iCloud, online LMS
6	User Interface (UI)	The component that enables interaction between the user and the device . Helps users see information and give commands.	Mobile apps, touch screens, voice commands



The six IoT components working as one system (data processing happens inside the cloud).

Remember: Sensor senses, Actuator acts. Both words give you the answer: sensor = see/sense (input), actuator = act (output). Remember the six components with "Some Ants..."

Exam: "Define IoT and explain its components" is one of the most frequent long questions. Write the definition with the three key ideas — physical devices, connected to the...

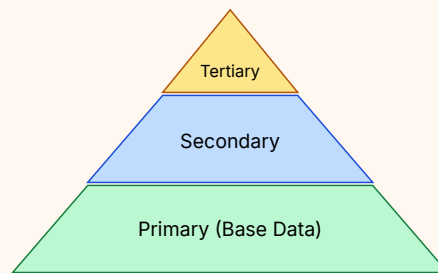
انٹرنیٹ آف تھنگز (IoT) ایک ایسا نظام ہے جس میں مختلف آلات (جیسے گھڑیاں، کیمرے، گاڑیاں) انٹرنیٹ سے جڑے ہوتے ہیں اور خودکار طریقے سے...

5.3 Information Sources (Primary, Secondary, Tertiary) ★ Golden

Information sources are ways to find knowledge and data.

- The internet has vast information, but not all of it is accurate.
- Sources are divided into three main categories.
- Primary sources give original, firsthand information straight from people, events or official records.

Primary (firsthand)	Secondary (explained by others)	Tertiary (summarised, quick reference)
Interview with a flood victim in Dadu	Newspaper report on the 2022 floods	Encyclopedia entry "Floods in Pakistan"
Photograph taken at an event	Textbook chapter explaining the event	Wikipedia article
Personal diary or letter	Biography written by a historian	Dictionary
NADRA official records (CNIC data)	Pakistan Bureau of Statistics report analysing census data	Handbook / almanac of facts
Original survey responses	Research paper analysing the survey	Index or list of research papers
Sensor readings from an IoT weather station	TV news analysis of the weather	A school "fact sheet" on climate
Quaid-e-Azam's original speech recording	Article discussing the speech	Encyclopedia summary of his life



The information pyramid: primary data is the base; each higher level is built from the levels below it and is more summarised.

Remember: Rule of thumb (Author – Date – References): If an online article does not state the author's name, the date of publication, or list its references, it should...

Exam: Typical questions: "Differentiate between primary, secondary and tertiary sources with two examples each" (long/short), "Classify the following sources..." (MCQ/short),...

معلومات کے ذرائع علم اور ڈیٹا تلاش کرنے کے طریقے ہیں، جنہیں تین اقسام میں تقسیم کیا جاتا ہے۔

5.5 Assistive Technologies ★ Golden

Assistive technologies are devices, software and systems that help people with disabilities or functional limitations (difficulty in seeing, hearing, moving, speaking, remembering, etc.).

Mobility → Sensory → Communication → Cognitive

- They allow more people to participate equally in school, work and society.
- There are four categories:

Category	Purpose	Examples
1. Mobility aids	Help people move from one place to another.	Smart wheelchairs, stair lifts
2. Sensory aids	Support vision and hearing to improve interaction.	Hearing aids, screen readers (VoiceOver), Braille displays
3. Communication aids	Assist individuals with speech difficulties to express themselves.	Speech-generating devices, text-to-speech apps
4. Cognitive / learning aids	Help people with memory, reading and understanding.	Audiobooks, spell-checkers, digital reminder apps

Remember: Four categories = "My Small Cat Cries": Mobility, Sensory, Communication, Cognitive. Link each to a body function: legs (mobility), eyes/ears (sensory), mouth...

Exam: Expect: "What are assistive technologies? Explain their categories with examples" (long), "Name two sensory aids" (short), or "A screen reader is an example of ___"...

معاون ٹیکنالوجیز (Assistive Technologies) وہ آلات، سافٹ ویئر اور نظام ہیں جو معذور افراد کی مدد کرتے ہیں تاکہ وہ اسکول، کام اور مع...

6.3 Types of Data ★ Golden

Data means facts, numbers, words or observations that we collect.

- The data gathered through a survey, an investigation or a classroom project can be described in two ways: qualitatively or quantitatively.
- Knowing the difference is critical, because each type is analysed in a different way.

Type of data	Nature	Examples	Best used for
Qualitative	Descriptive, non-numerical	Opinions, reasons, behaviour	Understanding why and how something happens
Quantitative	Numerical, countable or measurable	Hours, scores, percentages, totals	Comparing results and making graphs/charts

Remember: QUALitative = QUALity = words and feelings. QUANTitative = QUANTity = numbers and counts. Qualitative answers why/how; quantitative answers how many/how much.

Exam: Common questions: "Differentiate between qualitative and quantitative data" (4–5 marks) or "Classify the following data". Write the answer as a table with four rows:...

کیفیتی ڈیٹا (Qualitative Data) الفاظ پر مشتمل ہوتا ہے اور بتاتا ہے کہ لوگ کیا سوچتے، محسوس کرتے یا تجربہ کرتے ہیں، جیسے آن لائن کل...

6.4 Data-Collection Strategies ★ Golden

Digital literacy is not limited to finding information that already exists on the internet.

Interviews → Surveys → Prototypes → Observation → Simulations

- You should also learn how to gather original (new) information yourself.
- A data-collection strategy is a planned method for gathering this information.
- The five common strategies are: Interviews·Surveys·Prototypes·Observation·Simulations

Strategy	How it works (one line)	Usual data type	Best when...
Interview	Planned conversation with a person or small group	Mostly qualitative	You need deep, detailed answers from a few people
Survey	Same questions asked to many people	Quantitative and qualitative	You need answers from many people quickly
Prototype	Users test an early version and give feedback	Both	You want to improve a product or solution
Observation	Watch and record; no questions asked	Notes (qualitative) or counts (quantitative)	You want to see what people actually do
Simulation	Computer model; change variables and see what happens	Mostly quantitative	Real experiments are costly, risky or difficult

Remember: Mnemonic for the five strategies: "I S P O S" – "I Saw People On Screens": Interviews, Surveys, Prototypes, Observation, Simulations.

Exam: "List and explain the data-collection strategies" is a frequent long question. Name all five, give a one-sentence definition and one example for each, and say what type...

ڈیجیٹل خواندگی صرف موجود معلومات تلاش کرنے تک محدود نہیں، بلکہ نئی معلومات خود جمع کرنا بھی اس کا حصہ ہے۔

6.4.1 – 6.4.2 Interviews and Surveys ★ Golden

An interview is a planned conversation used to gather information directly from a person.

- It may be one-to-one (you and one person) or in a small group.
- Questions are planned in advance, so you do not forget anything important.
- Answers are usually qualitative (opinions, reasons, experiences).

Question type	What it looks like	Data it gives
Closed: Yes/No	Do you study with your phone nearby? (Yes / No)	Countable (quantitative)
Closed: Multiple choice	Which activity do you use your phone for most? (Lectures / Notes / Social media / Games)	Categories you can count
Number / scale	How many hours do you use your phone daily? ____	Quantitative
Rating scale	"My phone helps me study." 1 = Strongly disagree ... 5 = Strongly agree	Quantitative (you can average the ratings)
Open-ended	How does your phone help you in studies?	Qualitative (free text)

Remember: Interview = In-depth with few. Survey = Same questions to many. Think of a doctor talking to one patient (interview) vs a census form sent to every house...

Exam: You may be asked to "differentiate between an interview and a survey" or to "identify what is wrong with this survey question and rewrite it". For rewriting, name the...

انٹرویو ایک منصوبہ بند گفتگو ہے جس میں کسی ایک فرد یا چھوٹے گروہ سے براہ راست معلومات حاصل کی جاتی ہیں؛ اس کے جوابات عموماً کیفیتی ہوتے ہیں۔

6.4.3 – 6.4.5 Prototypes, Observation and Simulations ★ Golden

A prototype is a simple early version of a product, model or solution.

- It allows an idea to be tested before time and effort are spent on the final version.
- Users try the prototype and give feedback, and this feedback becomes useful data for improvement.
- Example: Before building a full school app, you draw the screens on paper (or make a few clickable screens in a design tool) and ask five...

Remember: Watch it → Observation. Ask it → Survey/Interview. Try it → Prototype. Model it → Simulation.

Exam: Scenario questions are popular: "Which data-collection method is most suitable for ... ? Give a reason."
Always name the method and justify it using its key feature (e.g....

پروٹوٹائپ کسی پروڈکٹ یا حل کا ابتدائی سادہ نمونہ ہے جسے صارفین آزما کر رائے دیتے ہیں تاکہ اسے بہتر بنایا جا سکے۔

6.5 Primary and Secondary Data ★ Golden

Whenever you use data, it is important to know where it came from.

- The source affects how much control the researcher had over collecting it, and how confidently the results can be used.

Aspect	Primary Data	Secondary Data
Source	Collected directly by the researcher.	Already collected by others.
Examples	Survey responses, interviews, observations.	Reports, articles, websites, databases.
Purpose	Prepared for a specific study.	Reused for a new purpose.
Control	High control over questions and collection process.	Limited control because the data already exists.

Primary Data

Fresh, direct collection
(You do the work)
Surveys · Interviews · Observation

Secondary Data

Existing archives
(Someone else did it)
Books · Reports · Databases

Primary data is collected by you; secondary data was collected by others.

Remember: Primary = "1st hand" (you collect it first). Secondary = "2nd hand" (like a second-hand book – someone else used it first).

Exam: "Differentiate between primary and secondary data" is one of the most-asked questions in this chapter. Draw the four-row table: Source, Examples, Purpose, Control. Learn...

بنیادی ڈیٹا (Primary Data) محقق خود کسی خاص مقصد کے لیے براہ راست جمع کرتا ہے، مثلاً سروے، انٹرویو یا مشاہدہ۔

6.8 Case Study: Digital Inquiry Project ★ Golden

This case study shows how to apply all the ideas of this chapter to solve a real-world problem from start to finish.

1 Question → 2 Search → 3 Collect → 4 Analyse → 5 Create

- An inquiry is an investigation to find the answer to a question.
- Research question: "How does daily mobile phone use affect the study habits of students?" Scenario: Students notice classmates using phones...
- Some say phones help them learn; others say phones waste time.

Workflow stage	Steps
Question & Search	1 Advanced searches · 2 Identify data types · 3 Select methods
Collect	4 Prepare the survey · 5 Choose the sample · 6 Collect primary data ethically · 7 Use secondary data
Analyse	8 Organise in a spreadsheet · 9 Create charts · 10 Analyse results
Create	11 Write the conclusion · 12 Recommendations → final digital artefact

Remember: Workflow in five words: Ask → Search → Collect → Analyse → Create. Every research project you ever do will follow this pattern.

Exam: Expect a long question: "Describe the steps of a digital inquiry project using the example of mobile phone use." Write the research question first, then the 12 steps...

اس کیس اسٹڈی میں تحقیقی سوال یہ ہے: "روزانہ موبائل فون کا استعمال طلبہ کی پڑھائی کی عادات پر کیسے اثر ڈالتا ہے؟"

6.8 Steps 1– 3: Advanced Search and Methodology ★ Golden

Students first search for background information using search engines (such as Google or Bing).

- They use advanced search operators – special symbols or words typed into the search box – to narrow down results and find reliable...
- Suggested queryOperator usedWhat it does "mobile phone use" "study habits" students" " (quotes)Finds the exact phrases "mobile phone use"...
- "academic performance" filetype:pdfFinds only PDF documents (often official reports and research papers).

Suggested query	Operator used	What it does
"mobile phone use" "study habits" students	" " (quotes)	Finds the exact phrases "mobile phone use" and "study habits", not the words scattered randomly.
"academic performance" filetype:pdf	filetype:	Finds only PDF documents (often official reports and research papers).
digital distraction site:.edu	site:	Searches only educational websites (addresses ending in .edu).

Remember: Three operators, three jobs: " " = exact, filetype: = format, site: = where. Note: there is no space after the colon (site:.edu, not site: .edu).

Exam: "Explain any three advanced search operators with examples" is a likely short question. For each, write the operator, an example query, and what it does. Also learn the...

پہلے قدم میں طلبہ سرچ انجن پر جدید سرچ آپریٹرز (Advanced Search Operators) استعمال کرتے ہیں: کوٹیشن مارکس (" ") بالکل وہی جملہ تلا...

6.8 Steps 4 – 6: Survey Preparation and Primary Data Collection ★ Golden

Design clear, unbiased questions. Each question is linked to the type of data it produces: QuestionType of data
How many hours do you use your phone daily?Quantitative Which activity do you use your phone for the most?
Quantitative...

- Design clear, unbiased questions.
- Each question is linked to the type of data it produces: QuestionType of data How many hours do you use your phone daily?Quantitative Which...

Question	Type of data
How many hours do you use your phone daily?	Quantitative
Which activity do you use your phone for the most?	Quantitative (categories)
How does your phone help you in studies?	Qualitative
Suggest one way to use phones responsibly.	Qualitative

Remember: Ethics in five words: Consent, No-Sensitive, Anonymous, Private, Educational – "C-NAPE". Think: "Cover the person's nape (neck) – protect them."

Exam: Questions often ask: "What ethical guidelines should be followed when collecting data?" List all five points. Use the words consent and anonymity – examiners look for...

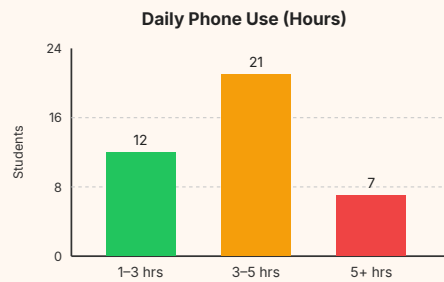
چوتھے قدم میں واضح اور غیر جانبدار سوالات تیار کیے جاتے ہیں۔

6.8 Steps 7 – 9: Secondary Data and Spreadsheet Organisation ★ Golden

Students include a reliable online article stating that "excessive screen time lowers concentration." They will compare this secondary data with their own primary survey results.

- If both agree, the conclusion is stronger.
- If they disagree, the students must think about why (for example, a different age group or country).

Student	Hours	Study Use	Main Activity	Distracted?
1	3	Yes	Lectures	Sometimes
2	5	No	Social Media	Yes
3	2	Yes	Notes	No



Bar chart of daily phone use for 40 students. The counts (12, 21, 7) are estimated from the bar heights in the source chart.

Remember: Chart chooser: Bar = Compare, Pie = Parts of a whole, Line = Change over time.

Exam: You may get the spreadsheet table and be asked to "write formulas to find the total and average hours" or "count students who use the phone for study". Write the full...

ساتویں قدم میں ایک قابل اعتماد آن لائن مضمون کا ثانوی ڈیٹا ("زیادہ اسکرین ٹائم توجہ کم کرتا ہے") اپنے سروے کے بنیادی ڈیٹا سے مواز...

6.8 Steps 10 – 12: Data Analysis, Conclusion and Recommendations ★ Golden

To analyse means to study the data carefully to find patterns and trends.

- Most students (the majority) use phones for 3–5 hours daily.
- Half use it for educational purposes (lectures/notes).
- A clear trend: students who use phones mostly for entertainment report much higher distraction levels than those who use them for study.

Remember: Conclusion formula: "The data shows ... However ... Therefore ..." – state the main finding, mention the other side, then give the overall answer.

Exam: You may be asked to "write a conclusion and two recommendations" for given data. Use the "shows ... however ... therefore" pattern and make sure each recommendation follows...

دسویں قدم میں چارٹس اور اسپریڈشیٹ سے رجحانات تلاش کیے جاتے ہیں: زیادہ تر طلبہ روزانہ تین سے پانچ گھنٹے فون استعمال کرتے ہیں،
نصف ا...

6.8 The Final Digital Artefact ★ Golden

A digital artefact is a final product created using digital technologies.

- Students must create one final artefact that answers the original research question.
- It is not just a decorated file; it must present evidence and conclusions.
- Artefact option Suitable digital tools Main purpose Digital Report MS Word, Google Docs Explain the full study in a structured text document.

Artefact option	Suitable digital tools	Main purpose
Digital Report	MS Word, Google Docs	Explain the full study in a structured text document.
Presentation	PowerPoint, Canva, Slides	Share key findings with a live audience.
Infographic	Canva, Adobe Express	Show the main results visually and quickly on one page.
Dashboard	Power BI, Tableau	Explore data through interactive charts and filters.

Remember: The structure follows the inquiry itself: What (title, question, intro) → How (method) → Found (primary, secondary, charts) → So what (conclusion,...)

Exam: Learn the four artefact options with tools and purpose (a table question), and the ten-part structure in order. "What is a digital artefact?" – stress that it must...

ڈیجیٹل آرٹیفیکٹ (Digital Artefact) ڈیجیٹل ٹیکنالوجی سے بنائی گئی آخری پروڈکٹ ہے جو تحقیقی سوال کا جواب ثبوت اور نتائج کے ساتھ پیش...

ERROR CLINIC

75–85 min. Collect the three most common wrong answers. Correct them in one box on the board.

- **Artificial Intelligence (AI):** Mistake: "AI means robots." Fix: Most AI is just software — YouTube suggestions, spam filters and face unlock have no robot body.
- **Internet of Things (IoT) & Components:** Mistake: Calling a fan controller or water valve a "sensor".
- **Information Sources (Primary, Secondary, Tertiary):** Mistake: Thinking "primary" means "most important" or "tertiary" means "useless".

تین عام غلطیاں بورڈ کے ایک خانے میں درست کریں۔

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Artificial Intelligence (AI)**. Write: Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding language and making decisions.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) Intelligence — it can perform tasks that require thinking. (2) Face recognition on a mobile phone to unlock the screen. — + example.

5 marks: 3-mark structure + four-point explanation + the worked example from this topic.

8–10 marks: 5-mark structure + a complete truth table and a labelled gate / logic diagram + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Internet of Things (IoT) & Components**. Write: The Internet of Things (IoT) is a system in which physical devices are connected to the Internet and can collect and share data to perform tasks automatically.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) The "things" are ordinary objects — watches, cameras, cars, fans, water pumps — that have been given sensors and an internet connection so... (2) Source example: A smartwatch collects your heart-rate data and sends it to your mobile phone, where it is displayed. — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + a fully labelled diagram copied from the board + a one-line conclusion. Method marks are awarded for working.

FULL-MARK ANSWER RECIPE

Train this wording every class: *Define. Explain. Example. Diagram. Working.*

1 mark: Define **Information Sources (Primary, Secondary, Tertiary)**. Write: Information sources are ways to find knowledge and data.

2 marks: definition + one local example (JazzCash, school portal, NADRA, load-shedding).

3 marks: definition + two points — (1) The internet has vast information, but not all of it is accurate. (2) Sources are divided into three main categories. — + example.

5 marks: 3-mark structure + labelled diagram + the worked example from this topic.

8–10 marks: 5-mark structure + a four-row comparison table and one rewritten survey question + a one-line conclusion. Method marks are awarded for working.

ACADEMY DRILL

Timed 20 minutes. Books closed. Mark at once with the answers in this box — this is class work, not homework.

1 mark. Why is AI called "artificial"?

3 marks. Explain Artificial Intelligence (AI) with two clear points and one example.

5 marks. Common questions: "Define AI and give two examples" (2–3 marks), and "Explain how AI, IoT and Data Analytics work together" (4–5 marks). To score full marks: (1) write the two-part meaning of the...

8 marks. Artificial Intelligence (AI): 5-mark answer plus a complete truth table and a labelled gate / logic diagram.

Answers: 1. Because it is created by humans 3. Intelligence — it can perform tasks that require thinking.; Face recognition on a mobile phone to unlock the screen. 5. Use the recipe in the blue box. Model opening: Artificial Intelligence (AI) is the ability of a computer system to perform tasks that normally need human thinking, such as recognising faces, understanding... 8. Same 5-mark body + a complete truth table and a labelled gate / logic diagram + one-line conclusion.

BATCH HOMEWORK

Write an 8-mark answer on **Artificial Intelligence (AI)** using the recipe in this topic. Include a complete truth table and a labelled gate / logic diagram. Then redraw the board work from memory on a fresh page — no notes.

How you will mark it tomorrow: Mark to the 8–10 mark recipe. Any answer with no working is half marks.

کل جمع کرائیں؛ بغیر ورکنگ کے جواب آدھے نمبر ہیں۔

Oral recap at the board

Say every chapter recap aloud, books closed · Recap

TODAY'S 90 MINUTES

90 minutes. 0–10 min settle · 10–80 min each chapter recap said aloud at the board, books closed · 80–90 min rewrite any line the batch could not say. No new topic.

Board today: Say every chapter recap aloud, books closed

★ **Must lock:** All chapters

Phones in the box at the door.

No new cards. One student per chapter at the board. The batch corrects missed lines.

Chapter 1 recap — Computer Systems

- Discrete = counted (stairs). Continuous = measured (ramp). Digital systems use bits 0 and 1; they are reliable, accurate, easy to design and programmable.
- Analog signal = smooth sine wave. Digital signal = square wave HIGH/LOW. Digital data resists noise.
- AND · output 1 only if all inputs 1. OR + output 1 if any input 1. NOT ' inverts. NAND = NOT-AND, NOR = NOT-OR (universal). XOR 1 when inputs differ. XNOR 1 when inputs same.
- Truth table rows = 2^n . SOP = sum of minterms where output is 1. POS = product of maxterms where output is 0.
- K-map: group 1, 2, 4, 8 adjacent 1s (power of 2, wrap-around allowed). Each group drops the variables that change.
- SDLC: requirement → design → implementation → testing → deployment → maintenance. Waterfall = sequential; Agile = short sprints and feedback.
- OSI 7: Physical, Data Link, Network, Transport, Session, Presentation, Application. Mnemonic: Please Do Not Throw Sausage Pizza Away. TCP/IP 4: Network Access, Internet, Transport, Application.

Chapter 2 recap — Computational Thinking & Algorithm

- CT pillars: decomposition, pattern recognition, abstraction, algorithms.
- Algorithm = finite, ordered, unambiguous steps. Pseudocode = structured English, not a programming language.
- Bubble Sort: compare neighbours, swap if out of order; largest "bubbles" to the end each pass. $O(n^2)$.
- Selection Sort: find the minimum in the unsorted part and swap it to the front. $O(n^2)$.
- Linear Search: check one by one. Works on unsorted data. $O(n)$.
- Binary Search: list MUST be sorted. Repeatedly test the middle. $O(\log n)$. Trace with low / mid / high.
- Judge an algorithm on correctness, efficiency (time and space) and clarity. Prefer binary search on large sorted lists; linear search on small or unsorted lists.

Chapter 3 recap — Programming Fundamentals

- Python: high-level, interpreted. Variables are names; types int, float, str, bool, list, dict. `input()` returns str.
- Arithmetic + - * / // % **. Relational. Logical and or not. Bitwise & | ^ ~ << >>.
- Selection: if / if-else / if-elif-else / nested. Test from the top; first True wins.
- for = known count + range. while = unknown count; always update the condition. break leaves; continue skips one.
- Libraries: import math / random / datetime. Third-party via pip: NumPy, Pandas, Matplotlib.
- Syntax = will not run. Runtime = crash + traceback. Logical = runs, wrong answer.

Chapter 4 recap — Data and Analysis

- Data = raw facts. Information = processed data used for decisions. Database = organised related data. DBMS = software that manages it (reduces redundancy, improves security, sharing, integrity).
- Table / relation; row / record / tuple; column / field / attribute. Primary key uniquely identifies a row. Foreign key refers to another table's primary key.
- ER model: entity (rectangle), attribute (oval), relationship (diamond), cardinality 1:1, 1:M, M:N.
- Referential integrity: you cannot orphan a child row. Cascade Update / Cascade Delete copy or remove matching child rows.
- Access objects: tables, forms, queries, reports. Queries filter with criteria, wildcards (*, ?), AND/OR. Totals: Count, Sum, Avg, Group By. Charts visualise the result.

Chapter 5 recap — Application and Impacts of Computing

- IoT = connected devices. Six parts: sensor, actuator, connectivity, processing, cloud, UI. AI = machines that learn and decide. Analytics = collect → store → clean → analyse → visualise.
- Together: IoT (eyes) → Analytics (sort) → AI (brain) → actuator (action).
- Sources: primary (firsthand), secondary (explained by others), tertiary (summaries). No author / date / references → do not trust.
- Assistive tech: mobility, sensory, communication, cognitive. Goal: equal access, independence, inclusion.

Chapter 6 recap — Digital Literacy

- Digital literacy = find, evaluate, create and share information with digital tools, safely.
- Qualitative = words/descriptions. Quantitative = numbers. Strategies: interview, survey, prototype, observation, simulation.
- Primary data you collect; secondary data already exists. Good survey questions are clear, single-idea, unbiased.
- Present with spreadsheets (formulas such as `=AVERAGE()`), slides, infographics and reports. Cite sources. Follow the 12-step inquiry project from question to artefact.

Mock paper — academy hall

2 hours 30 minutes · 75 marks · no notes · Mock

TODAY'S 90 MINUTES

90 minutes. Academy mock morning: sit the paper in 2 hours 30 minutes. Hall conditions — no notes, no phones, no Day 29 peeking. Use leftover time to pack scripts. Do not mark today.

Board today: 2 hours 30 minutes · 75 marks · no notes

★ **Must lock:** Whole syllabus

Phones in the box at the door.

Invigilate as the board will. No phone, no notes, no Day 29 peeking.

Mock paper — Computer Science XI

Time: 2 hours 30 minutes | Total: 75 marks | Answer all questions. Calculators allowed. Write Python with correct indentation.

Section A — Multiple choice (15 × 1 = 15)

1. Which quantity is discrete?
 - a. Room temperature
 - b. Number of desks
 - c. Car speed
 - d. Time of day
2. NAND is called universal because:
 - a. it has two inputs
 - b. any gate can be built from it
 - c. its output is always 1
 - d. it is faster than AND
3. The Boolean expression for XOR is:
 - a. $A \cdot B$
 - b. $A + B$
 - c. $A \oplus B$
 - d. $(A \cdot B)'$
4. OSI layer that chooses the best path (IP addresses) is:
 - a. Transport
 - b. Network
 - c. Data Link
 - d. Session
5. Which SDLC model delivers working software in short sprints?
 - a. Waterfall
 - b. Agile
 - c. Spiral only
 - d. Big Bang
6. Binary Search requires the list to be:
 - a. unsorted
 - b. sorted
 - c. all unique and unsorted
 - d. of even length
7. Worst-case time of Bubble Sort is:
 - a. $O(n)$
 - b. $O(\log n)$
 - c. $O(n^2)$
 - d. $O(1)$
8. `input()` in Python returns a:
 - a. int
 - b. str
 - c. float
 - d. bool
9. `17 // 5` equals:
 - a. 3.4
 - b. 3
 - c. 2
 - d. 3.0
10. A program that runs but prints the wrong total has a:
 - a. syntax error
 - b. runtime error
 - c. logical error
 - d. hardware fault
11. A field that uniquely identifies each record is a:
 - a. foreign key
 - b. primary key
 - c. composite description
 - d. query
12. An actuator in IoT:
 - a. collects temperature
 - b. opens a water valve
 - c. stores cloud files
 - d. draws a chart
13. Wikipedia is a:
 - a. primary source
 - b. secondary source
 - c. tertiary source
 - d. sensor
14. Marks out of 100 are:
 - a. qualitative data
 - b. quantitative data
 - c. tertiary data
 - d. metadata only
15. Which Access object is best for printing a formatted class list?
 - a. Table
 - b. Form
 - c. Query
 - d. Report

Section B — Short questions (10 × 3 = 30)

1. Differentiate analog and digital signals with one example of each.
2. Draw the symbol and truth table of a 2-input NOR gate.
3. List the four pillars of computational thinking in one sentence each.
4. Why can you not use Binary Search on an unsorted attendance list?
5. Write a Python `if-elif-else` that prints Pass if marks ≥ 40 , else Fail.
6. Define primary key and foreign key.
7. Name the six components of IoT.
8. Classify: (a) a photograph you took, (b) a textbook, (c) an encyclopedia.
9. Give two good and two bad survey questions about mobile-phone use in college.
10. State three advantages of a DBMS over paper files.

Section C — Long questions (3 × 10 = 30). Attempt any three.

1. Simplify $F(A,B,C) = \Sigma(1,3,5,7)$ using a 3-variable K-map. Show the grouping and the final expression. Draw the logic diagram.
2. Dry-run Bubble Sort on the list [5, 1, 4, 2]. Show every pass. Then search for 4 using Binary Search on the sorted result, with a low/mid/high table.
3. Write a complete Python program that reads n subject marks, prints the average, and prints "A" if average ≥ 80 , "B" if ≥ 70 , "C" if ≥ 60 , else "D". Include a sample run of four marks: 80, 70, 90, 60.
4. Design an ER diagram and relational schema for a school library (Student, Book, Issue). Mark keys. Write one Access/SQL query that lists all books issued to roll number 101.
5. Explain how IoT, Data Analytics and AI work together on a Sindh farm. Then discuss two positive and two negative impacts of computing on education in Pakistan.

Mark the mock · repair list

Mark to the key. List every ★ Golden the batch missed. · Mark

TODAY'S 90 MINUTES

90 minutes. 0–70 min mark the mock to the sealed key using 1 / 3 / 5 / 8–10 recipes · 70–85 min each student fills the repair list · 85–90 min one missed Golden rewritten from memory.

Board today: Mark to the key. List every ★ Golden the batch missed.

★ **Must lock:** Missed Golden

Phones in the box at the door.

An answer with no working, no diagram or no example cannot score full marks.

Mock paper — answers

Section A

1. b
2. b
3. c
4. b
5. b
6. b
7. c
8. b
9. b
10. c
11. b
12. b
13. c
14. b
15. d

Section B

1. Analog: continuous, sine wave, e.g. human voice. Digital: two levels 0/1, square wave, e.g. computer data. Digital resists noise.
2. NOR symbol: OR-gate shape with a bubble on the output. $Y = (A+B)'$. Truth: 00→1, 01→0, 10→0, 11→0.
3. Decomposition = split a problem. Pattern recognition = reuse similarities. Abstraction = ignore extra detail. Algorithms = write ordered steps.
4. Binary Search repeatedly halves a sorted range. If the list is unsorted, the middle value does not tell you which half to keep, so the method is invalid.
5. `m = int(input("Marks: "))` then `if m ≥ 40: print("Pass") else print("Fail")`.
6. Primary key: unique identifier of a row (e.g. StudentID). Foreign key: a field that refers to the primary key of another table (e.g. StudentID in Issuance).
7. Sensors, actuators, connectivity, data processing, cloud computing, user interface.
8. (a) primary (b) secondary (c) tertiary.
9. Good: "How many hours per day do you use a phone for study?" / "Which apps do you use for class notes?" Bad: "Don't you agree phones are destroying students?" (leading) / "Do you like phones and also use laptops?" (double-barrelled).
10. Less redundancy, easier search and update, better security and multi-user access, integrity constraints. (Any three.)

Section C — outline

1. K-map 1s at m1, m3, m5, m7 form one group of 4 covering B=1 (or C=1 depending on labelling — with standard ABC order the 1s are all cells where C=1). F = C. Diagram: a single wire C to output (or a buffer). Show the map with the group circled.
2. Pass 1: [1,4,2,5]; Pass 2: [1,2,4,5]; Pass 3: already sorted. Sorted [1,2,4,5]. Search 4: low=0 high=3 mid=1 (2<4) → low=2; mid=2 (4=4) found at index 2.
3. Read n, loop n times adding `float(input())`, avg = total/n, then if-elif for A/B/C/D. Sample: total 300, avg 75.0, grade B.
4. Entities Student(StudentID PK, Name, Class), Book(BookID PK, Title, Author), Issue(IssueID PK, StudentID FK, BookID FK, Date). Query: `SELECT Book.Title FROM Issue INNER JOIN Book ON Issue.BookID = Book.BookID WHERE StudentID = 101;`
5. Soil sensor (IoT) → cloud spreadsheet of moisture (Analytics) → model decides "too dry" (AI) → pump (actuator). Education +: LMS, remote classes. Education -: screen addiction, urban/rural device gap.

Sealed key. Open only after the mock is collected.

REPAIR LIST

Repair list. Copy every Golden scored 0 or half. Tomorrow the clinic redos only this list.

#	Golden topic missed	What the batch will rewrite from memory
1		
2		
3		
4		
5		
6		

Repair clinic · then stop

Redo the Day 29 list only. Dismiss by 6 pm. · Repair

TODAY'S 90 MINUTES

90 minutes. 90-minute repair clinic. Redo only the Day 29 list. Say the definition, draw the table, write one 5-mark answer. Dismiss by 6 pm. No new topic.

Board today: Redo the Day 29 list only. Dismiss by 6 pm.

★ **Must lock:** Missed Golden only

Phones in the box at the door.

No new chapter. No new non-Golden card. Each missed Golden: say the definition, draw the table, write one 5-mark answer on the board.

Dismiss by 6 pm. The batch walks into the hall with Golden wording in their heads — not in a book.

نیا باب نہ کھولیں۔ صرف چھوٹی گولڈن دہرائیں۔ شام چھ بجے کلاس ختم کریں۔